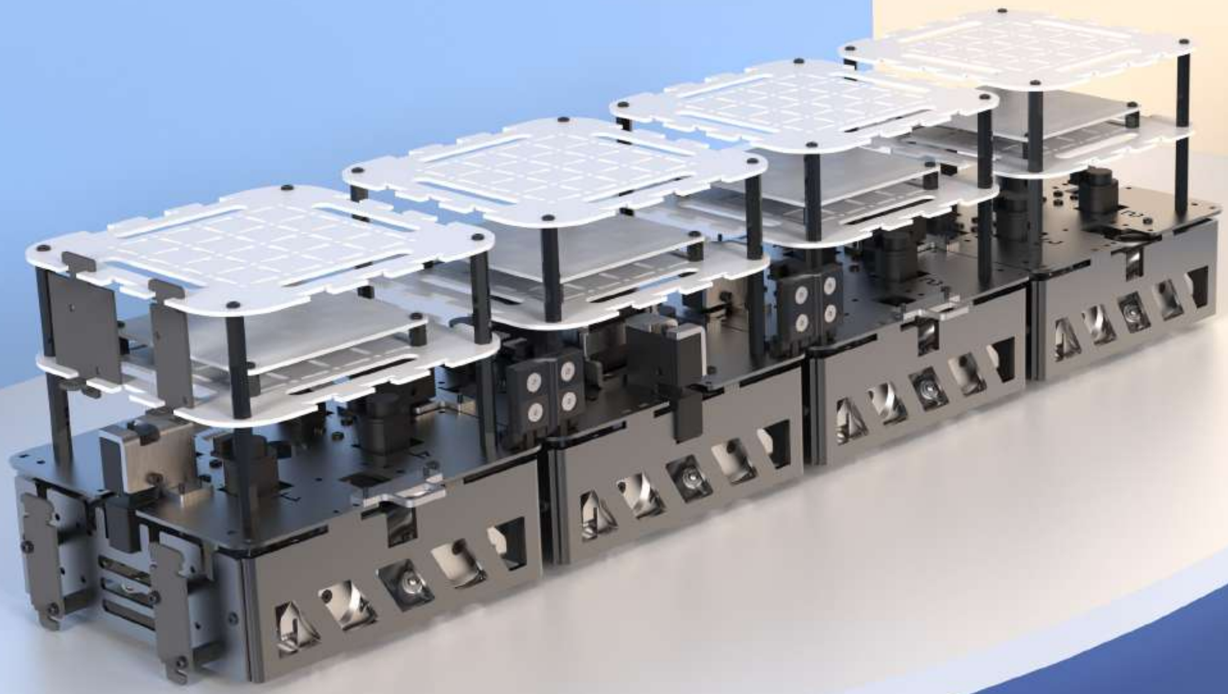
wefaa robotics



smorphi 2

transforming learning with
transformer robots

assembly & info

(contents)

(part list)

All colors of parts are represented accurately here.
In the assembly steps, colors of some parts will be changed for diagram clarity.

- 8 x Acrylic Base Plate
- 4 x Aluminium Base Plate
- 8 x Base Skirt Panel A
- 8 x Base Skirt Panel B
- 8 x Mecanum Wheel (Right)
- 8 x Mecanum Wheel (Left)
- 16 x Mecanum Motor
- 16 x Motor Shaft Sleeve
- 16 x Motor Mount
- 6 x Solenoid
- 6 x Solenoid Latch Mount
- 6 x Solenoid Latch Guide
- 6 x Solenoid Catch
- 6 x Hinge Mount

- 3 x Hinge Mechanism
- 1 x Battery
- 2 x Battery bracket
- 1 x Masterboard (ESP32)
- 4 x Slaveboard
- 1 x Sound Sensor
- 1 x Temperature Sensor
- 4 x IR Sensor
- 1 x Husky Camera
- 1 x Husky Mount
- 10 x Sensor Lock
- 32 x Hex M-F M3 Nylon 45mm
- 28 x Hex F-F M3 Nylon 10mm
- 200 x Cap Screw M3x5

- 45 x Cap Screw M3x10
- 50 x Cap Screw M3x25
- 15 x Countersunk Screw M4x8
- 45 x Hex Nut M3
- 2 x Wing Screw M3x5
- 1 x USB-C Cable
- 8 x 4-pin Connector
- 4 x 8-pin Connector
- 1 x Battery Charger
- 1 x Ceramic Screwdriver CD-25
- 1 x HEX Key 1.5mm
- 1 x Husky Cam Cable
  (included in huskycam package)

(basic assembly tips)

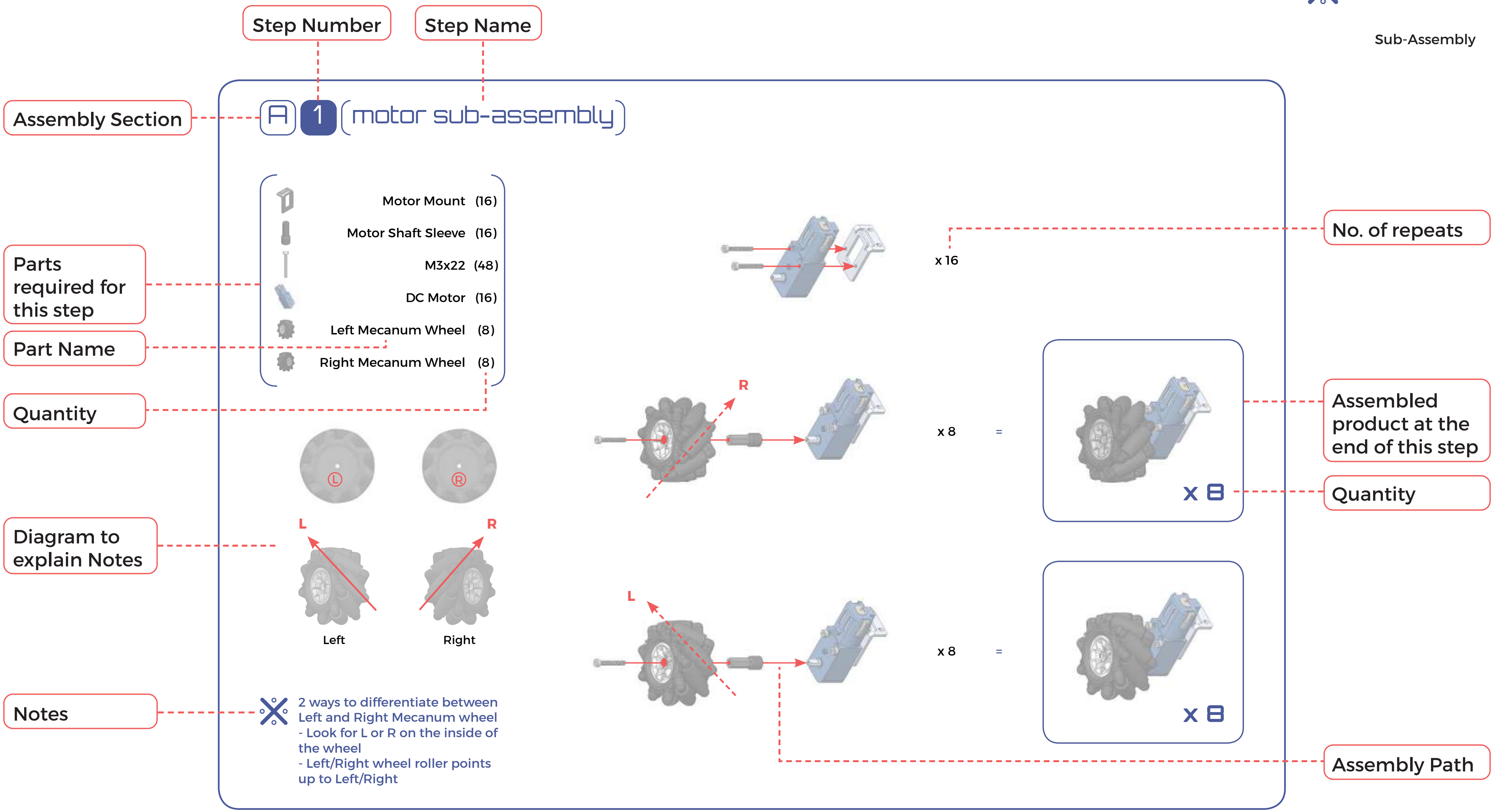
Symbols Used

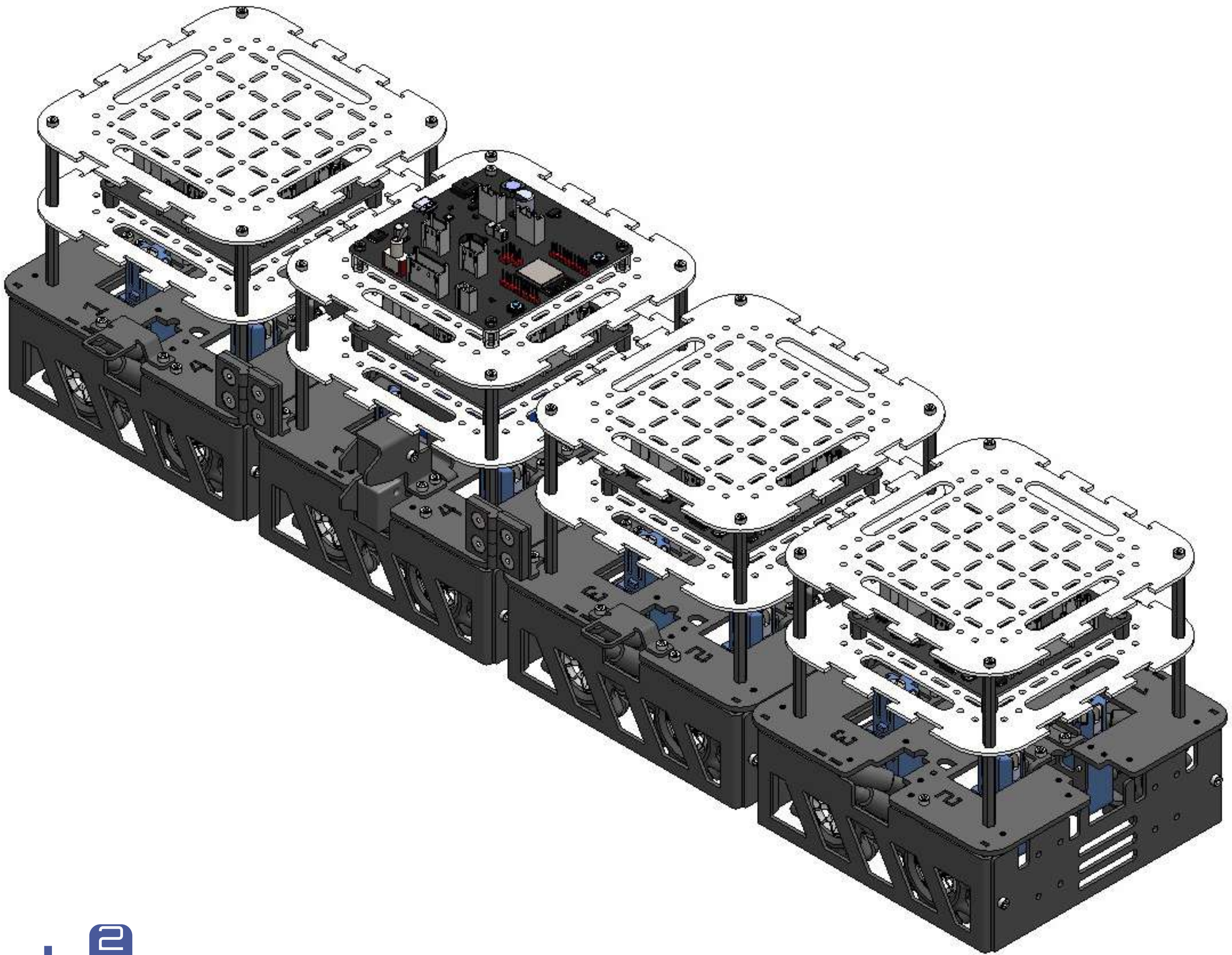
Rotate Part

Assembly Path

Important Note

Sub-Assembly



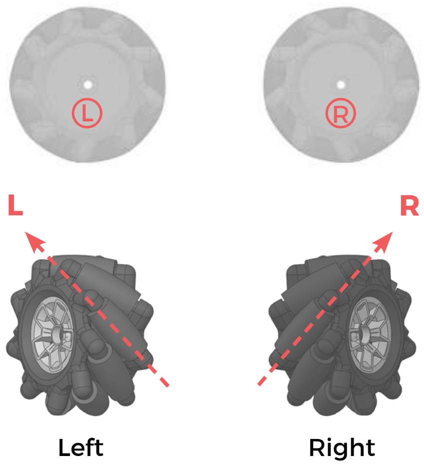


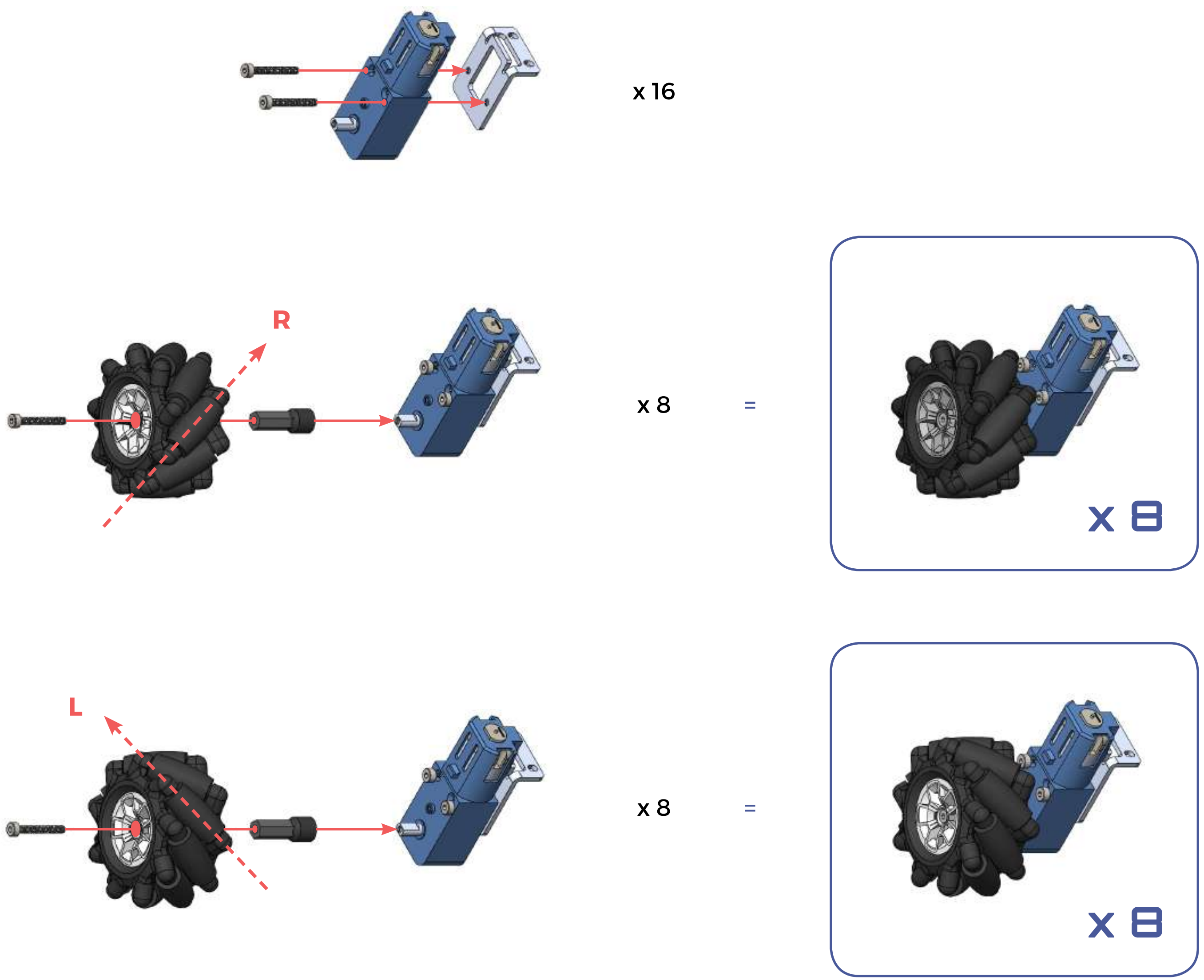
smorphi

assembly start

A 1 motor sub-assembly

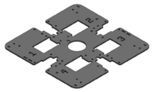
	Motor Mount	(16)
	Motor Shaft Sleeve	(16)
	M3x22	(48)
	DC Motor	(16)
	Left Mecanum Wheel	(8)
	Right Mecanum Wheel	(8)

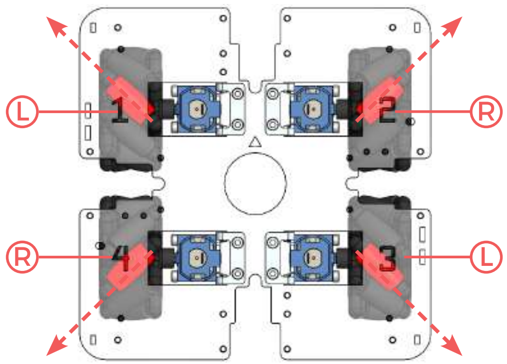




2 ways to differentiate between Left and Right Mecanum wheel
- Look for L or R on the inside of the wheel
- Left/Right wheel roller points up to Left/Right

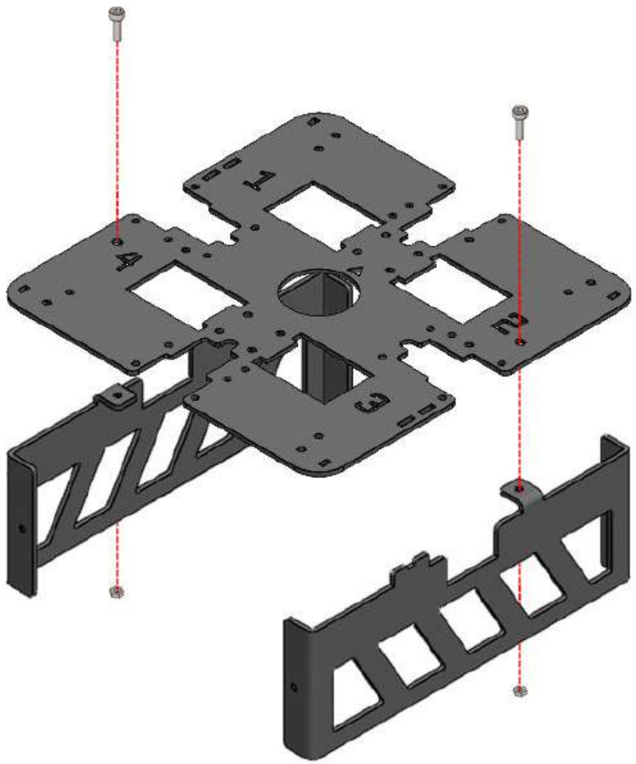
A 2 (base module sub-assembly)

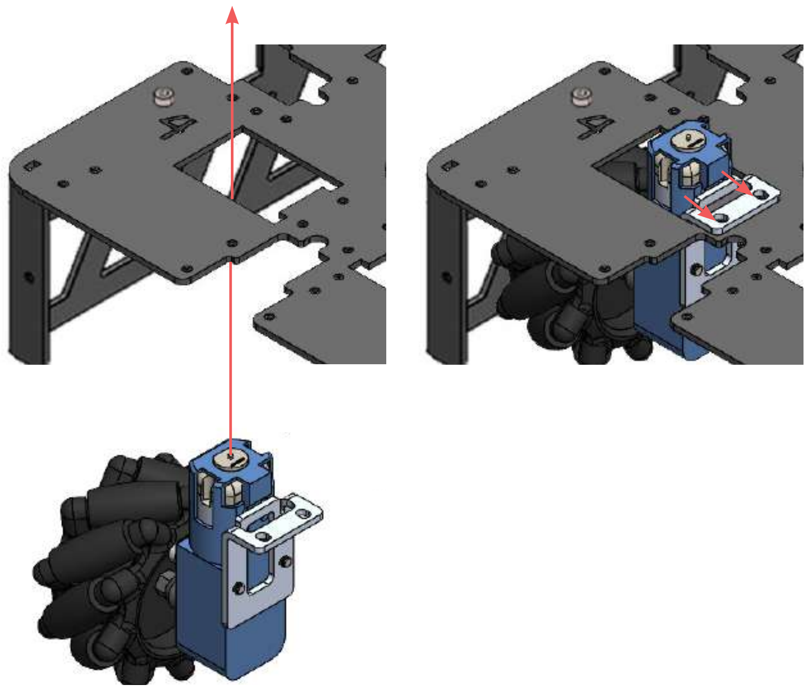
	Al Base Plate	(4)
	Hex M-F M3 45mm	(16)
	M3x5	(16)
	M3x10	(45)
	M3 Hex Nut	(45)
	Left Wheel SA from A.1	(8)
	Right Wheel SA from A.1	(8)

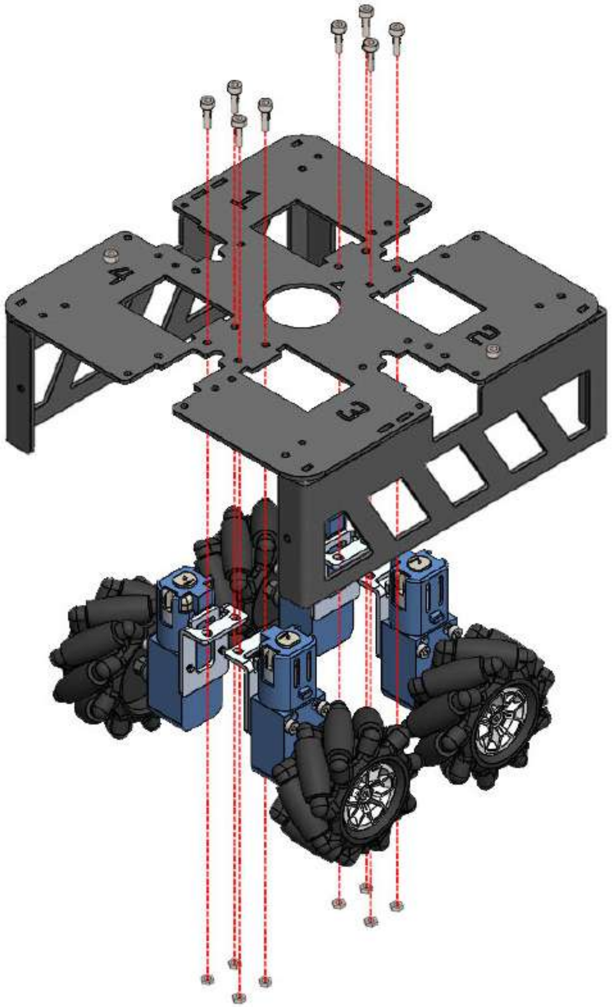


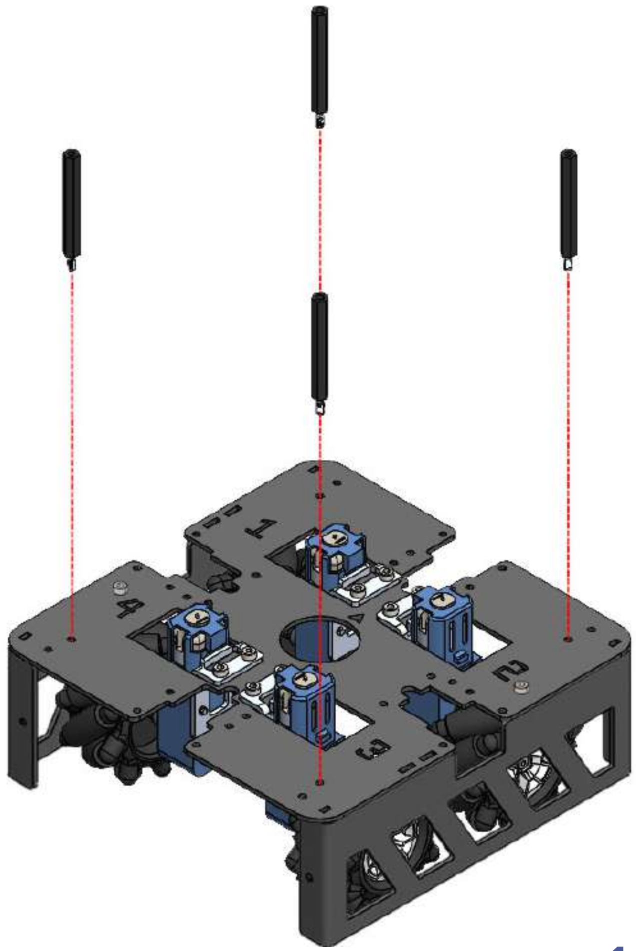
Make sure that the numbers are facing the right way up as shown in the plan view above.

Before attaching each wheel, check that the wheel is of the correct orientation for each numbered slot.

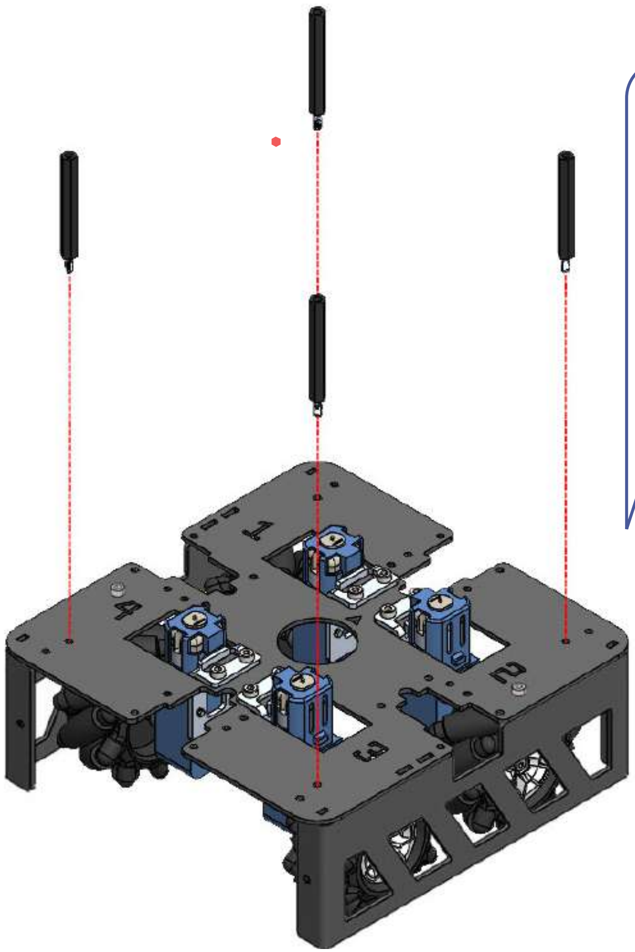




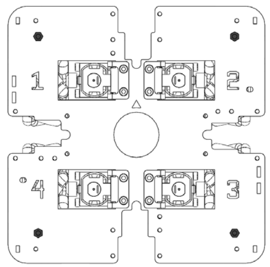



x 4

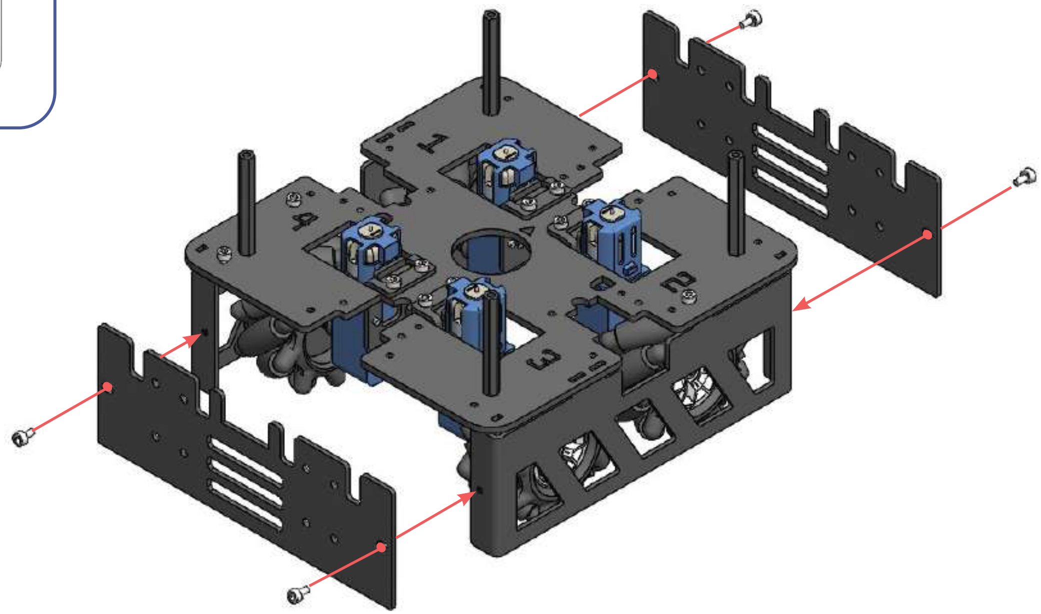
Base Skirt Panel A	(8)
Base Skirt Panel B	(8)
M3 x 5	(24)

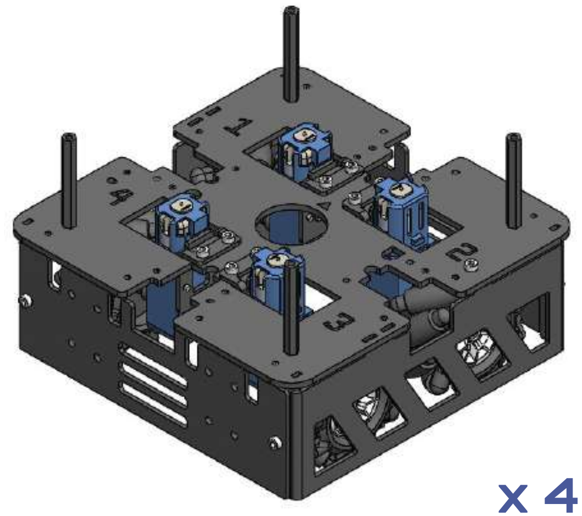


plan view



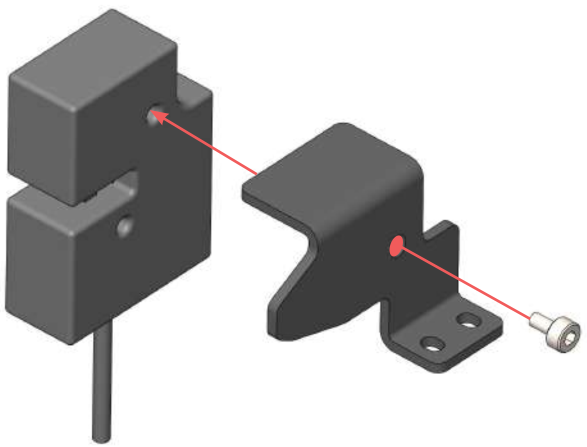
x 4 =

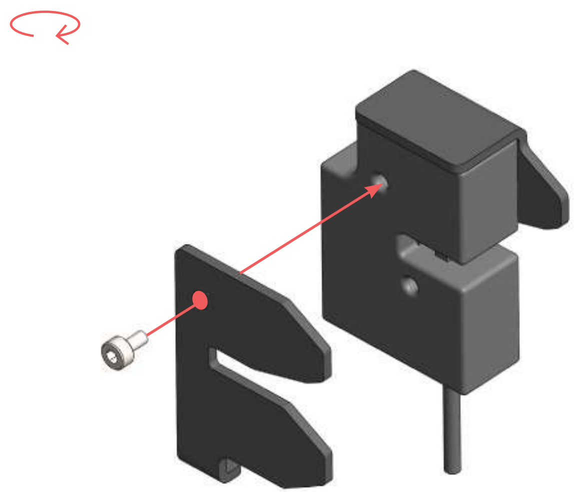


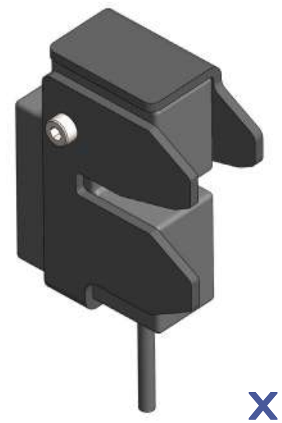


A 3 (solenoid latch sub-assembly)

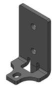
	Solenoid latch	(6)
	Solenoid latch mount	(6)
	Solenoid latch guide	(6)
	M3 x 5	(12)

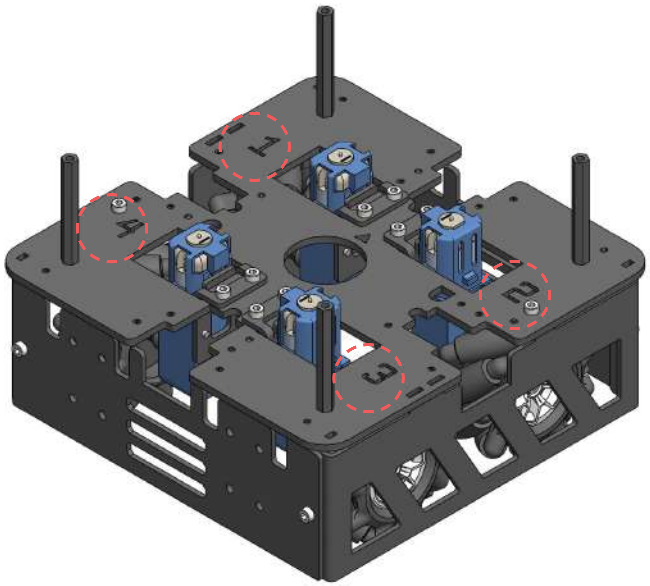


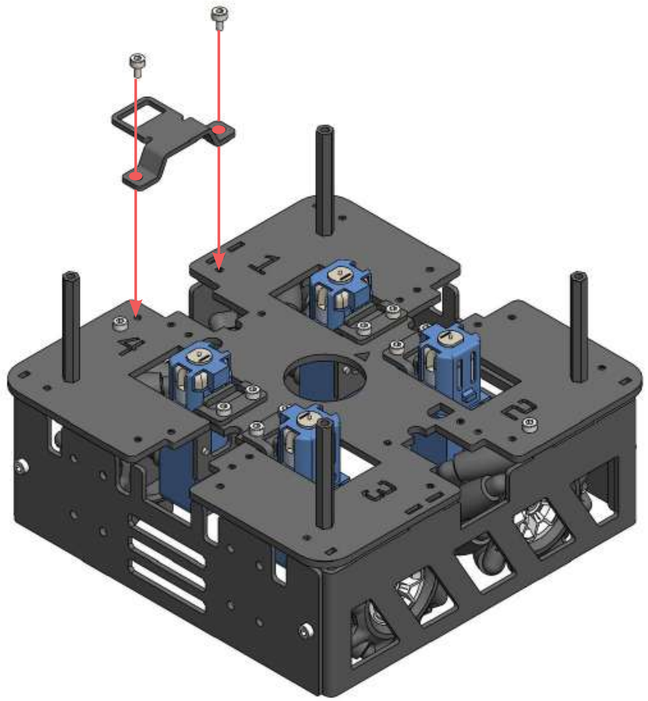


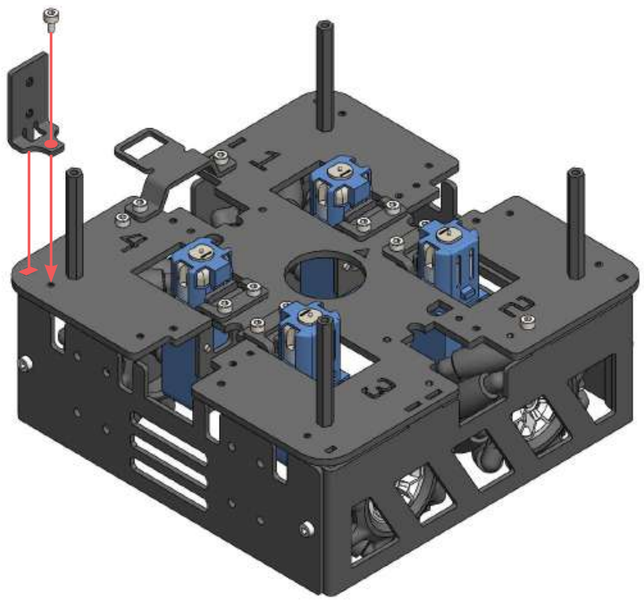


A 4 (module 1 mechanical sub-assembly)

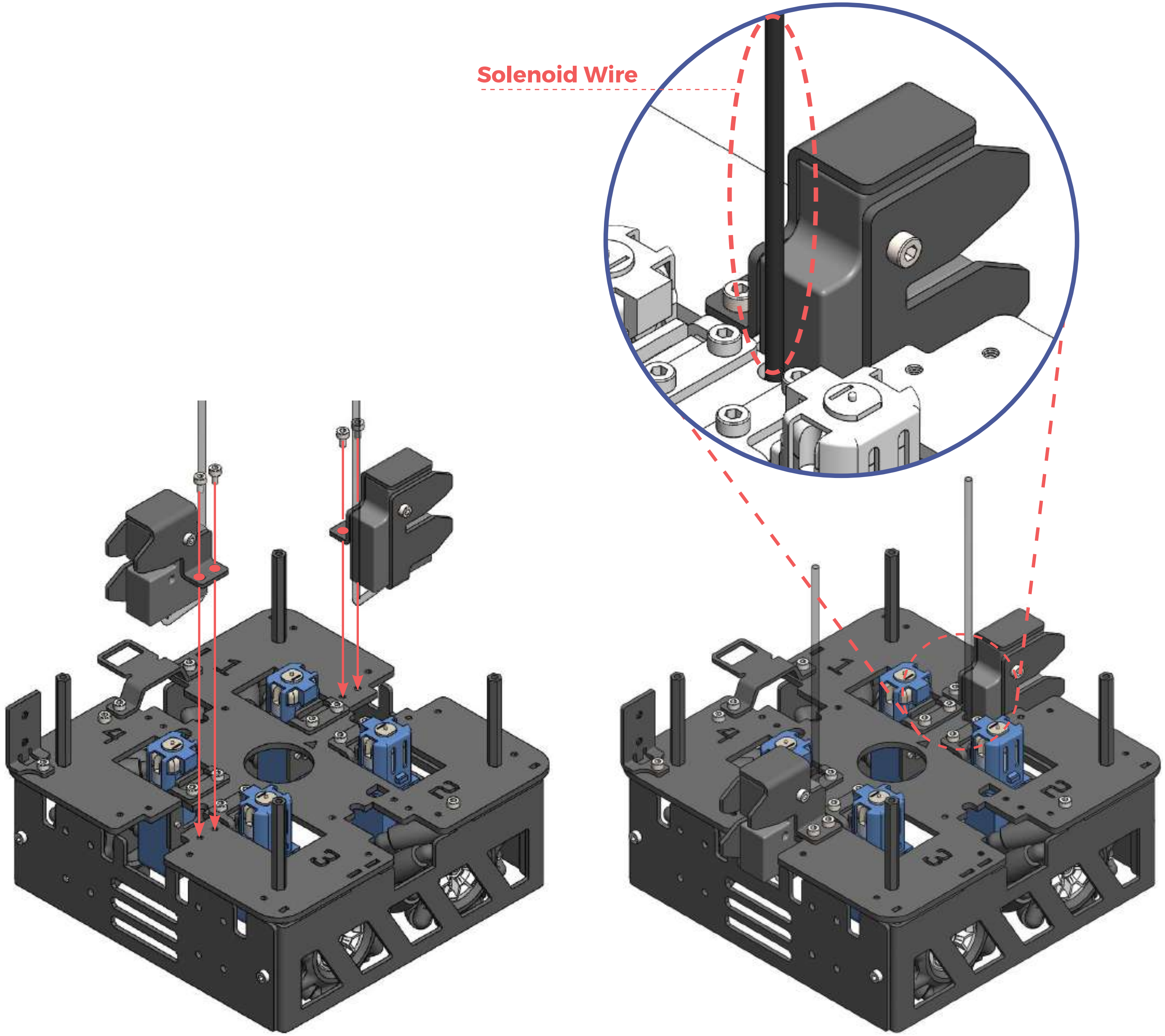
	Hinge mount	(1)
	Solenoid catch	(1)
	Solenoid latch SA from A.3	(2)
	Base Module SA from A.2	(1)
	M3 x 5	(7)

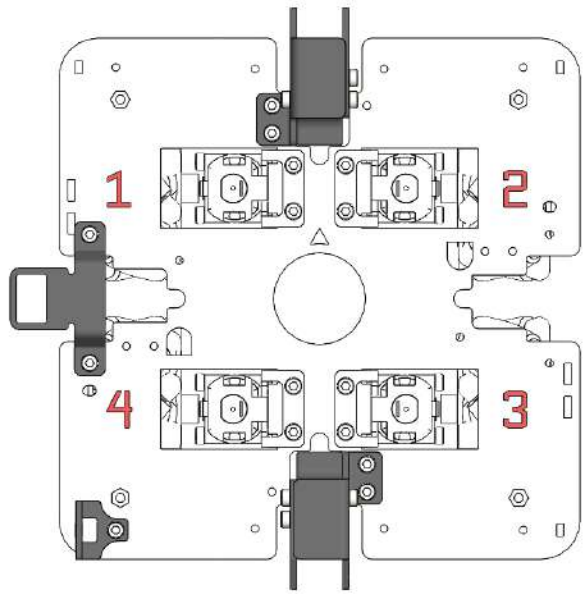






Pay attention to the numbers and their positions in relation to the parts being attached.

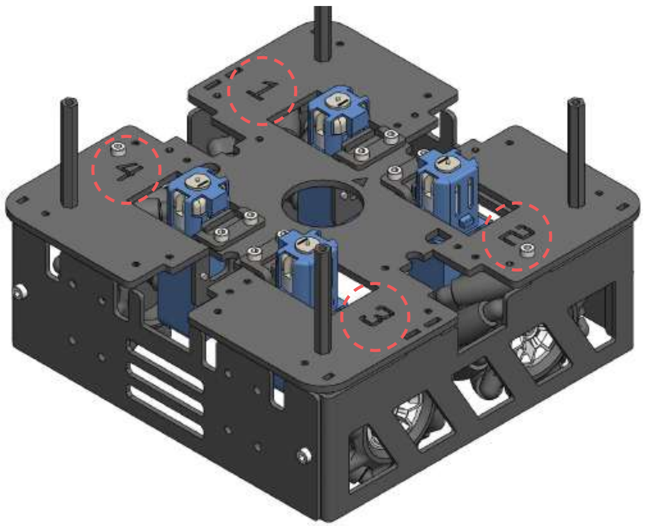


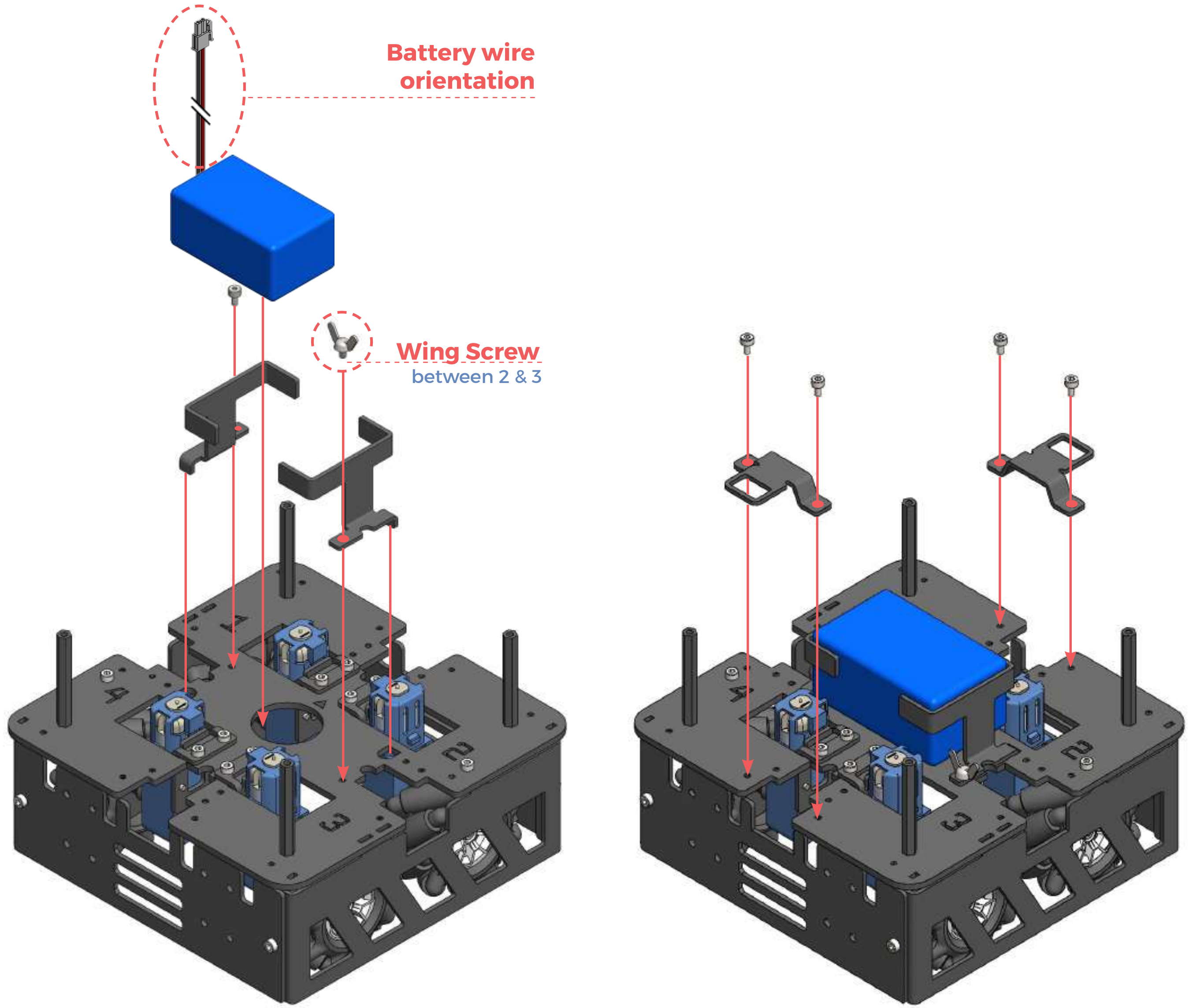


Check that the parts have been attached in the right location, in relation to the numbers.

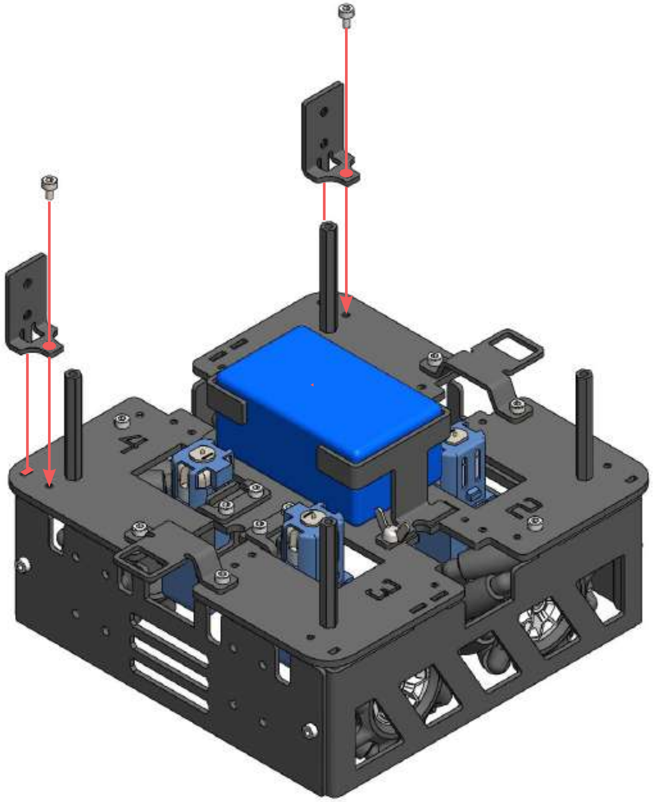
A 5 (module 2 mechanical sub-assembly)

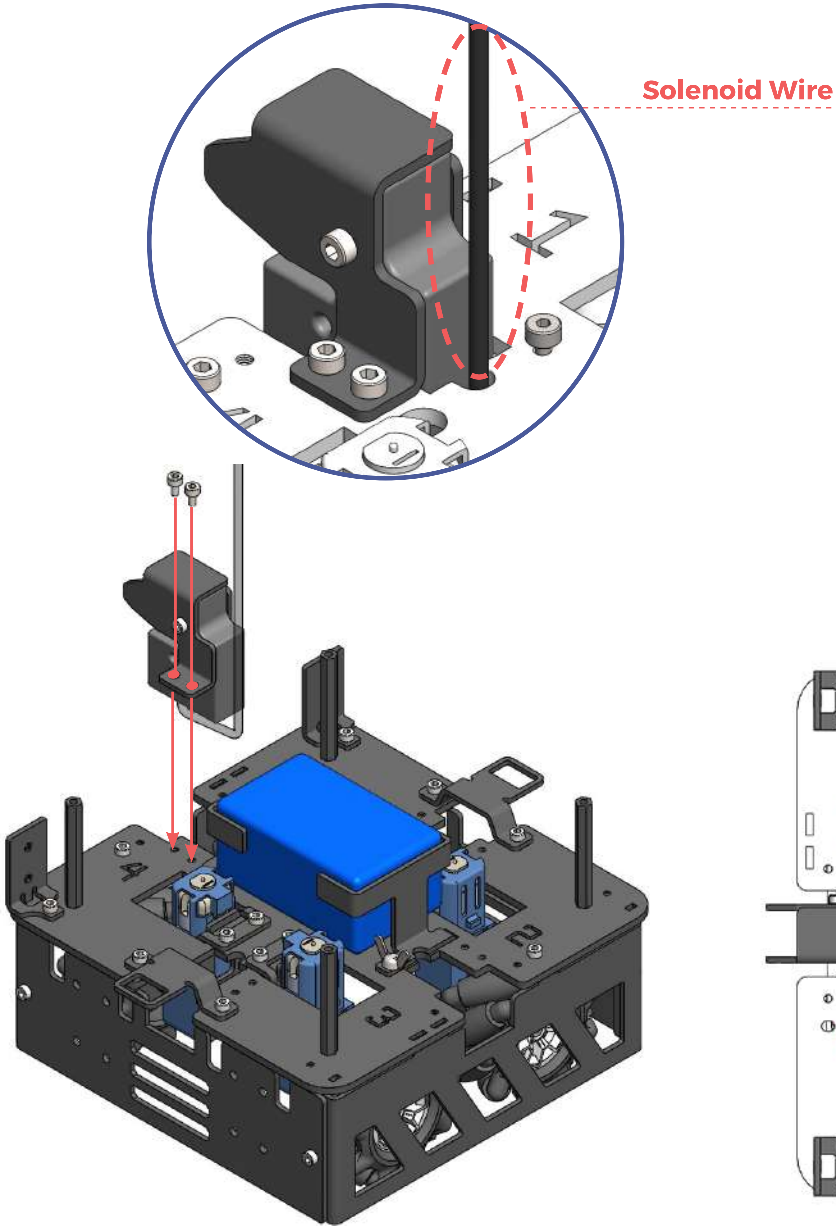
- Hinge mount (2)
- Solenoid catch (2)
- Solenoid latch SA (1) from A.3
- Battery bracket (2)
- LiPo Battery (1)
- Base Module SA (1) from A.2
- M3 x 5 (9)
- Wing Screw M3 x 5 (1)

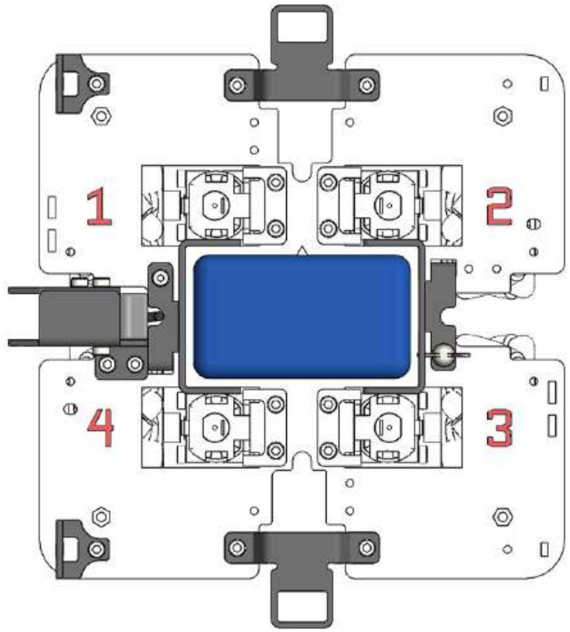




Pay attention to the numbers and their positions in relation to the parts being attached.

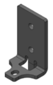






Check that the parts have been attached in the right location.

A 6 (module 3 mechanical sub-assembly)

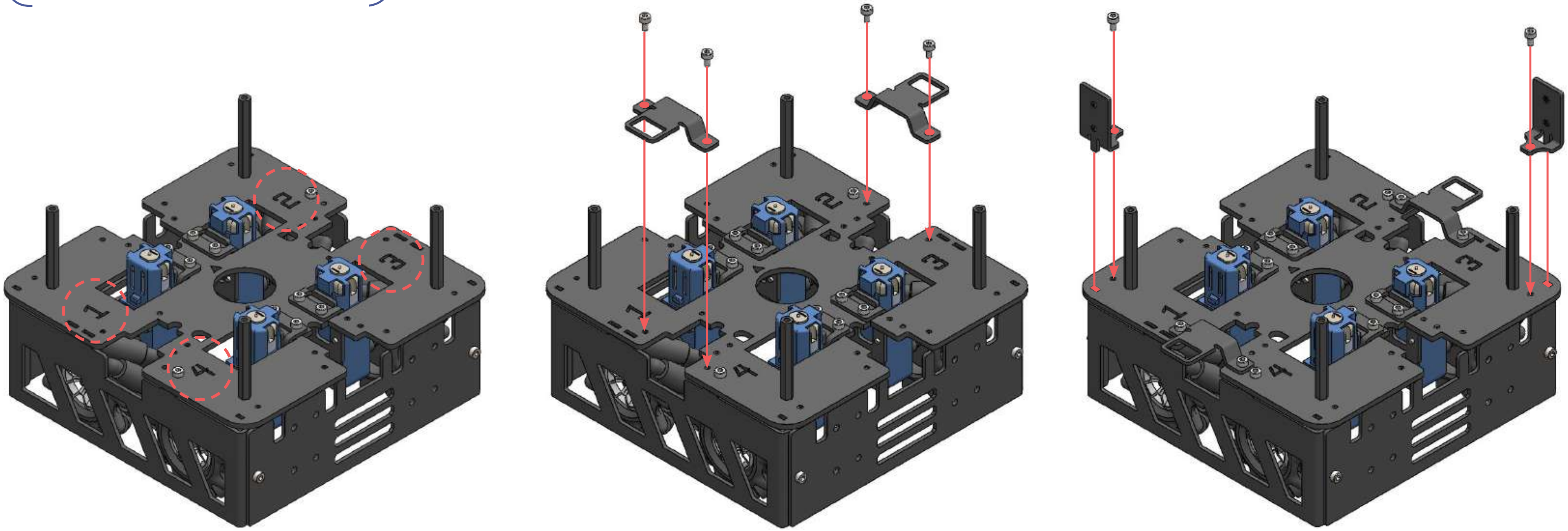

Hinge mount (2)


Solenoid catch (2)

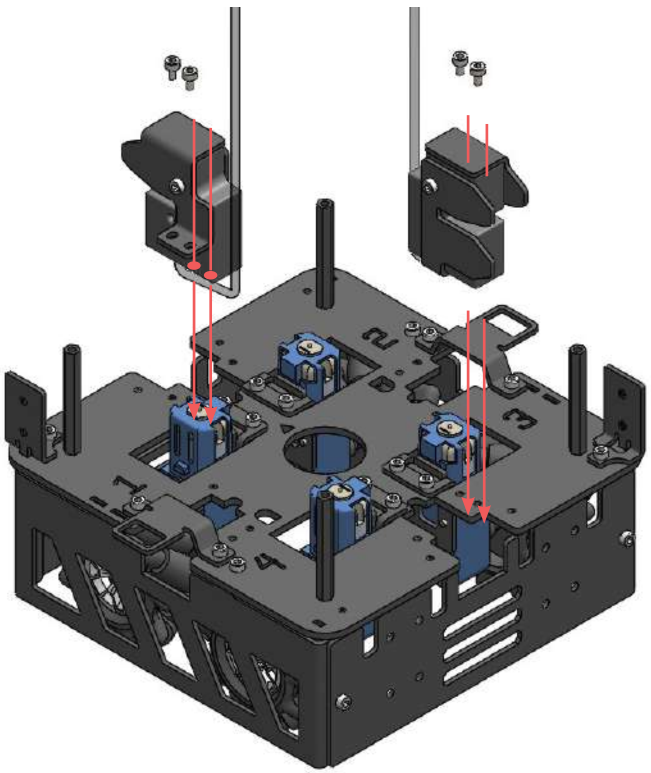

Solenoid latch SA (2)
from A.3

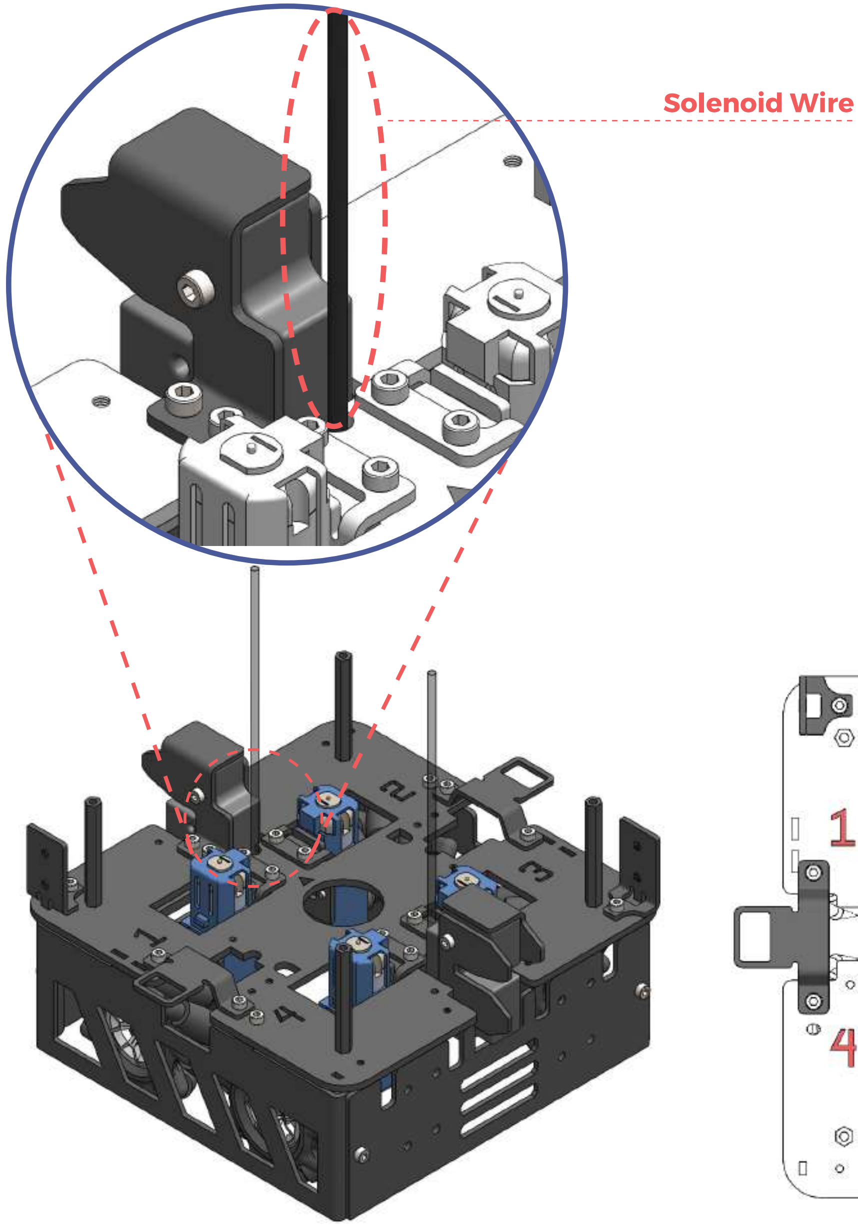

Base Module SA (1)
from A.2

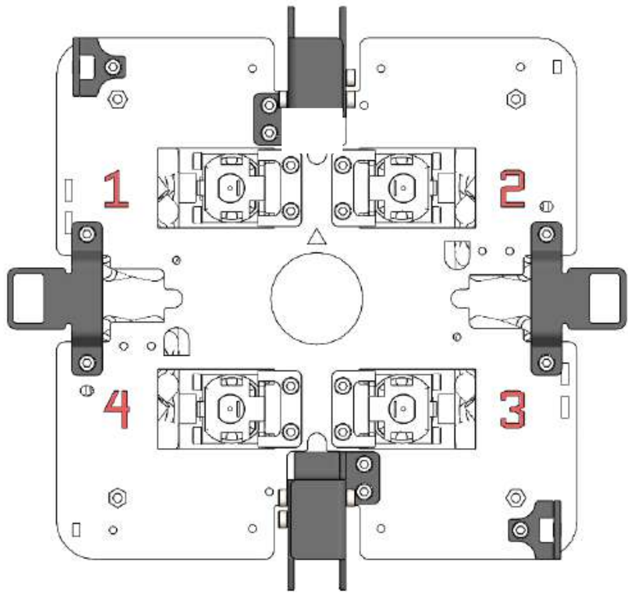

M3 x 5 (10)



Pay attention to the numbers and their positions in relation to the parts being attached.



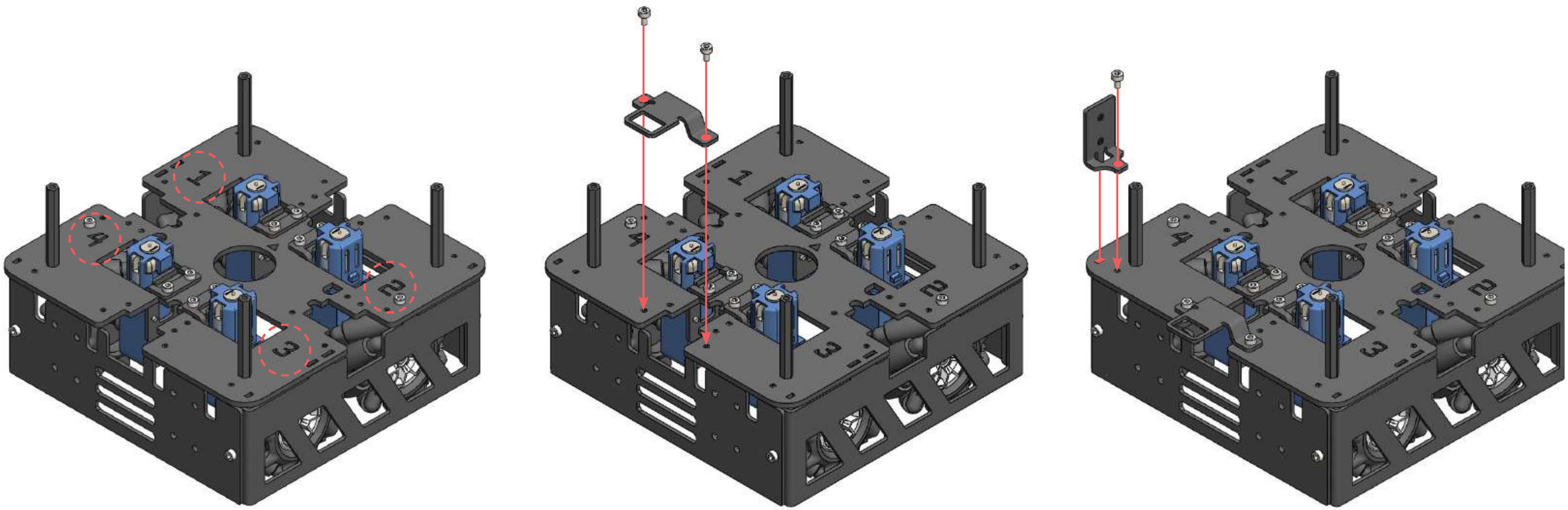




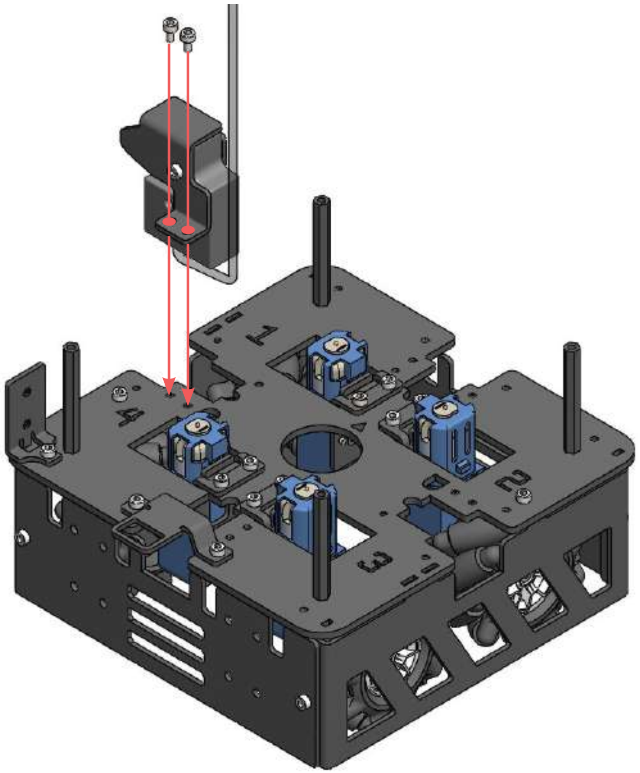
Check that the parts have been attached in the right location, in relation to the numbers.

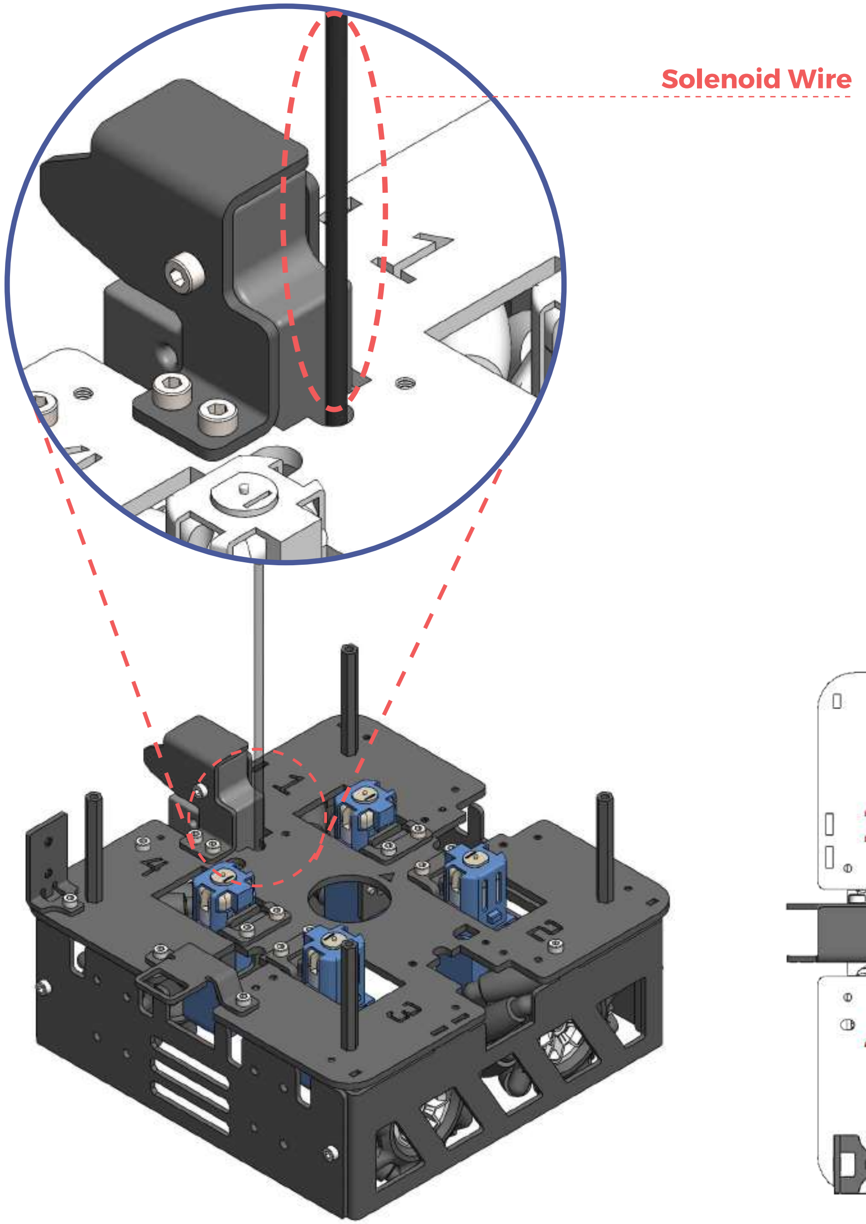
A 7 module 4 mechanical sub-assembly

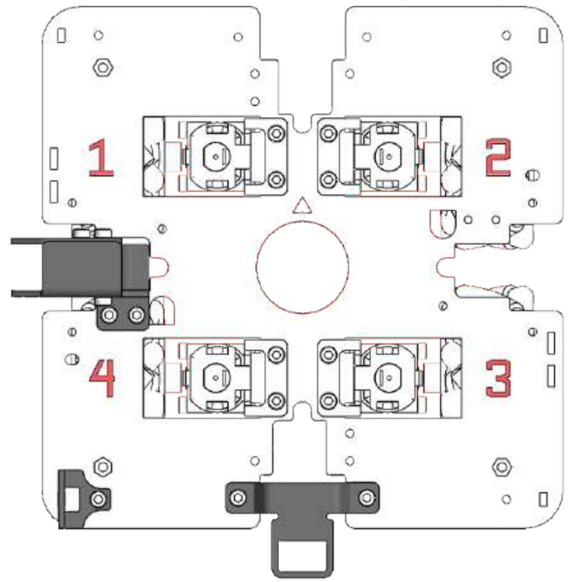
Hinge mount	(1)
Solenoid catch	(1)
Solenoid latch SA from A.3	(1)
Base Module SA from A.2	(1)
M3 x 5	(5)



Pay attention to the numbers and their positions in relation to the parts being attached.

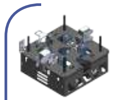


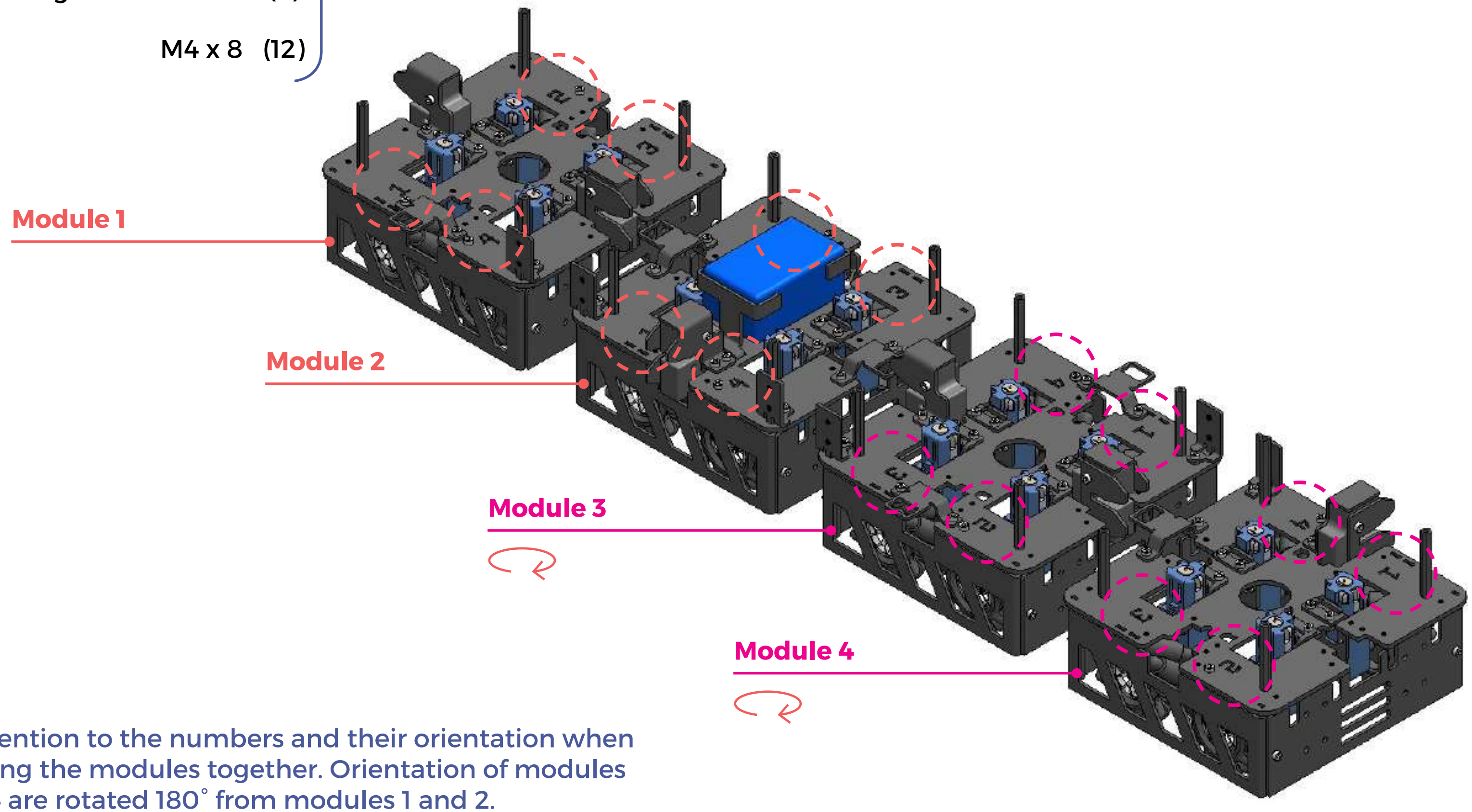


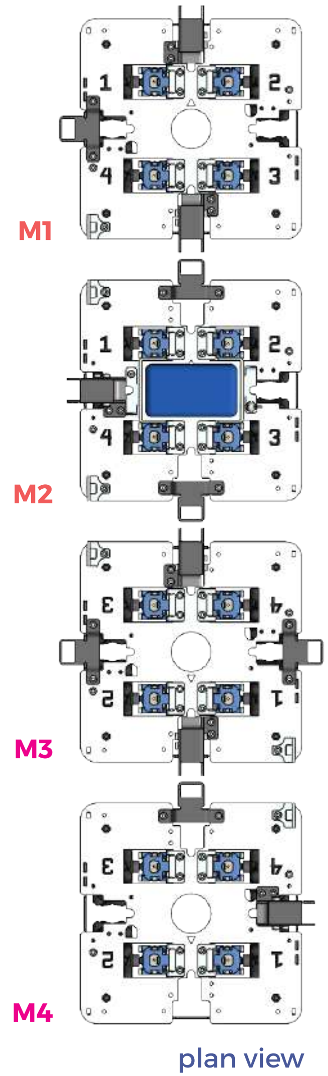


Check that the parts have been attached in the right location.

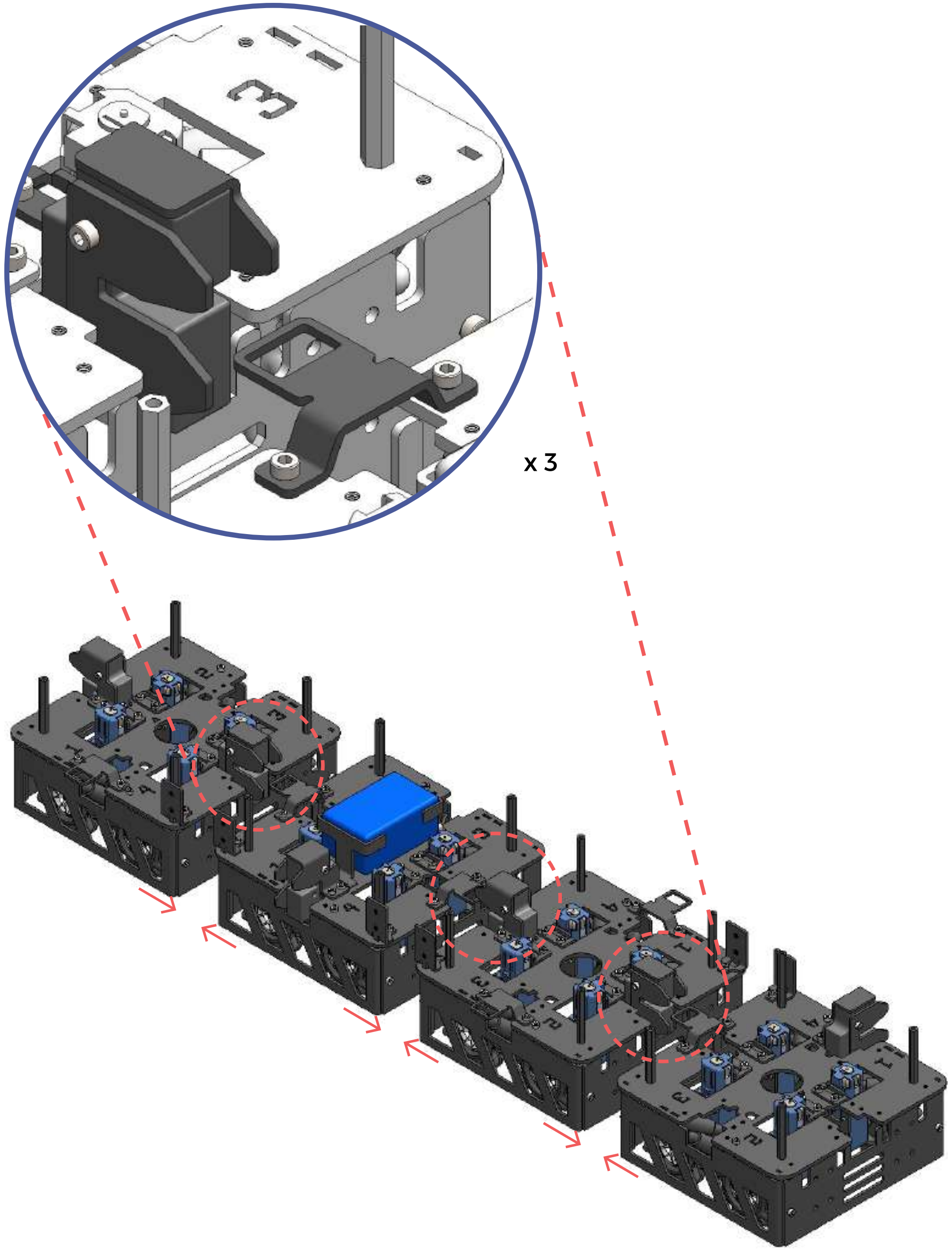
A 8 (full mechanical assembly)

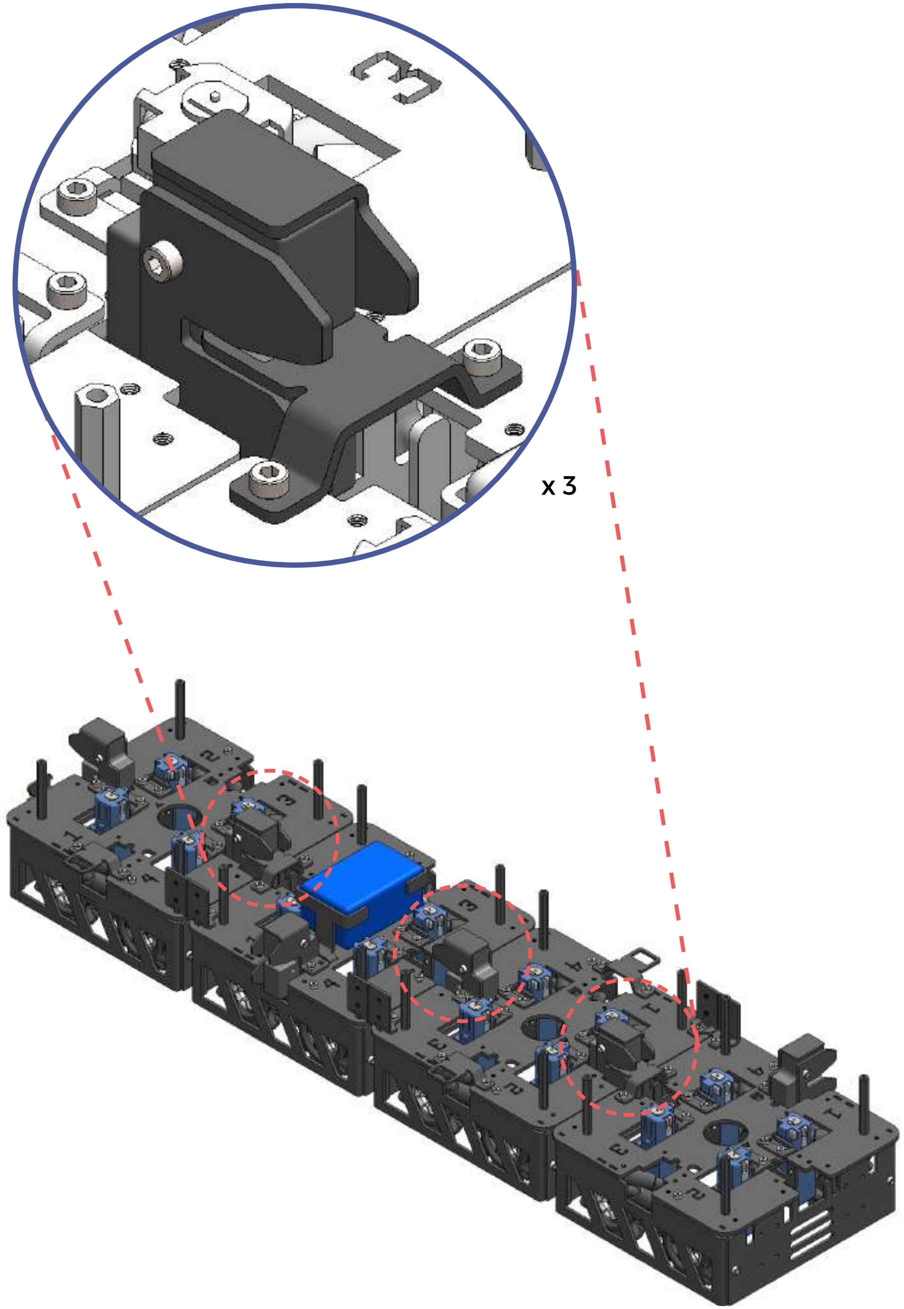
	Module 1 from A.4	(1)
	Module 2 from A.5	(1)
	Module 3 from A.6	(1)
	Module 4 from A.7	(1)
	Hinge mechanism	(3)
	M4 x 8	(12)

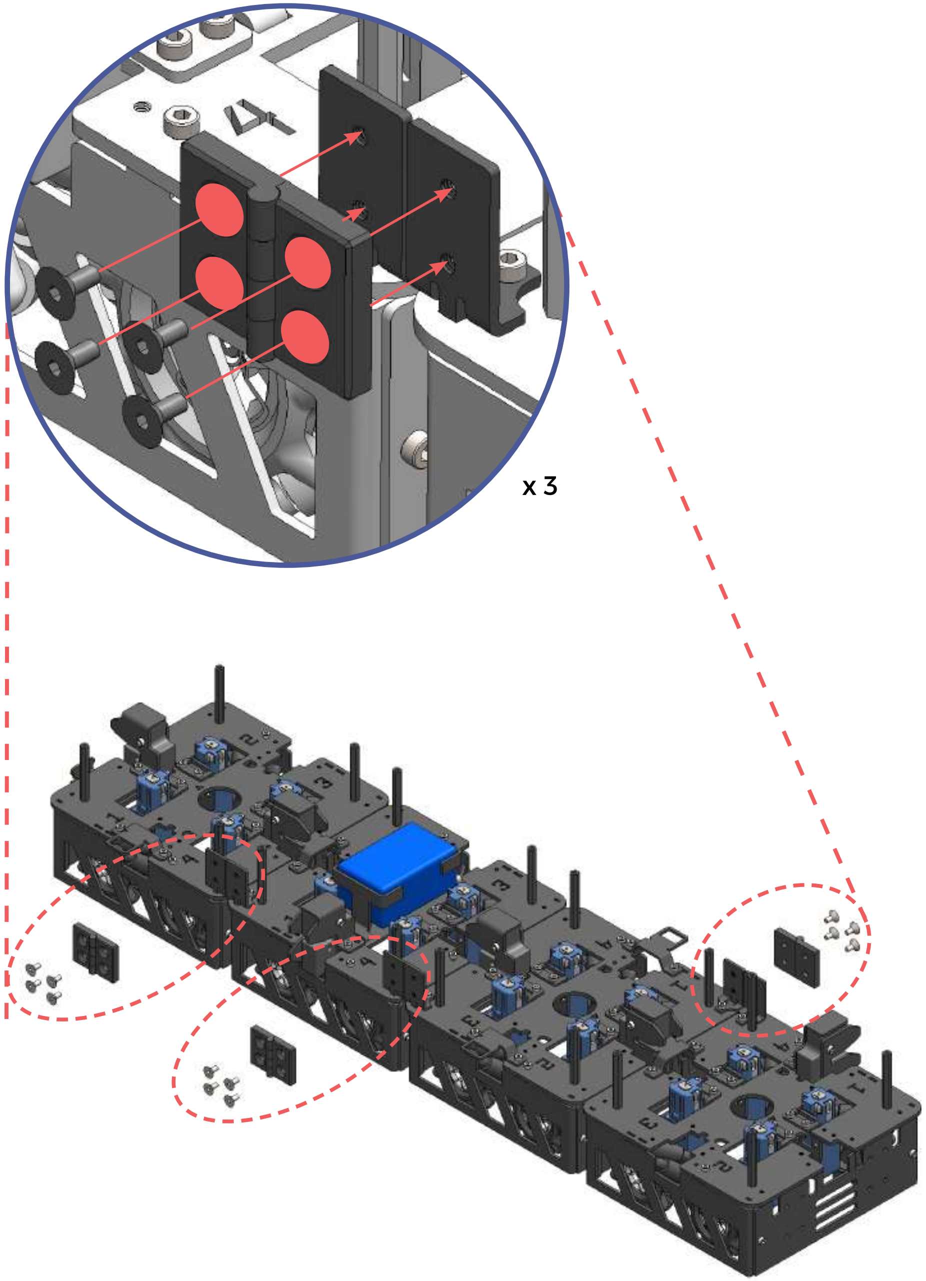


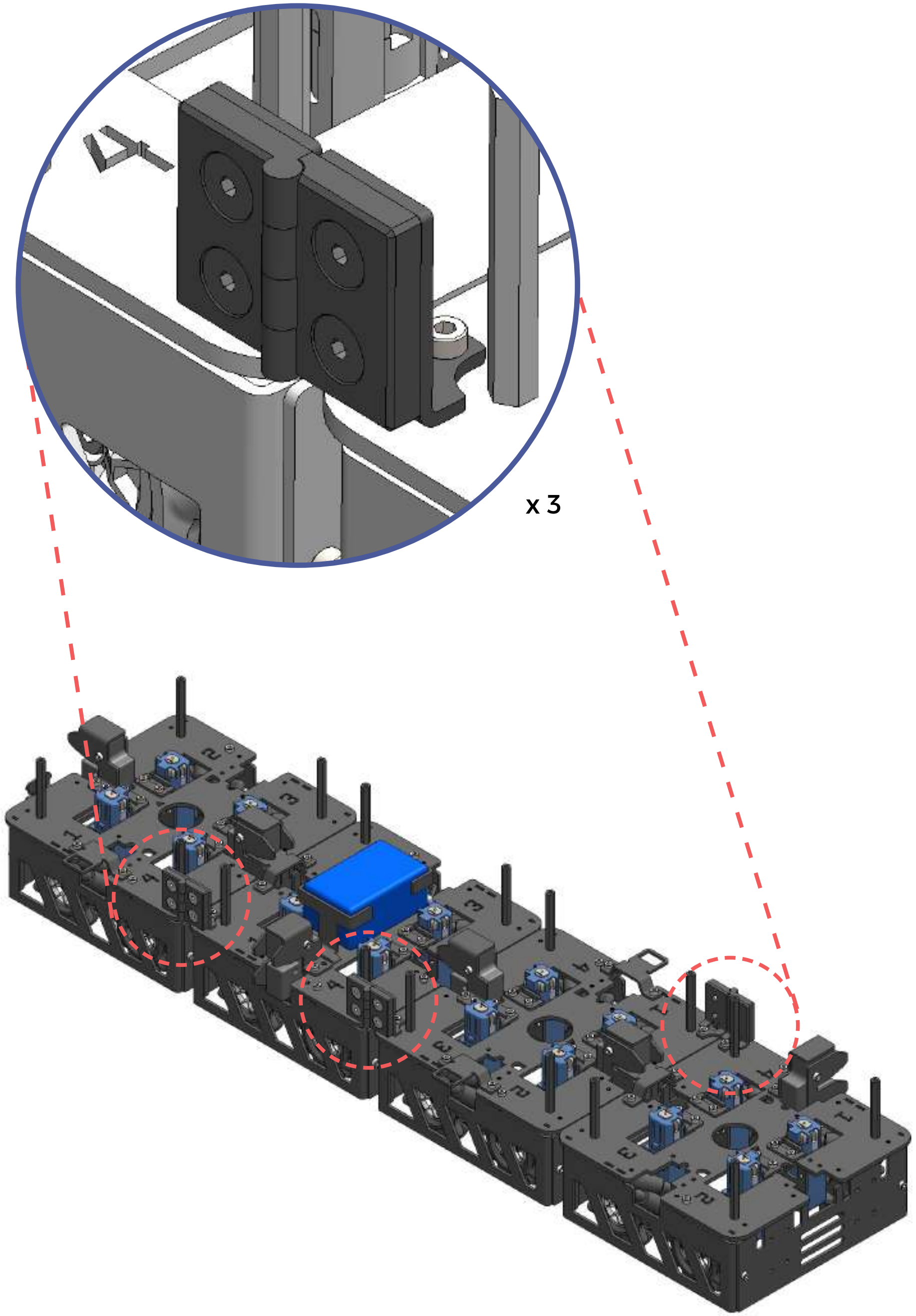


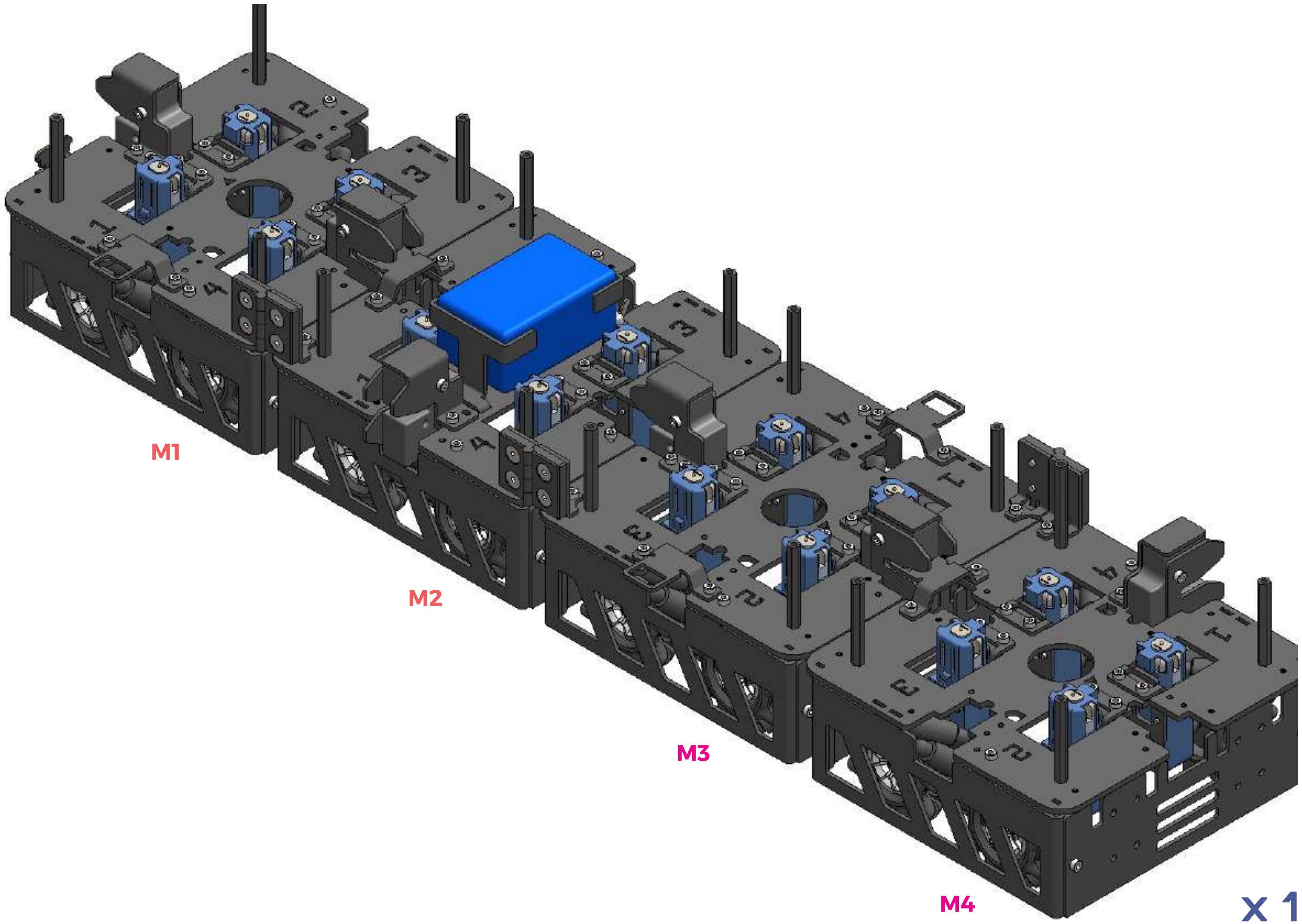
Pay attention to the numbers and their orientation when attaching the modules together. Orientation of modules 3 and 4 are rotated 180˚ from modules 1 and 2.


x 3


x 3


x 3


x 3

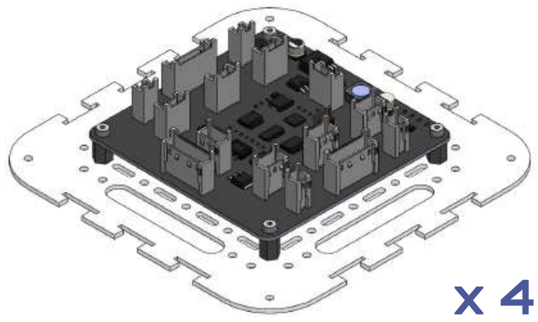


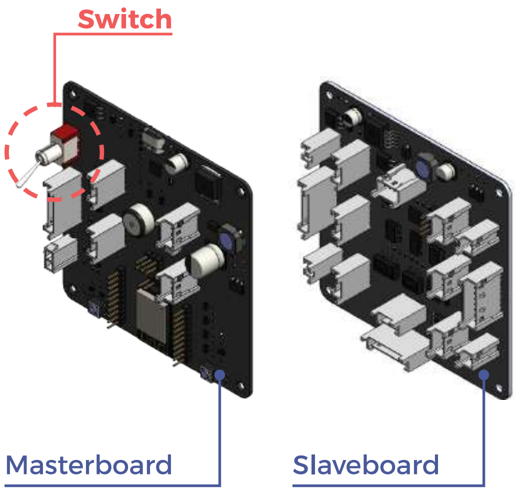
Check all the positions of solenoids, solenoid catches, hinge mounts before proceeding to section B.

A mechanical assembly completed

B 1 (e-tray sub-assembly)

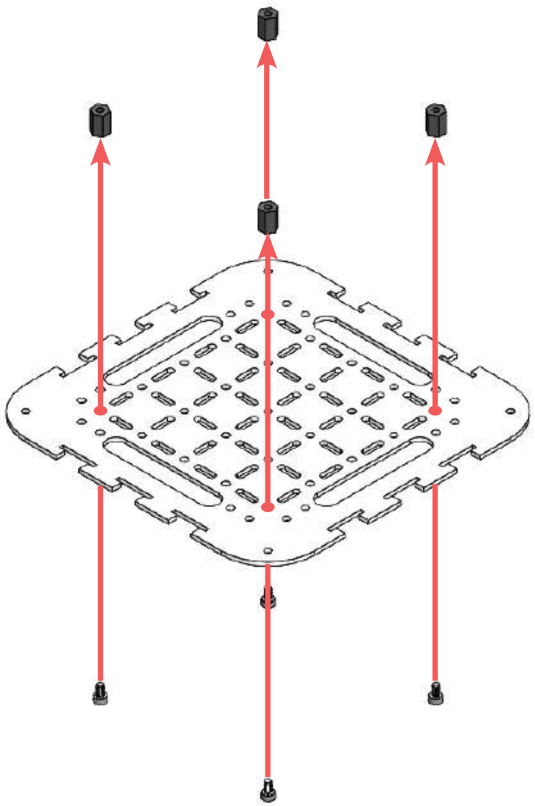
Acrylic base plate	(4)
Slaveboard	(4)
Hex F-F M3 10mm	(16)
M3 x 5	

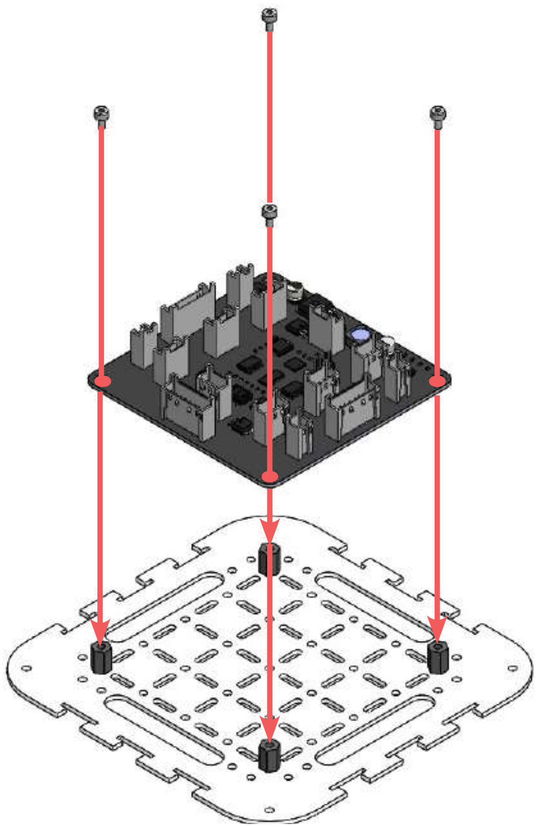




1 unique piece

4 identical pieces





Make sure that you are attaching the slaveboards and not the masterboard.

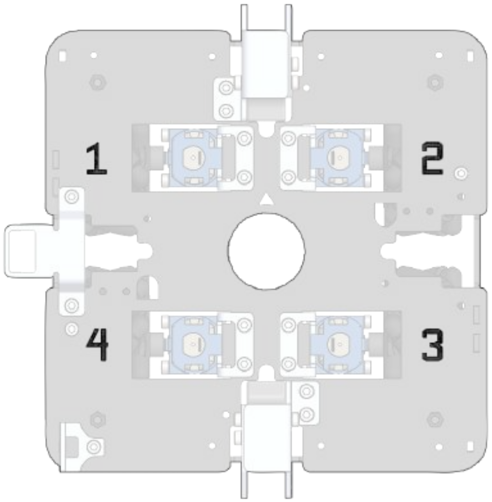
How to differentiate between masterboard and slaveboards:
- Masterboard has a special switch

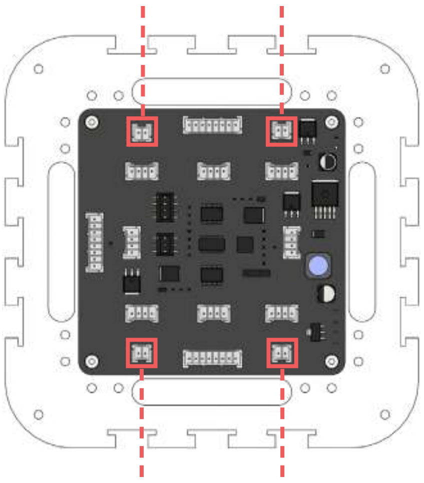
B 2 (e-tray onto mechanical assembly)

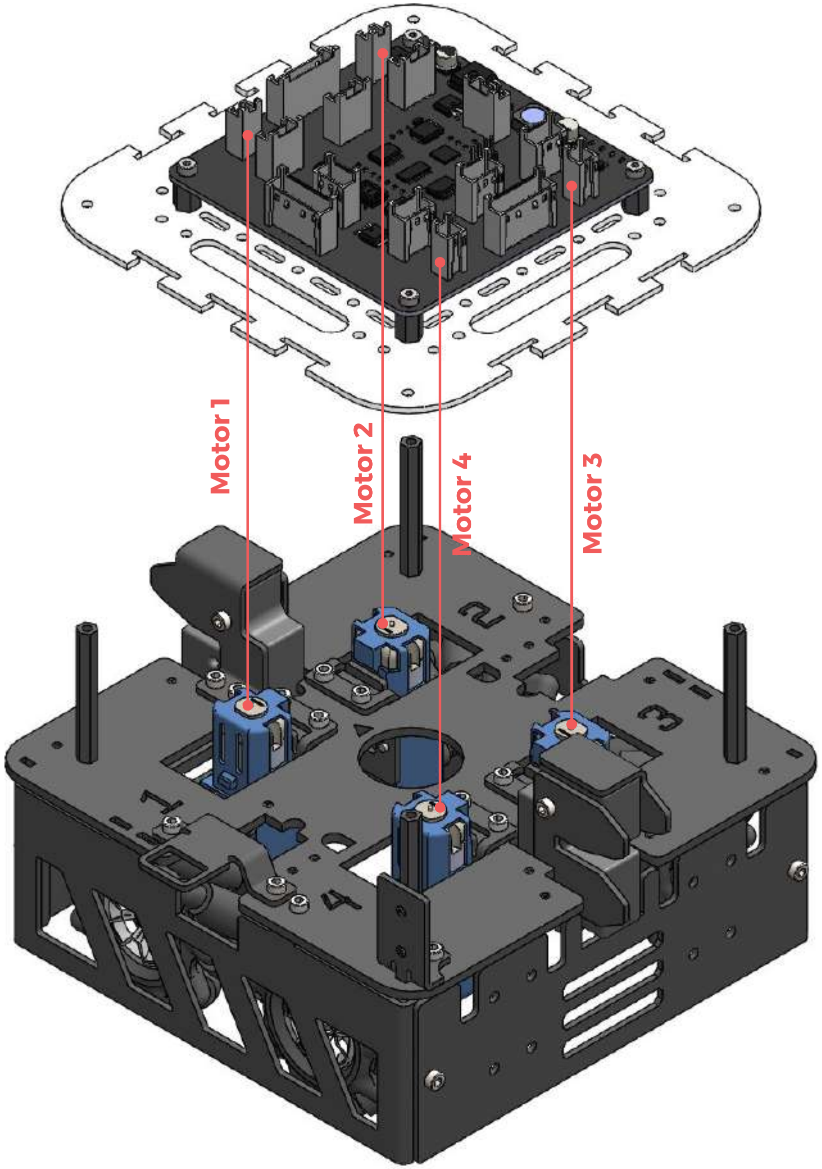

e-tray SA (4)
from B.1


full mechanical SA (1)
from A.8

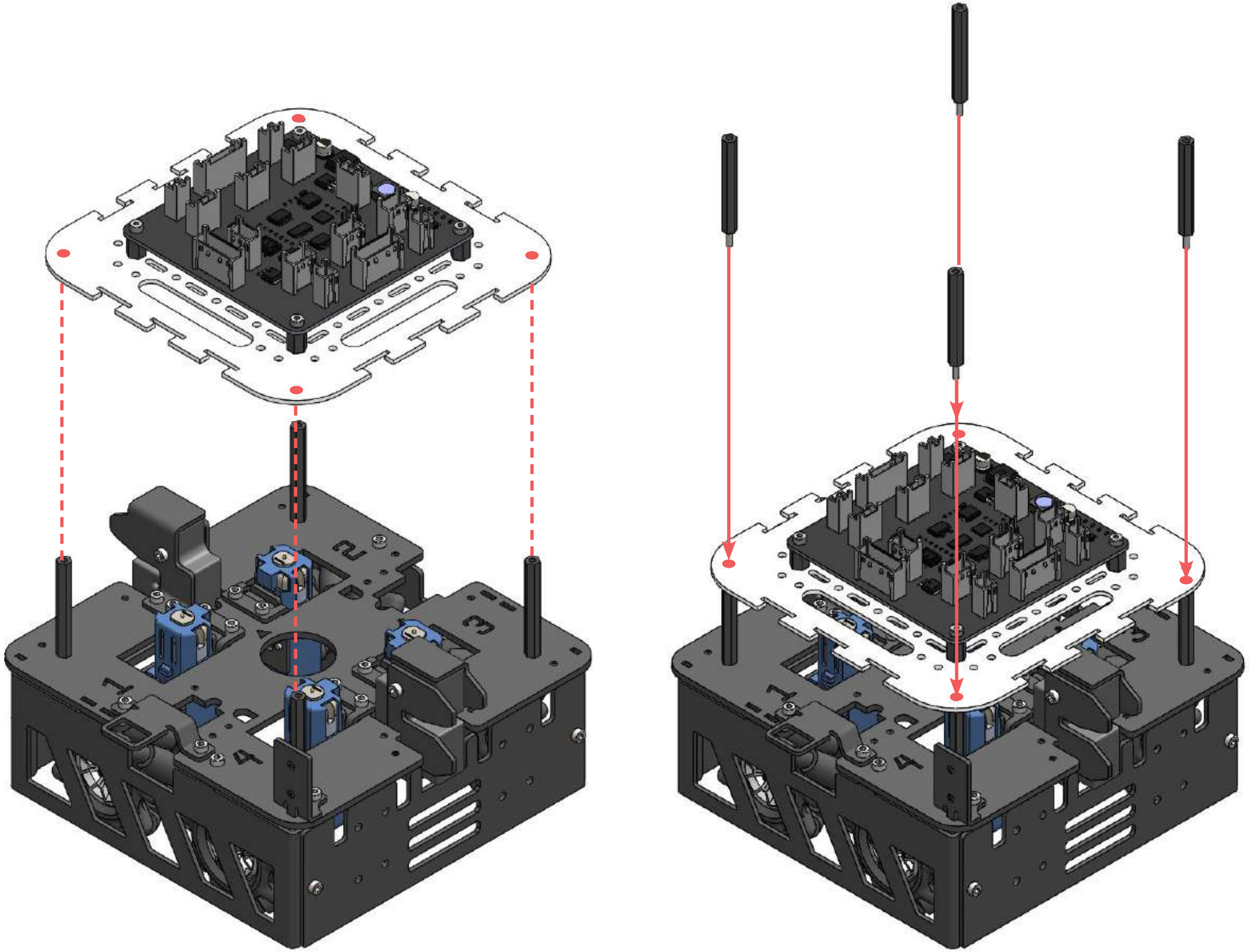

Hex M-F M3 45mm (16)



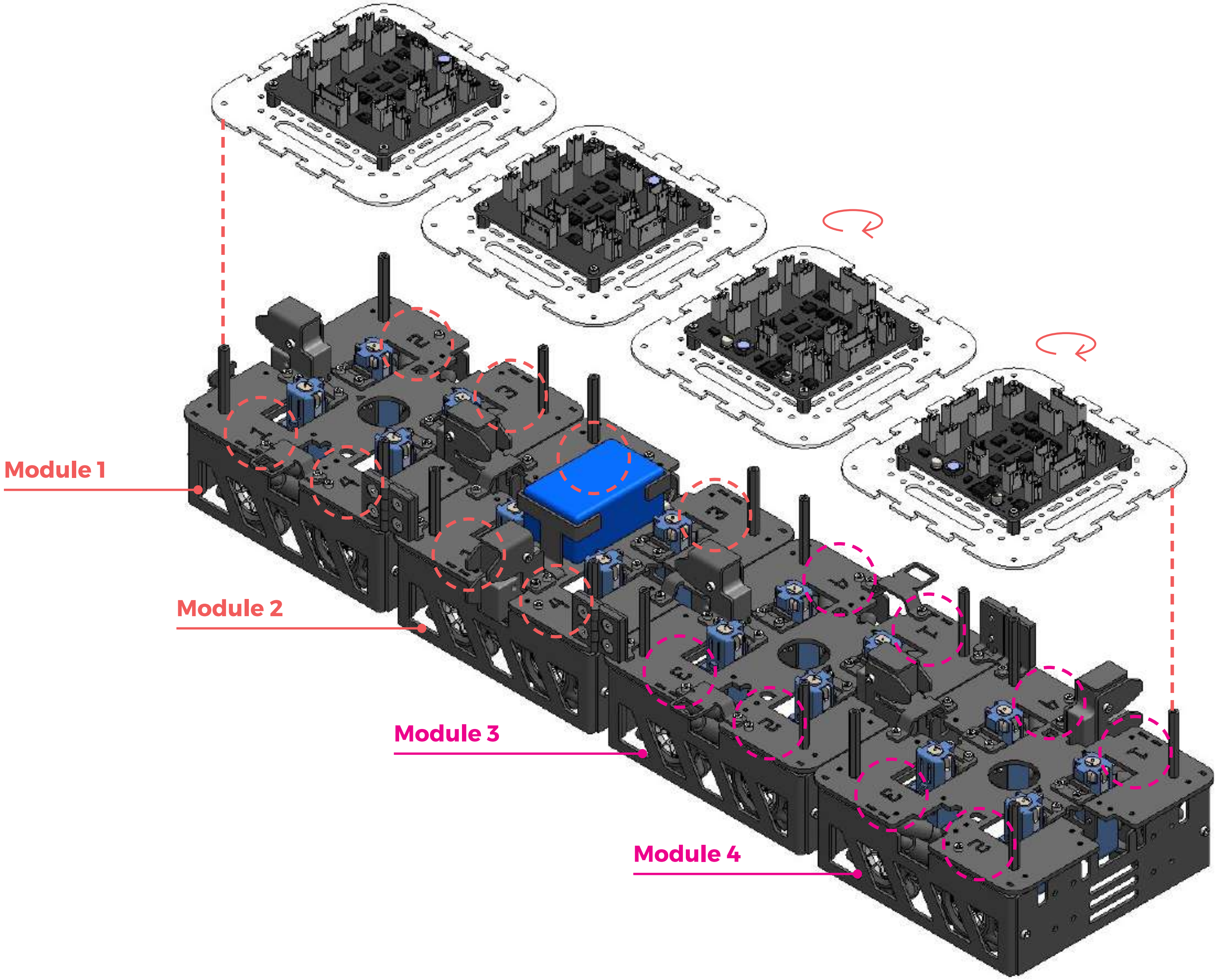


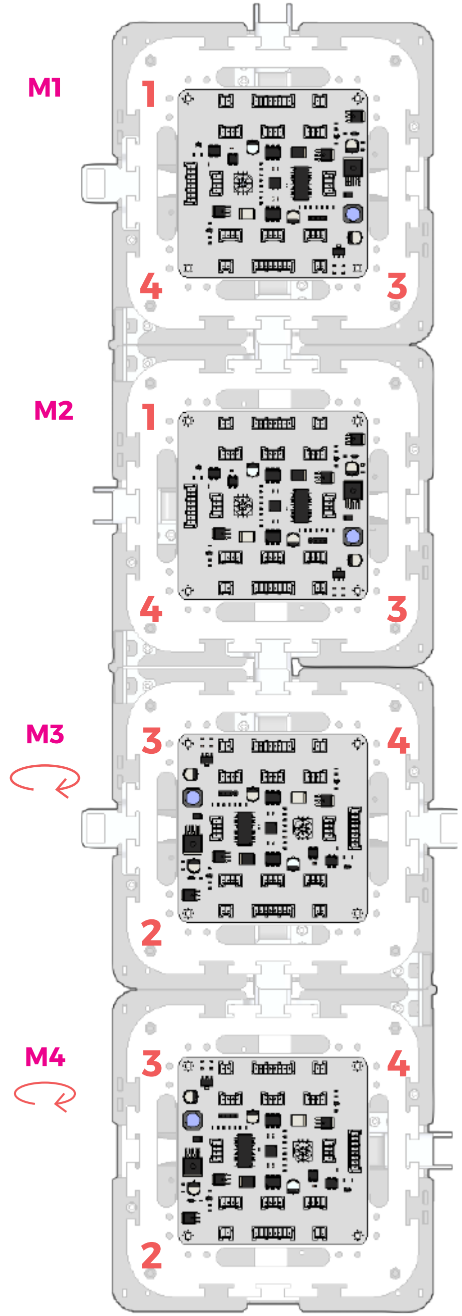


Orientate e-tray SA and base module as shown on the right.
Motor 1 connector on e-tray should be on top of Motor 1 of base module.
Same goes for Motor 2, 3 and 4.

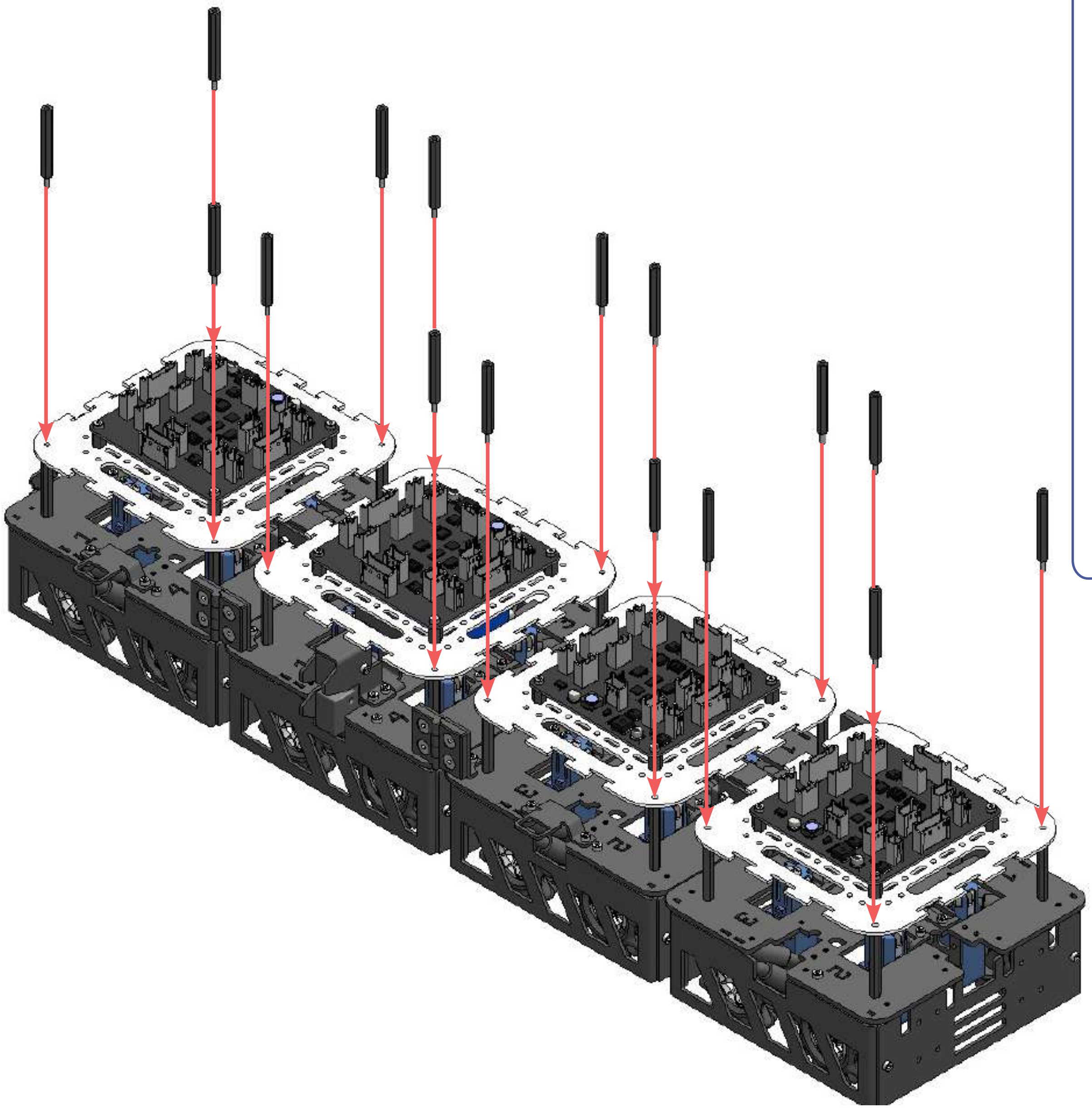


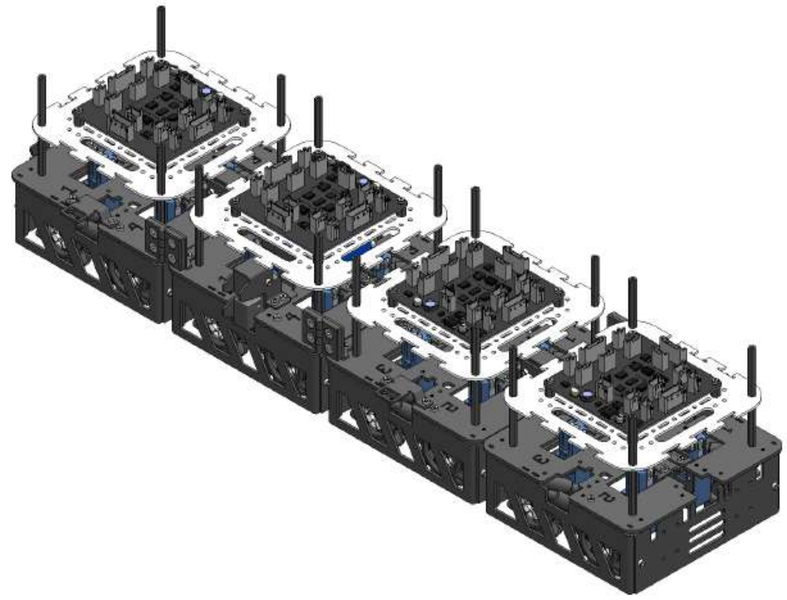
Before attaching the e-tray SA, please flip all the cables and connectors for motors and solenoids (not shown in diagram for clarity) out, for easy attachment later on.





Orientation of modules 3 and 4 are rotated 180˚ from modules 1 and 2. Therefore, make sure that the e-tray SAs for module 3 and 4 are also rotated 180˚ to correspond to the motor orientation as numbered.





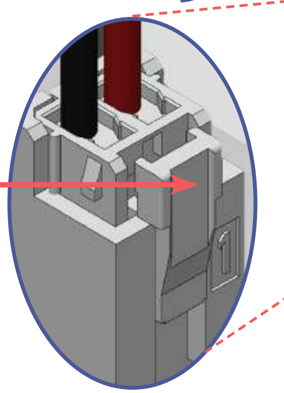
x 1

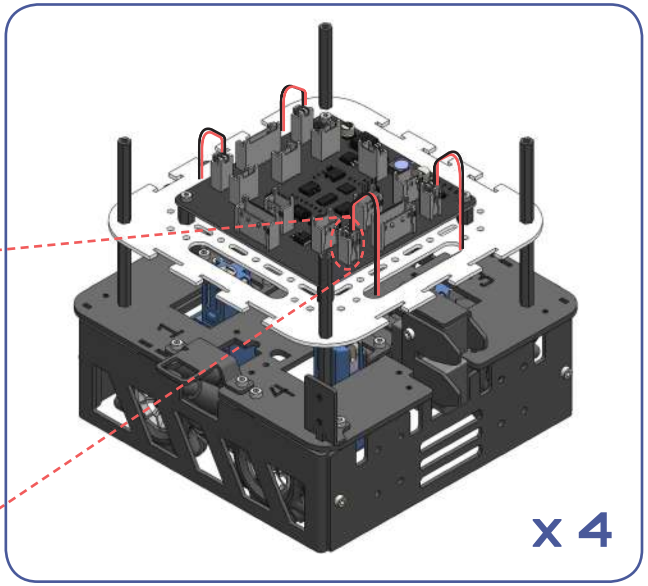
B3 (all module motor cable connection)


motor cables (16)
attached to motor

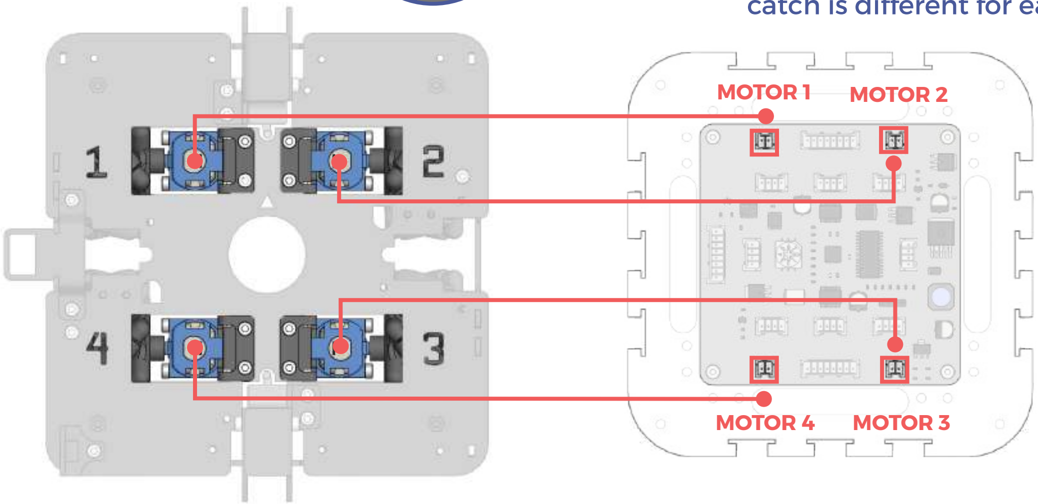

module SA (4)
attached to full assembly

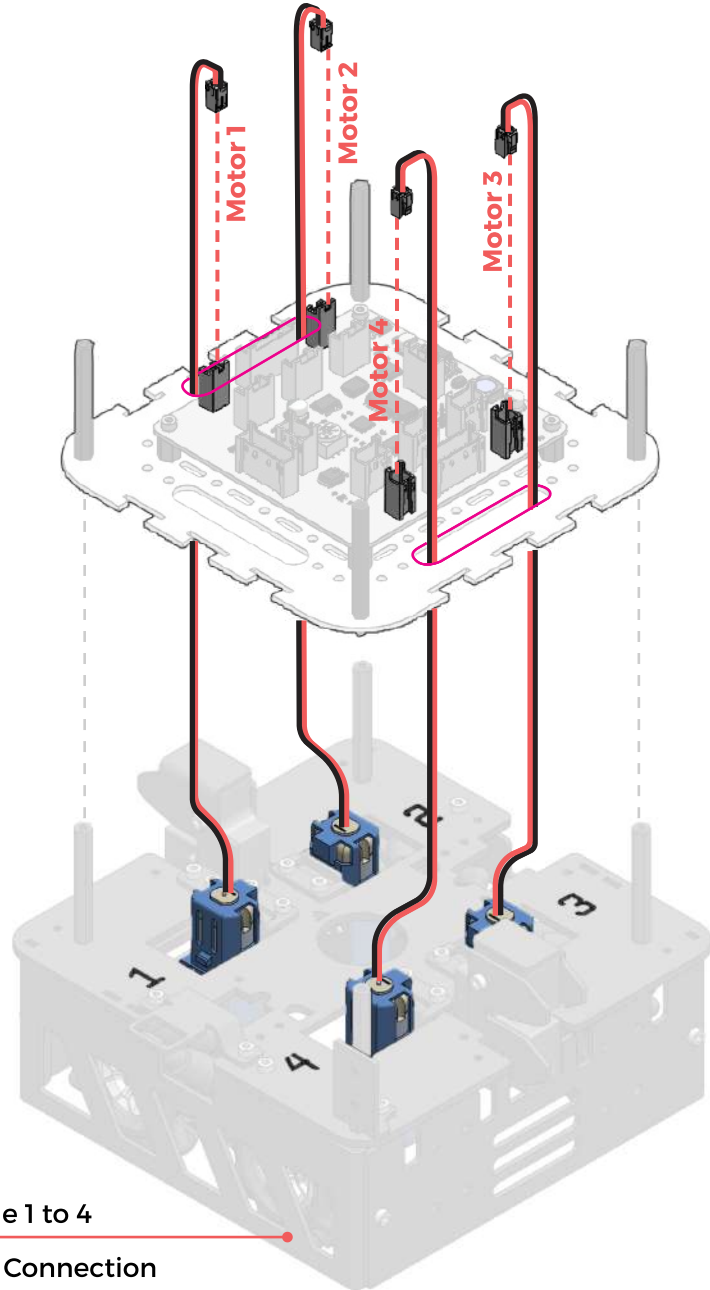
To unplug cable:
Press and pull out
the connector





Note: Position of solenoid latch and solenoid catch is different for each module





Module 1 to 4

Motor Connection

Opening to pass wire through

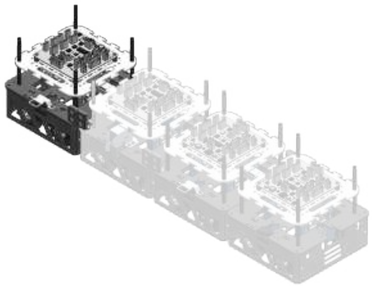
Make sure that Motor 1 is connected to Motor 1 connector on Slaveboard; the same goes for Motor 2, 3 and 4.

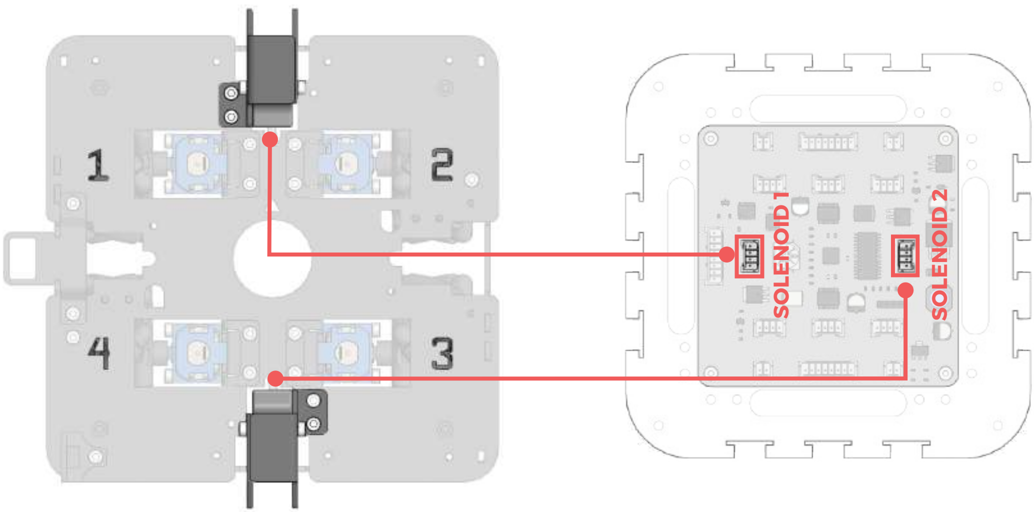
Motor cable connection is the same for all 4 modules

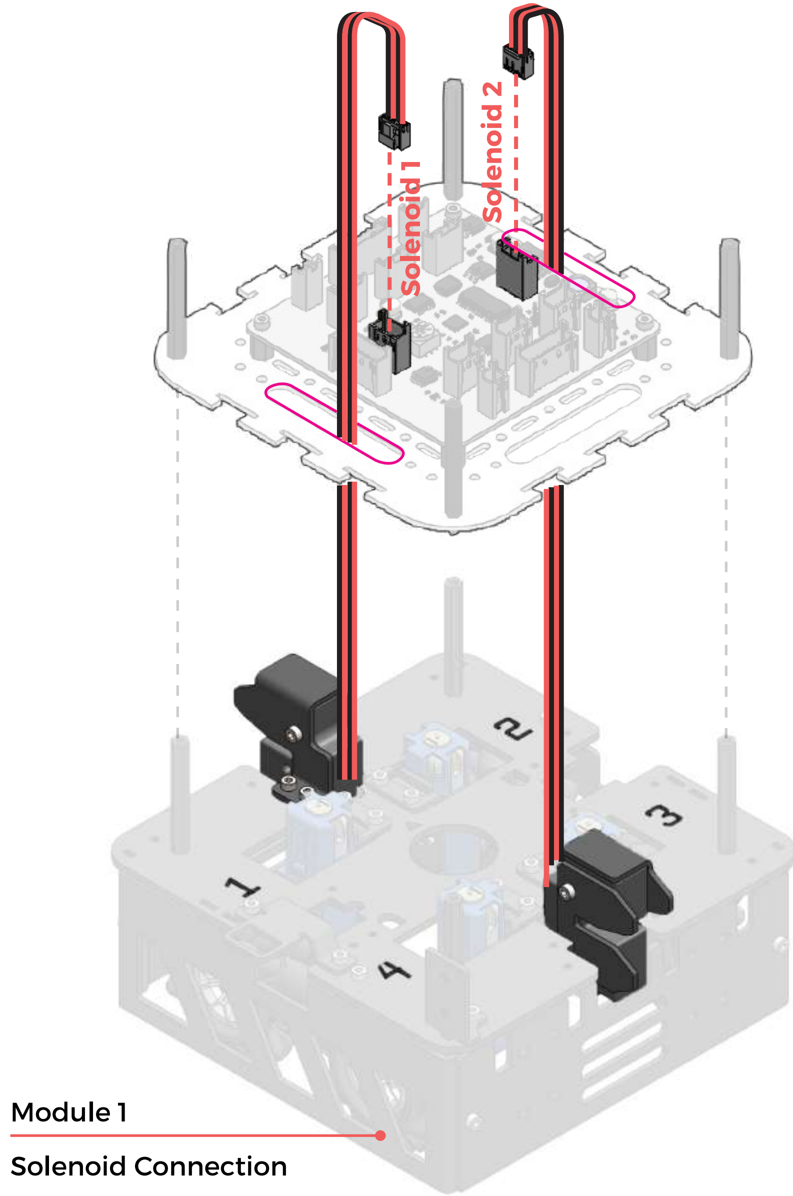
B 4 module 1 solenoid cable connection


solenoid cables (2)
attached to solenoids


module 1 SA (1)
attached to full assembly







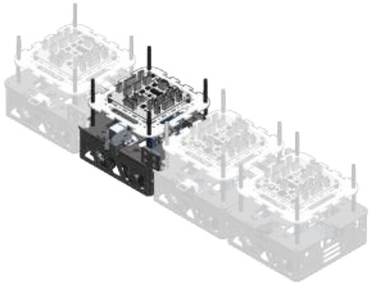
Opening to pass wire through

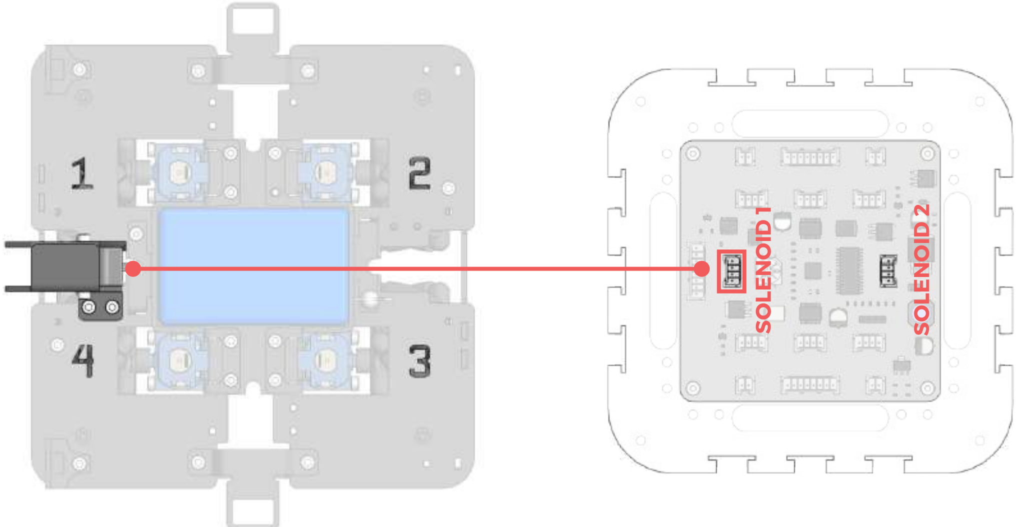
Make sure to connect the solenoid to the labelled solenoid connector exactly as in the diagram below.

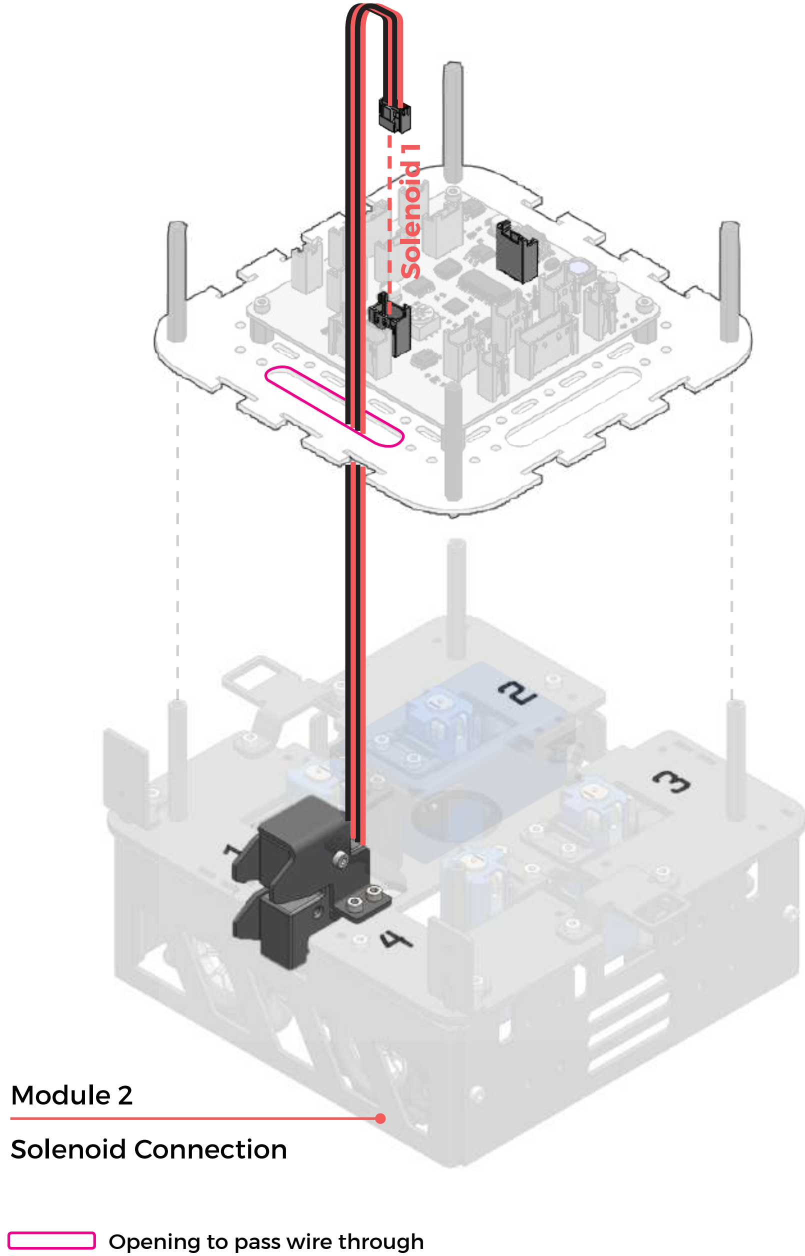
B 5 module 2 solenoid cable connection

solenoid cables (1)
attached to solenoids

module 2 SA (1)
attached to full assembly





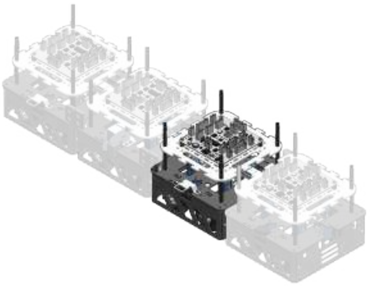


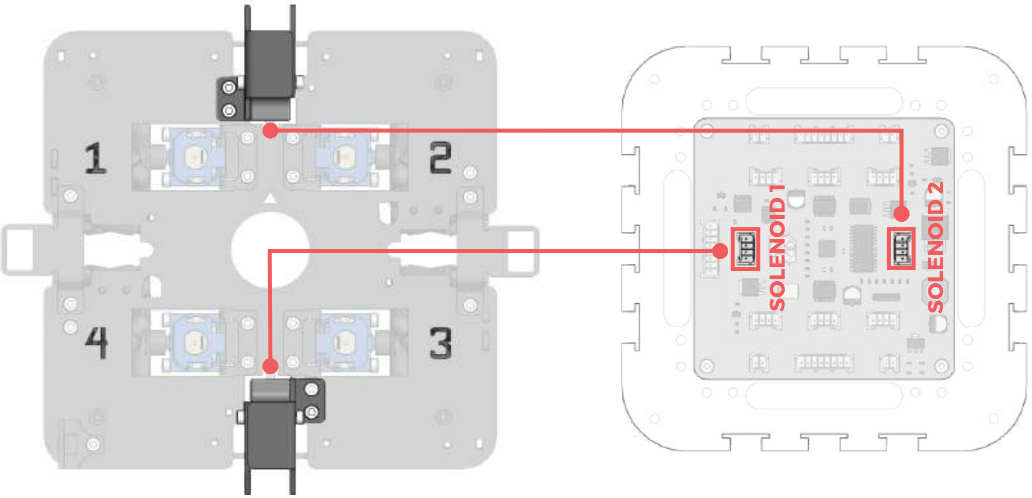
Make sure to connect the solenoid to the labelled solenoid connector exactly as in the diagram below.

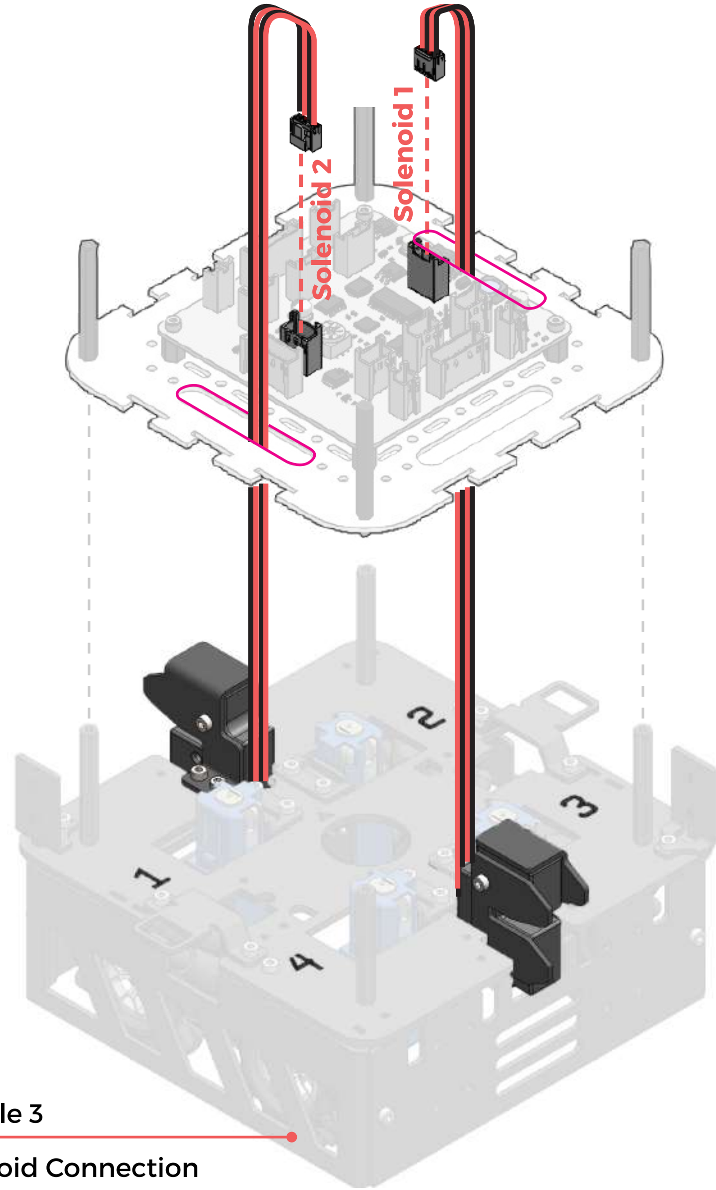
B 6 (module 3 solenoid cable connection)

solenoid cables (2)
attached to solenoids

module 3 SA (1)
attached to full assembly







Module 3

Solenoid Connection

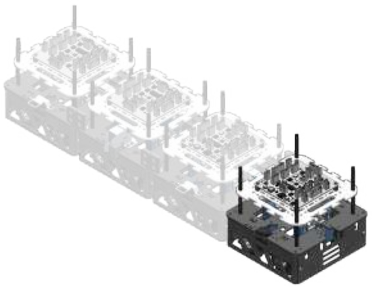
Opening to pass wire through

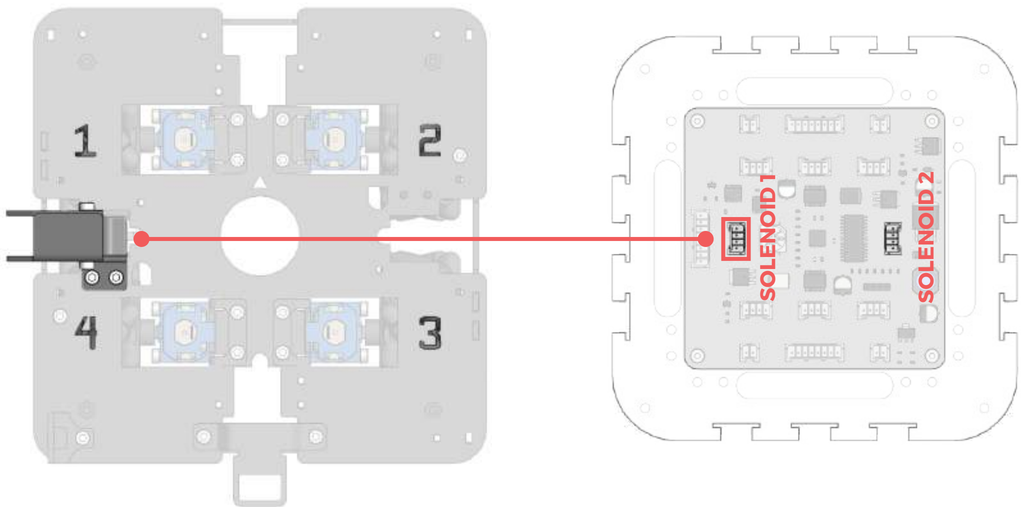
Make sure to connect the solenoid to the labelled solenoid connector exactly as in the diagram below.

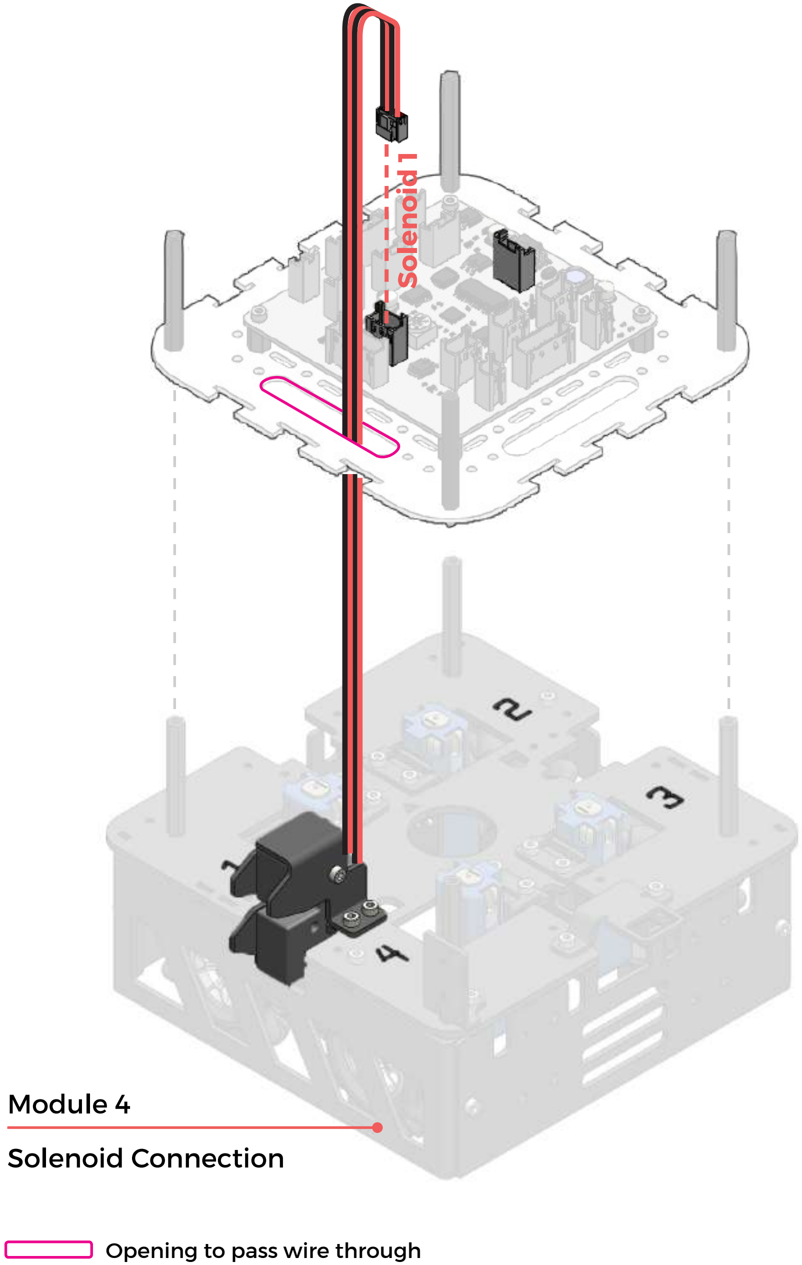
B 7 (module 4 solenoid cable connection)

solenoid cables (1)
attached to solenoids

module 4 SA (1)
attached to full assembly





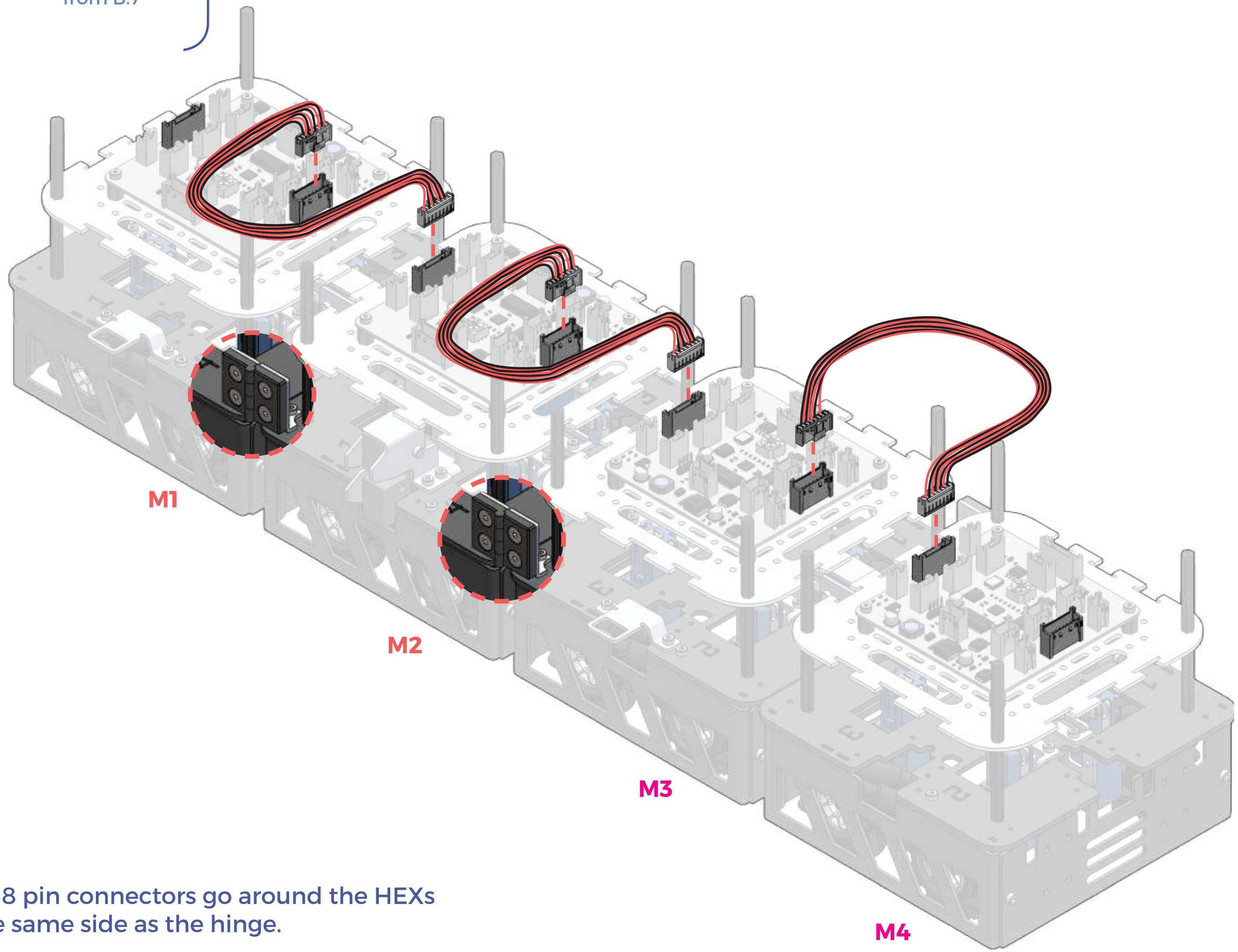


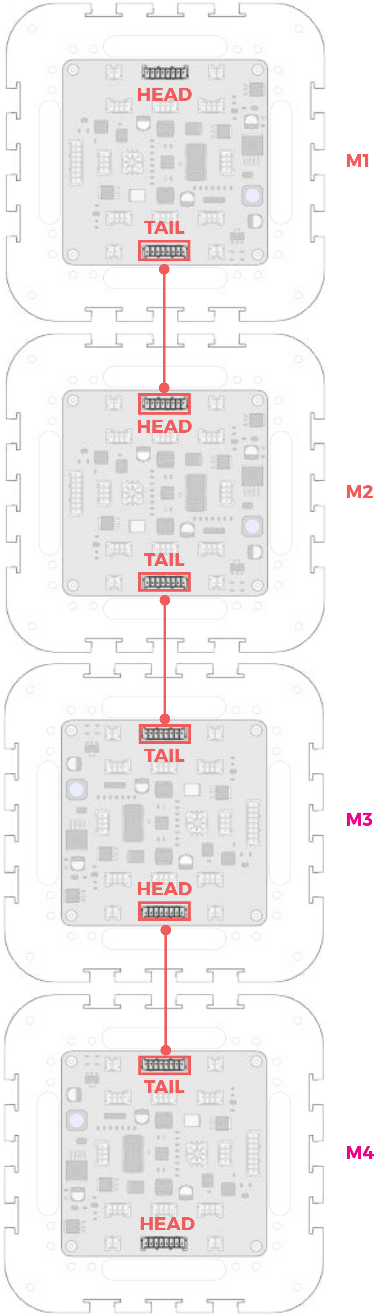
Make sure to connect the solenoid to the labelled solenoid connector exactly as in the diagram below.

B 8 (inter-module cable connection)

8-pin connectors (3)

full assembly (1)
from B.7



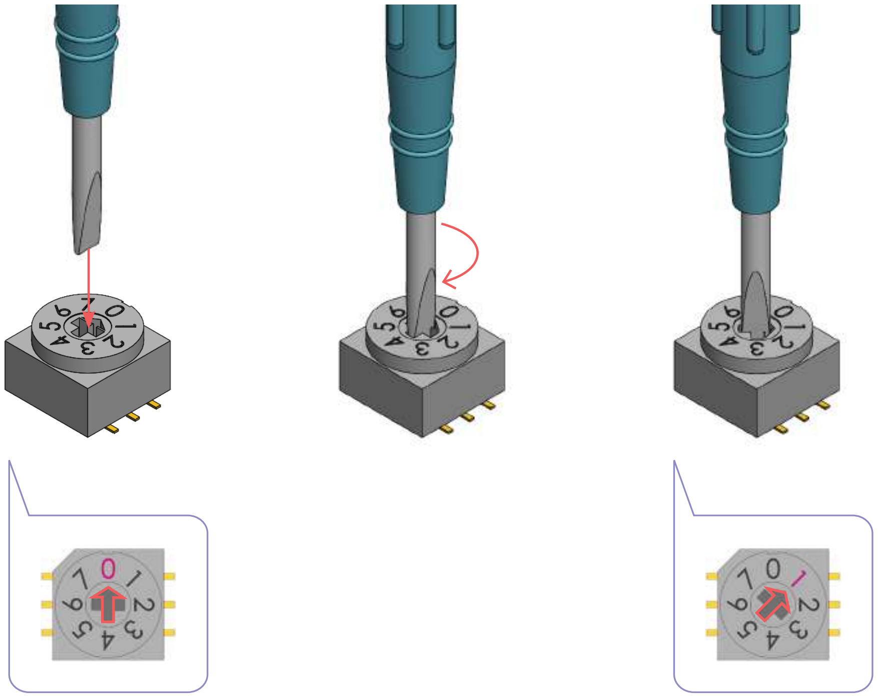


Make sure the 8 pin connectors go around the HEXs that are on the same side as the hinge.

B 9 (address selection)

full assembly (1)
from B.8

ceramic screwdriver (1)

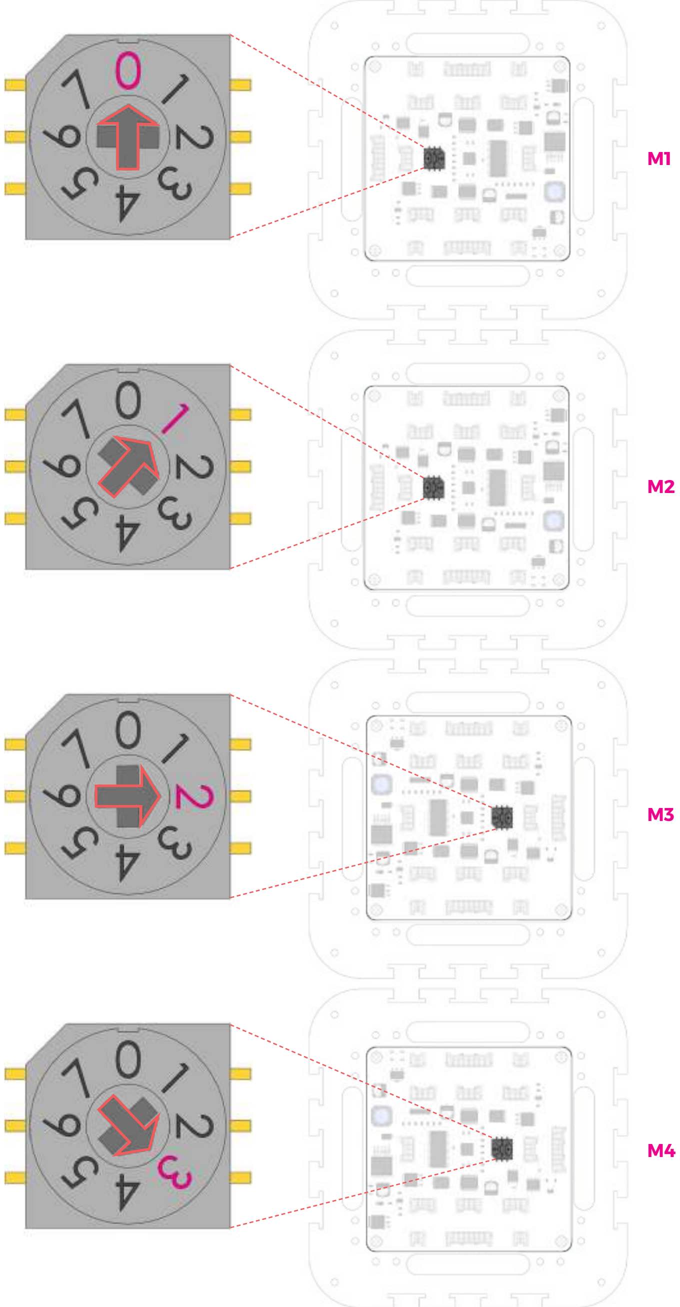


plan view

plan view

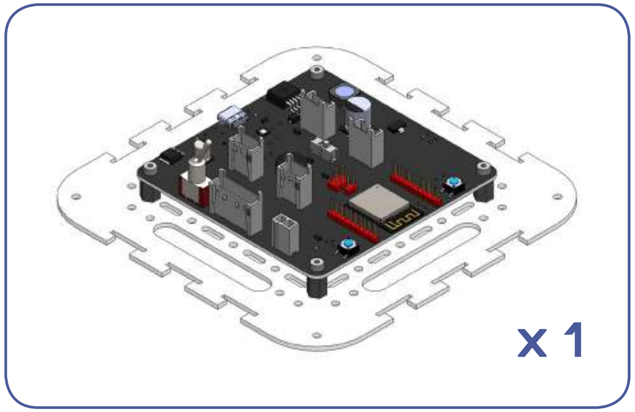
Use the ceramic screwdriver to adjust the rotary switch and select the address for all 4 modules.
Module 1: Address 0, Module 2: Address 1,
Module 3: Address 2, Module 4: Address 3

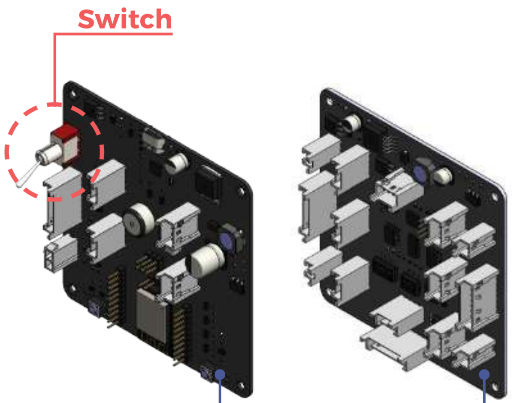
Address Select



B 10 (masterboard e-tray sub-assembly)

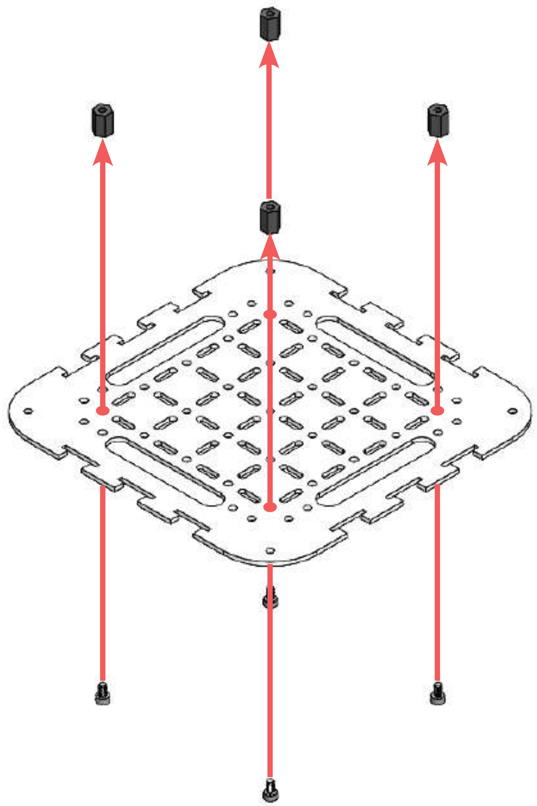
Acrylic base plate	(1)
Masterboard (ESP32)	(1)
Hex F-F M3 10mm	(4)
M3 x 5	(8)

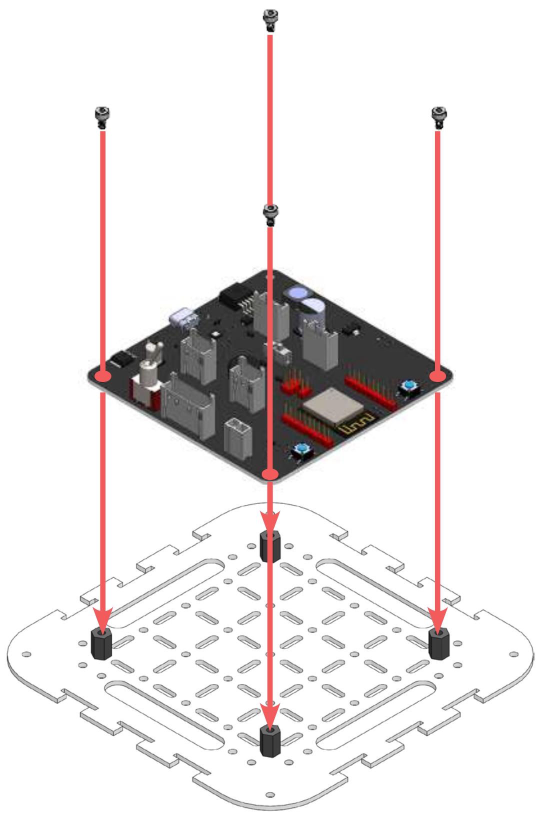




Masterboard
1 unique piece

Slaveboard
4 identical pieces

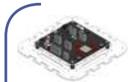




Make sure that you are attaching the masterboard and not the slaveboards.

How to differentiate between masterboard and slaveboards:
- Masterboard has a special switch

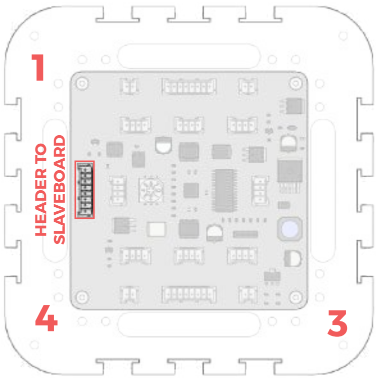
B 11 (masterboard e-tray onto main assembly)


masterboard e-tray SA (1)
from B.10

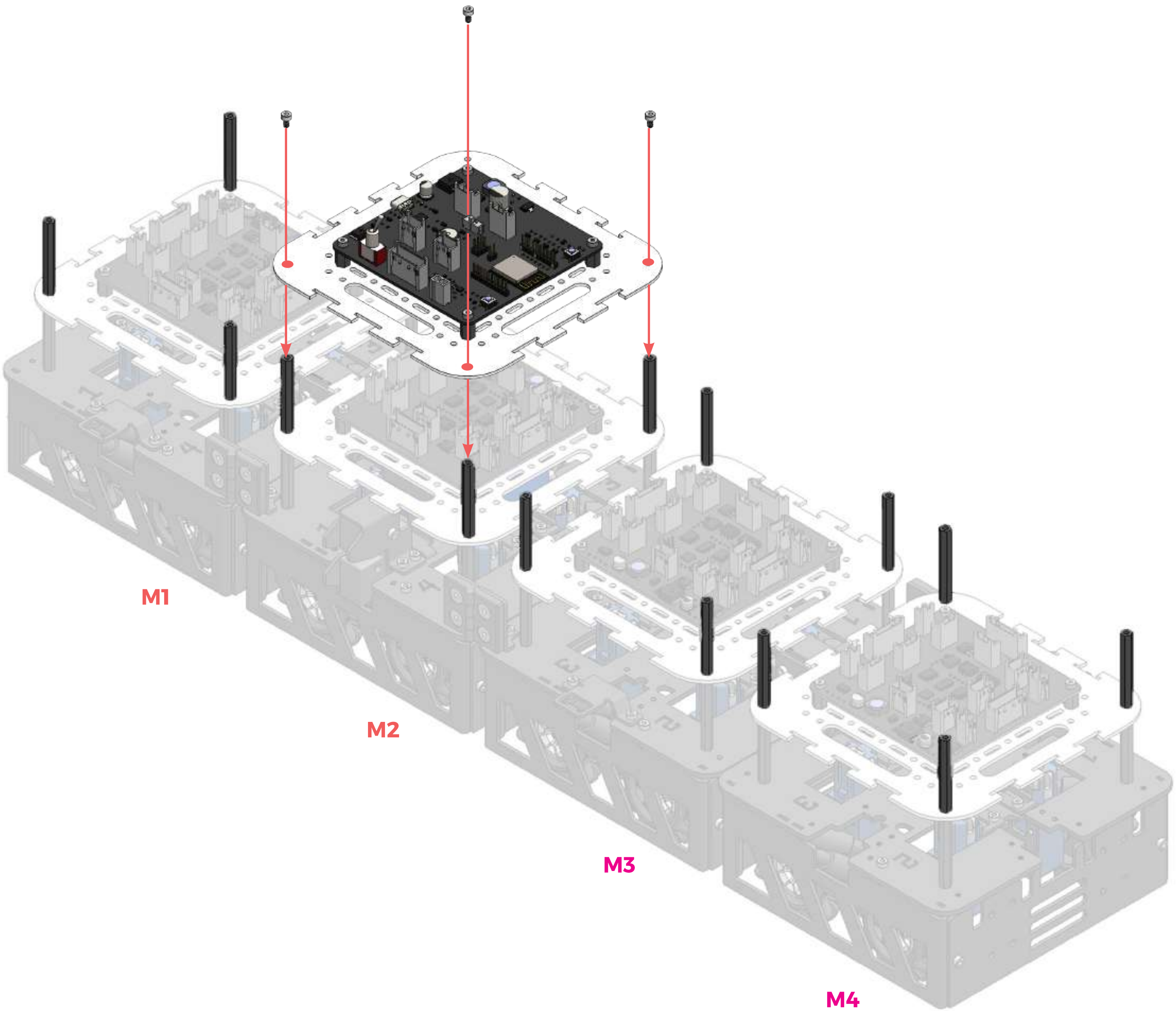

full mechanical SA (1)
from B.9

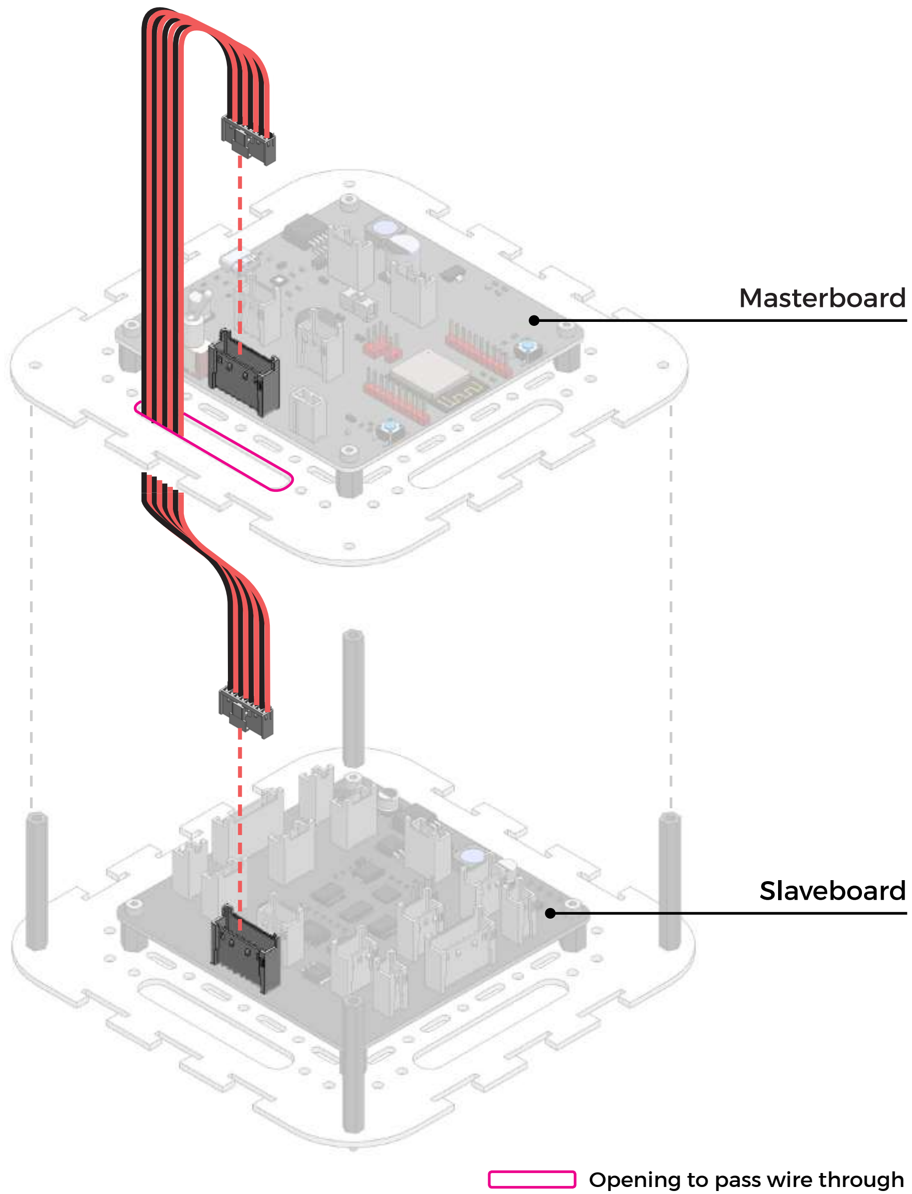

8-pin connector (1)

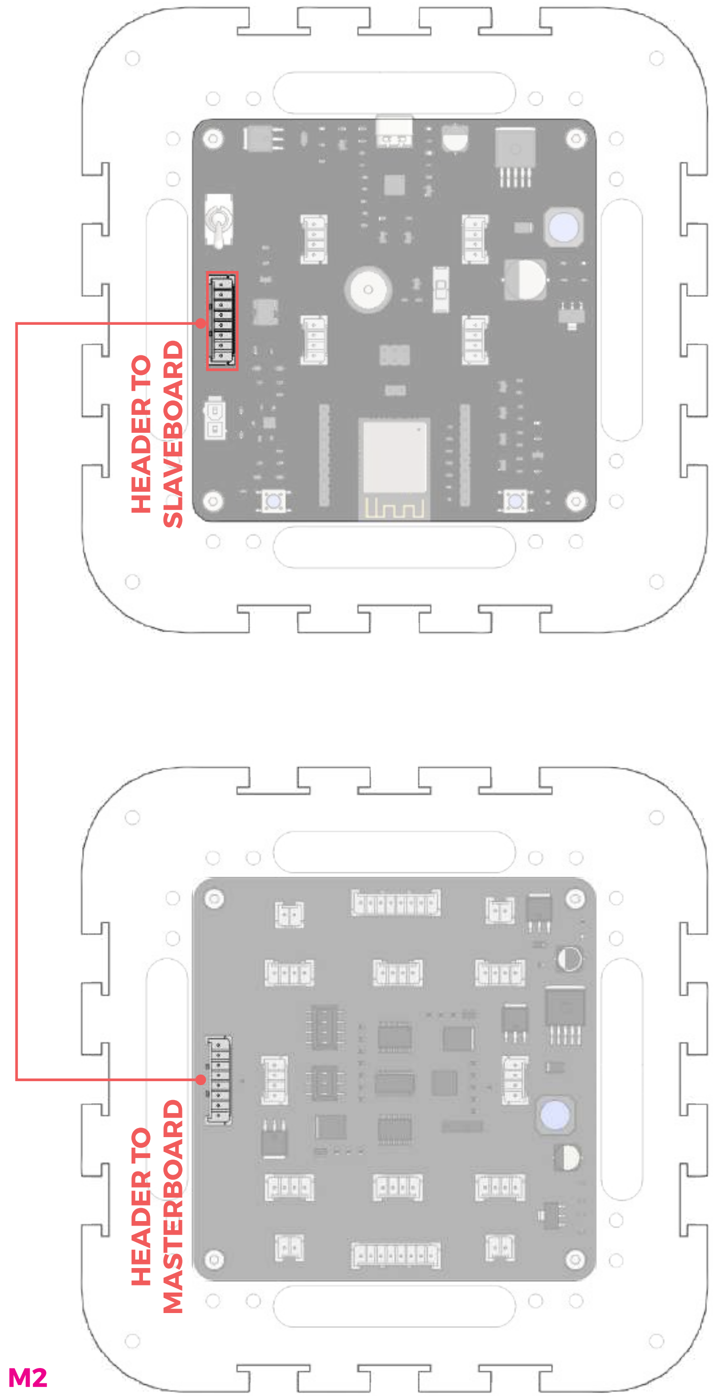

M3 x 5 (4)



Align the header to slaveboard connector on the masterboard to face the side connecting Motor 1 and 4.




Masterboard
Slaveboard
Opening to pass wire through

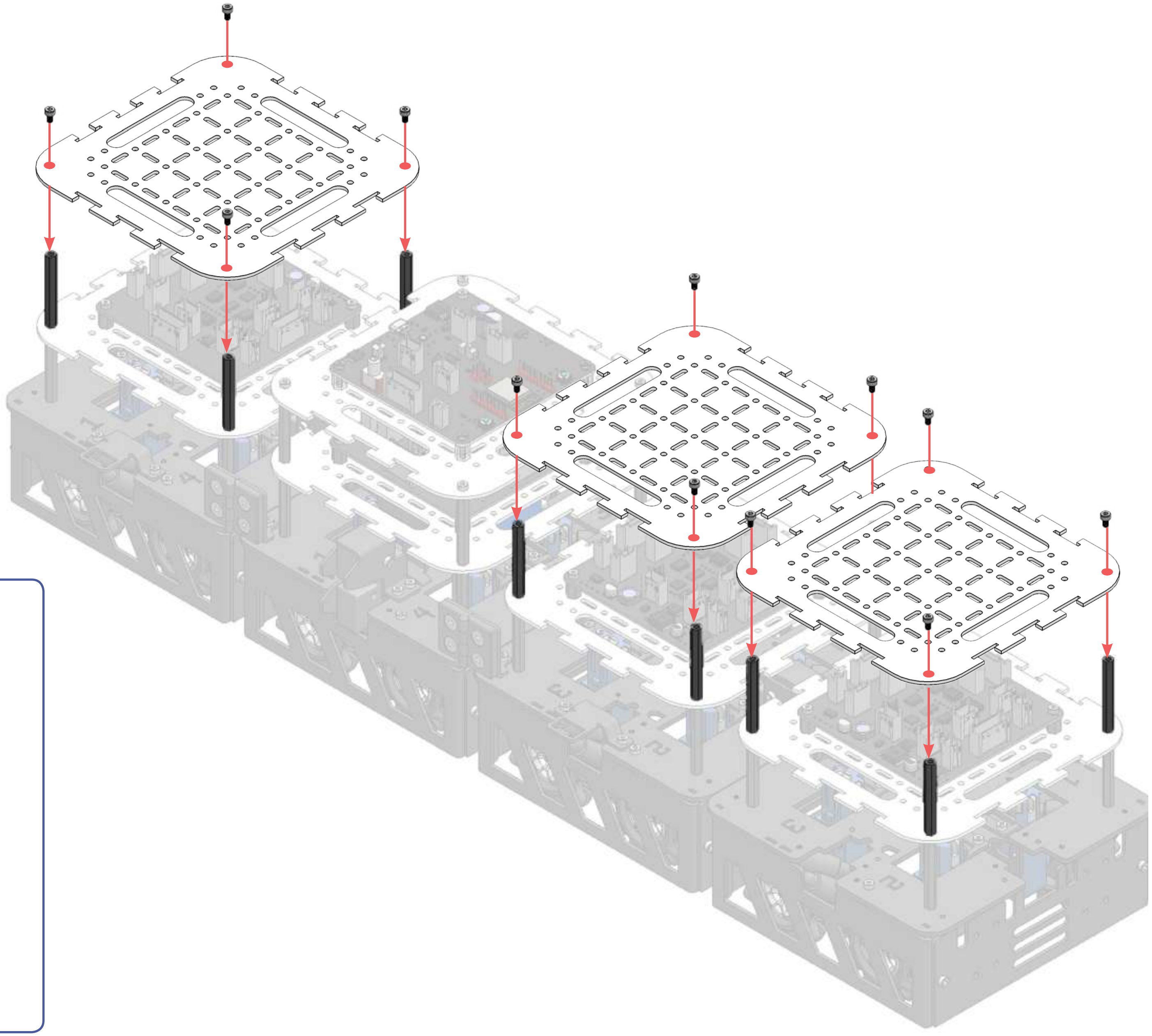

HEADER TO SLAVEBOARD
HEADER TO MASTERBOARD
M2

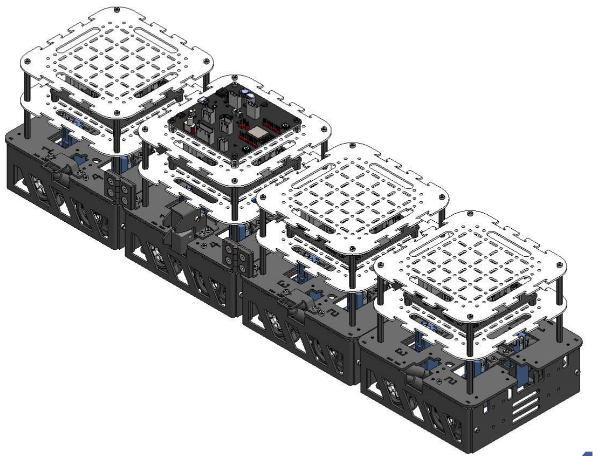
B 12 (attach acrylic covers)


Acrylic base plate (3)


full mechanical SA (1)
from B.11

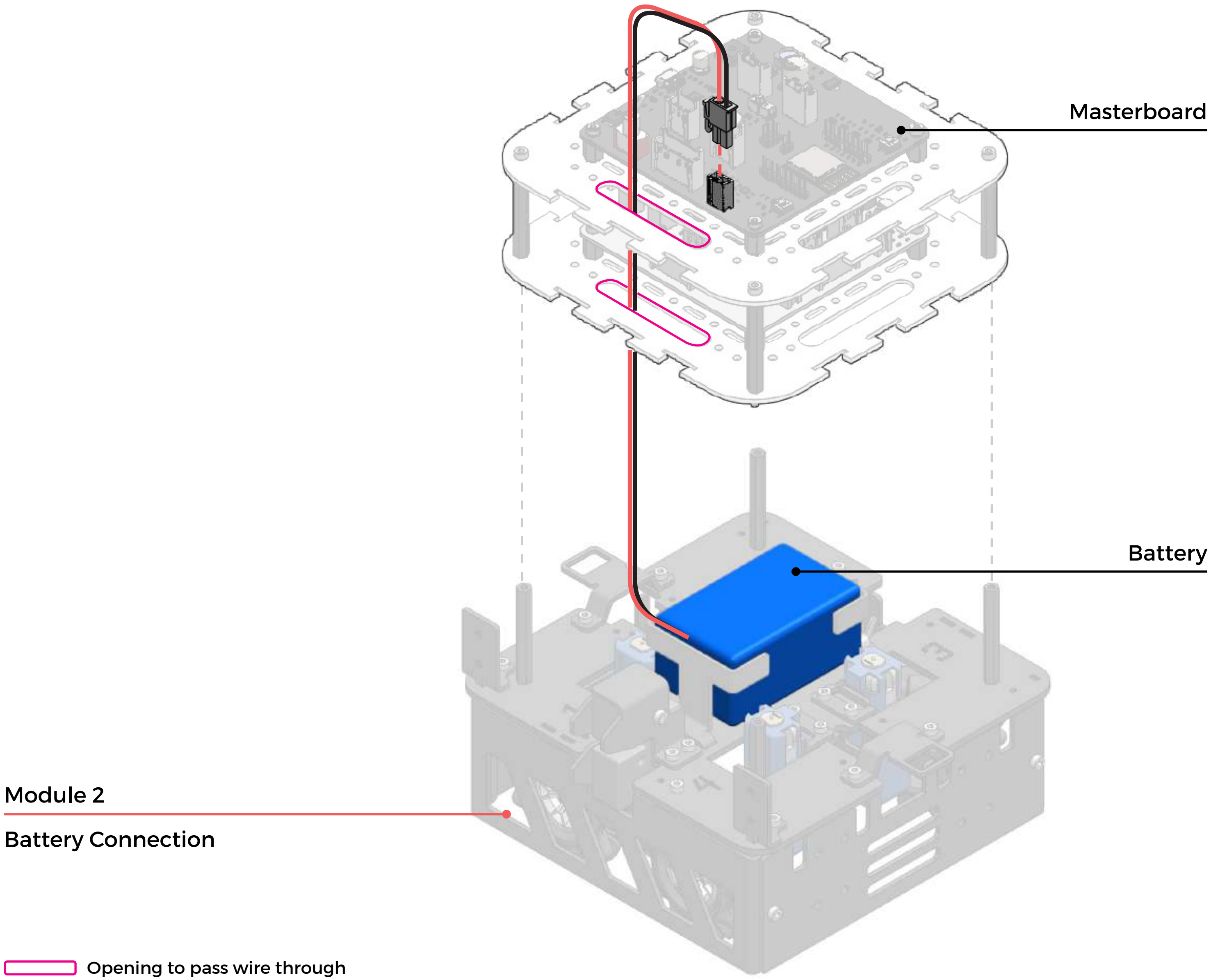

M3 x 5 (12)



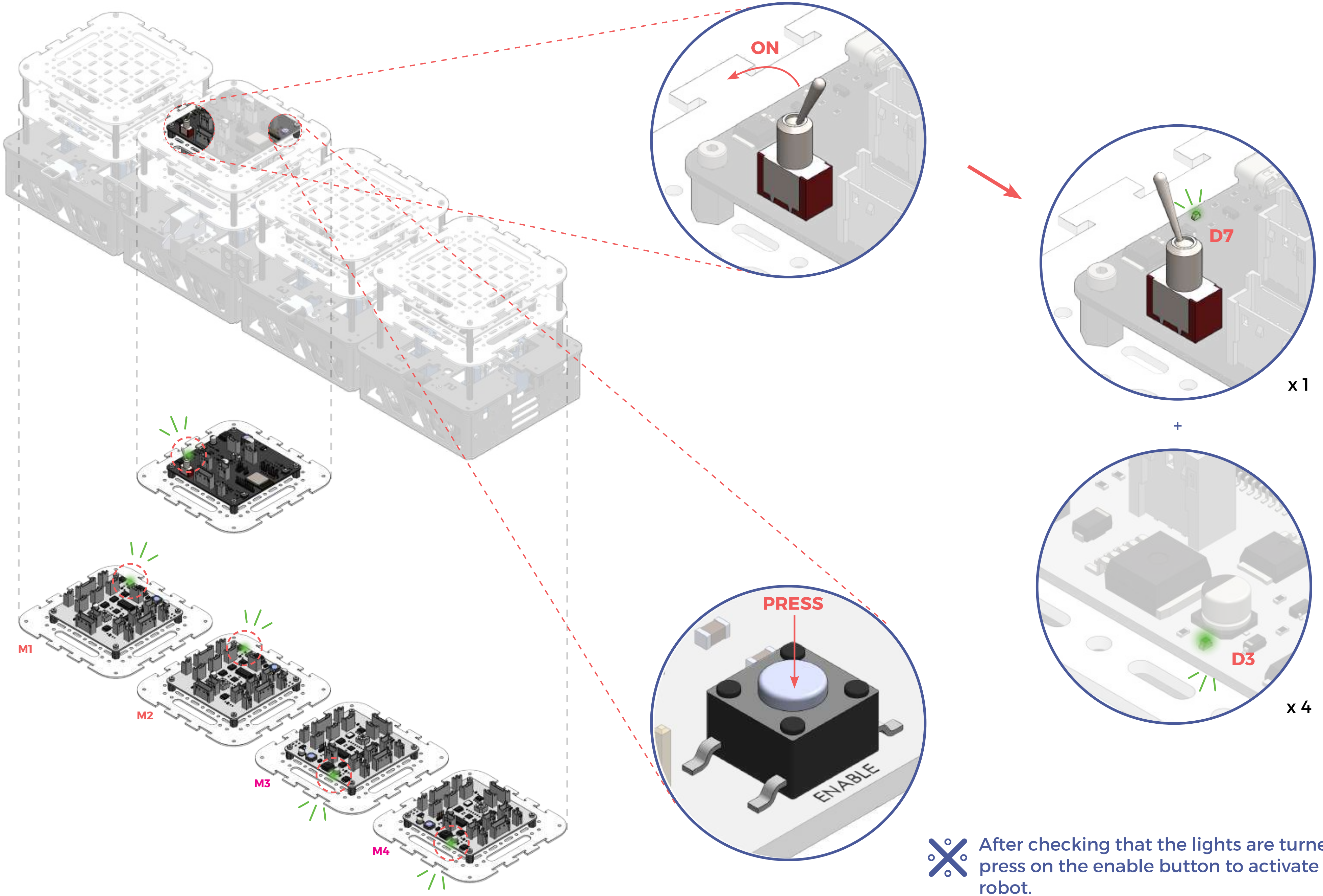

x 1

B 13 (connect battery to masterboard)

LiPo Battery	(1)
full mechanical SA	(1)



B 14 (check connection)

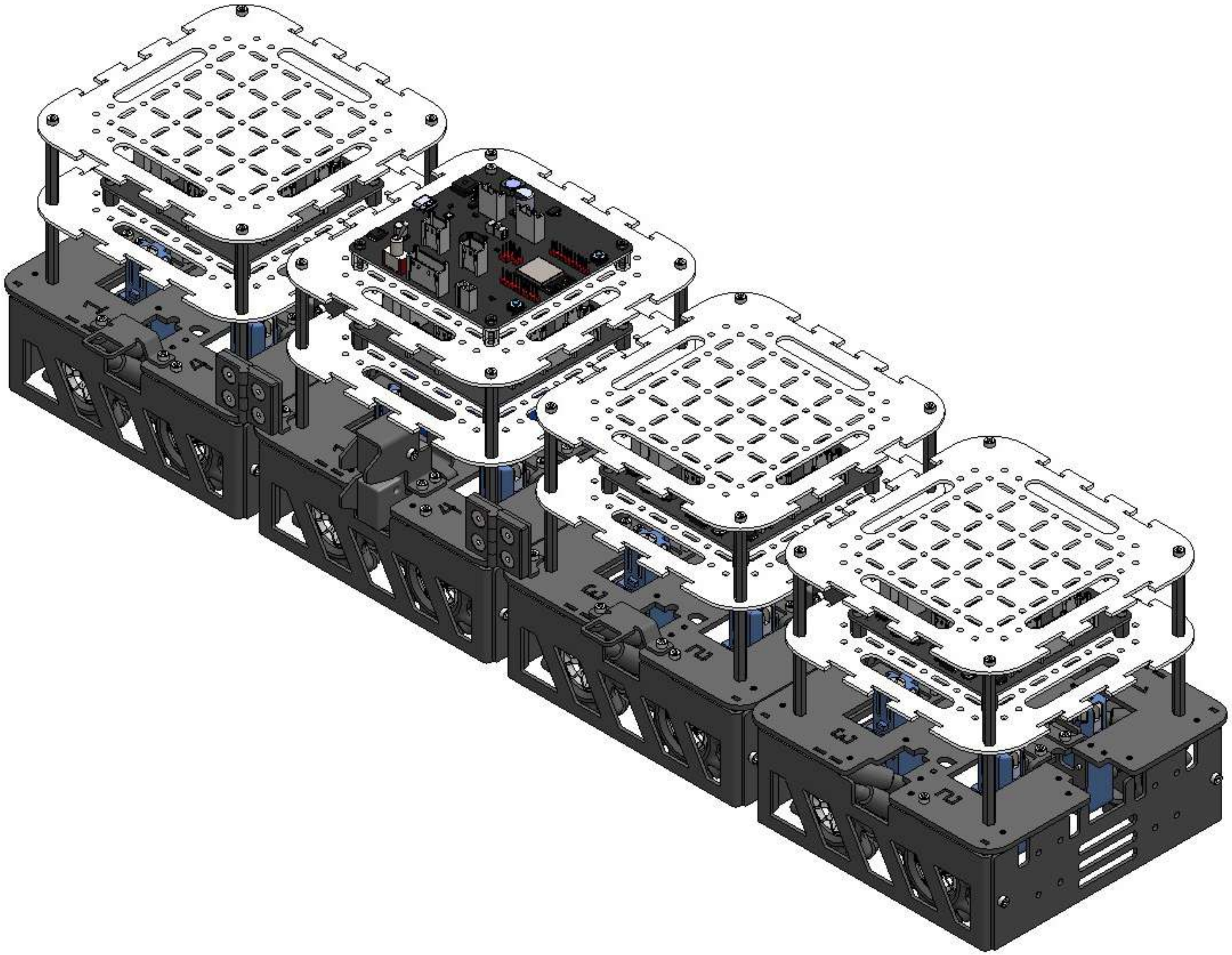


After checking that the lights are turned on, press on the enable button to activate the robot.

B electronic assembly completed

smorphi is ready!

next step: connect with app



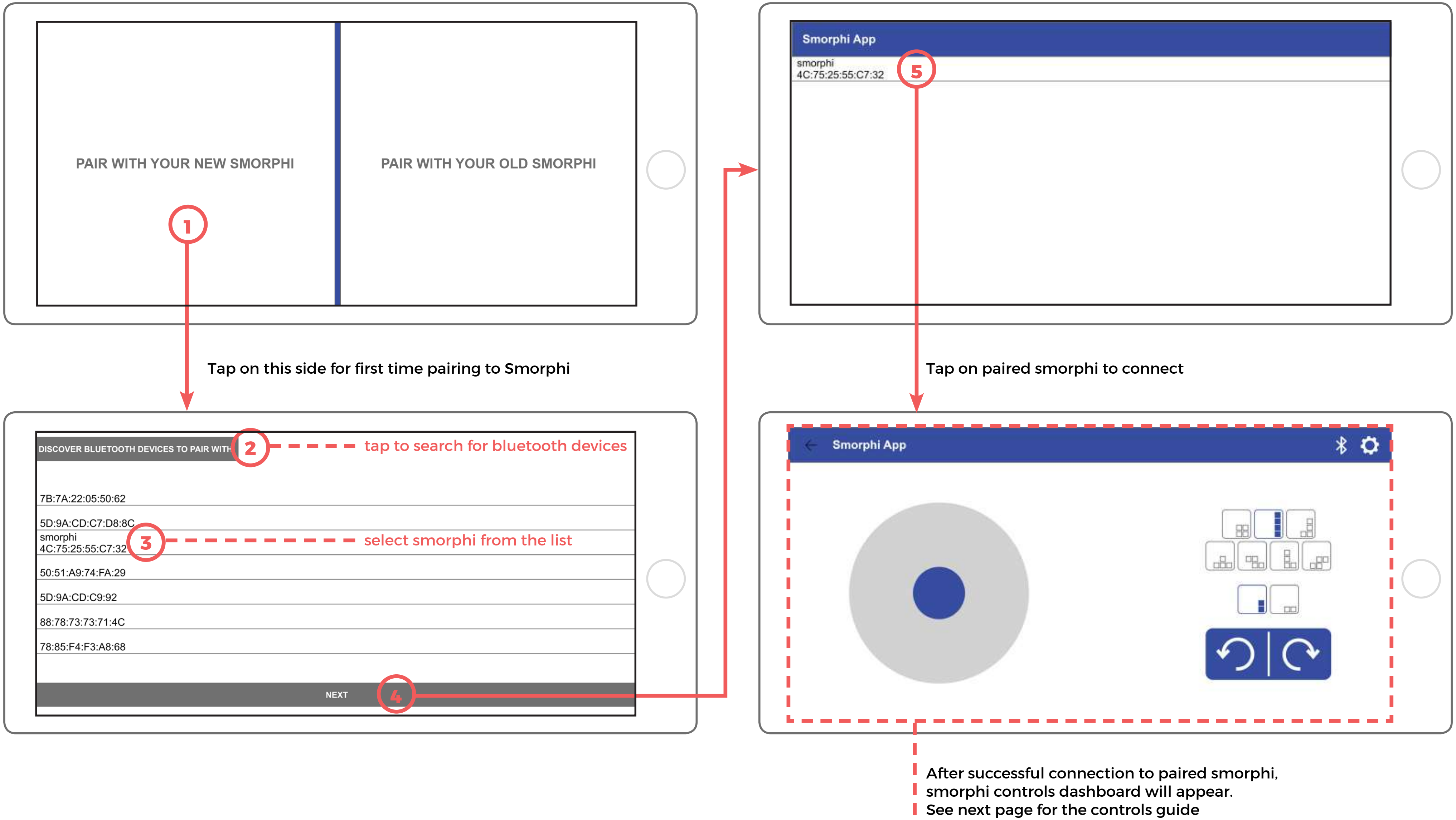
app

1. App Download.

Not all devices are compatible now.
See the table below to check if your device is compatible with the Smorphi app.

Smorphi App Information	
Available Platforms	Android
Download from	Google Play
App Icon & Name	smorphi
System Requirements OS Requirements	Android 6+ Bluetooth 4.0+

2. Bluetooth Connection. Turn on Smorphi and the Bluetooth of your smart device. Tap on Smorphi app icon to launch application.



3. Smorphi controls dashboard guide.

Direction & Speed Joystick Control

Direction Control | Move Joystick in intended direction
Smorphi will move in the direction that the user intend

Speed Control | Speed of smorphi is controlled by how much the user push the joystick from centre

Shape Transformation Control (Smorphi Mini)

Shape Transformation Control (Smorphi Square)

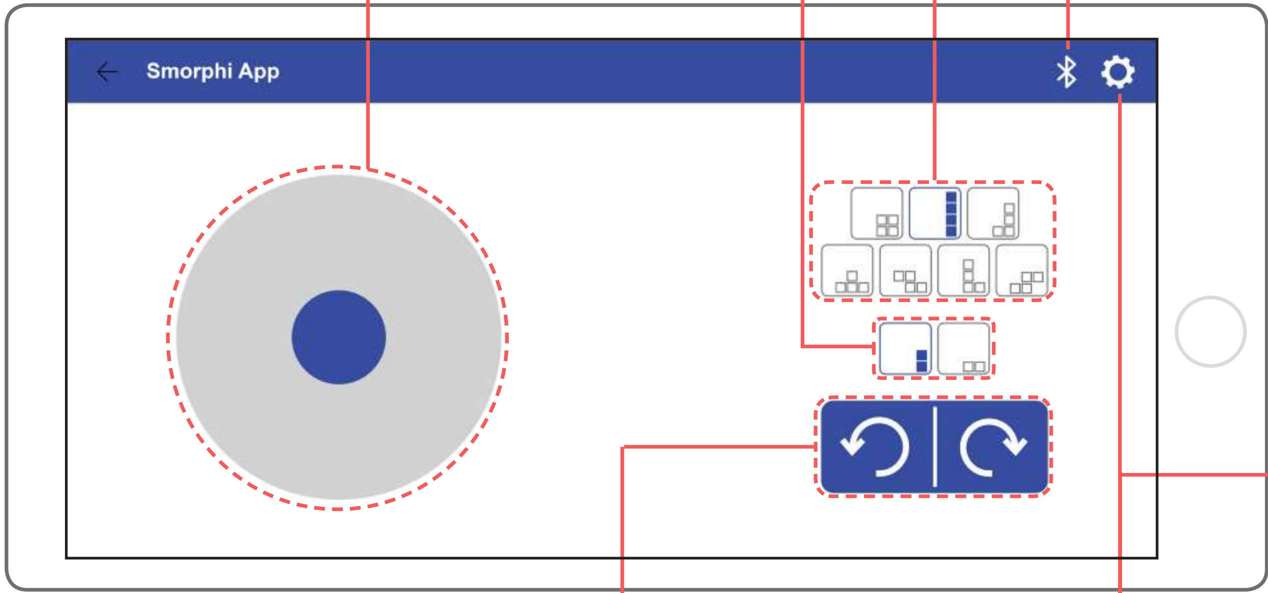
Current Shape | Highlighted Shape on Highlighted Button

Shape Transformation | Click on the button to transform Smorphi into the intended shape

Note: Wait for Smorphi to finish transforming before attempting another shape transformation

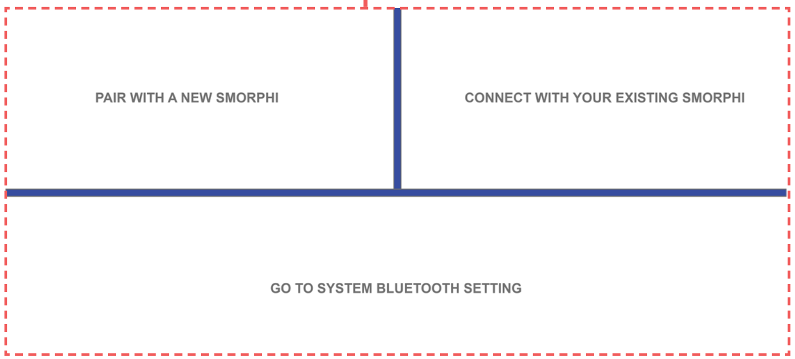
Bluetooth

After pairing the robot, you can click on the bluetooth button to reconnect the robot to the app.



Setting

Tap on the setting button to access settings to the robot-device bluetooth connection



Pivot Turning

Press and hold to pivot the robot around centre

For tutorial to Smorphi activites and exercises, please refer to the online class exercises by following the link below:
https://github.com/WefaaRobotics/Smorphi/wiki/

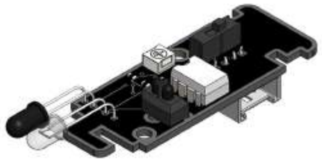
sensors

sound sensor (x2)

Sound sensor measures volume of sound. Onboard potentiometer* can be used to tune the range of sensing.
Possible applications: sound-triggered shape transformation or sound-triggered locomotion.

temperature sensor (x1)

Temperature sensor measures surrounding temperature, with a range of -55˚C to 125˚C.



IR sensor (x4)

IR sensors comes with 2 different modes, toggled by the switch onboard the sensor itself. One IR is front-facing and can be used to detect obstacles ahead. The other IR faces the ground and can be used for path tracking.

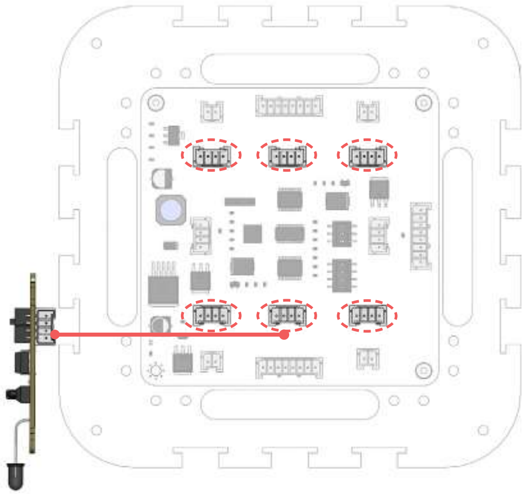
Husky camera (x1)

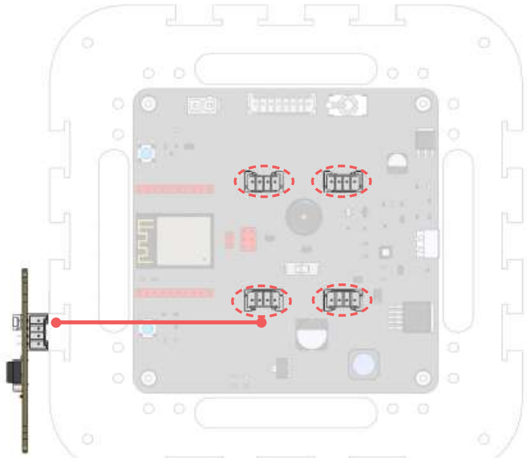
Husky Camera is an AI camera that is able to colour code, detect and track lines and intersections, and learn to detect objects taught to it.

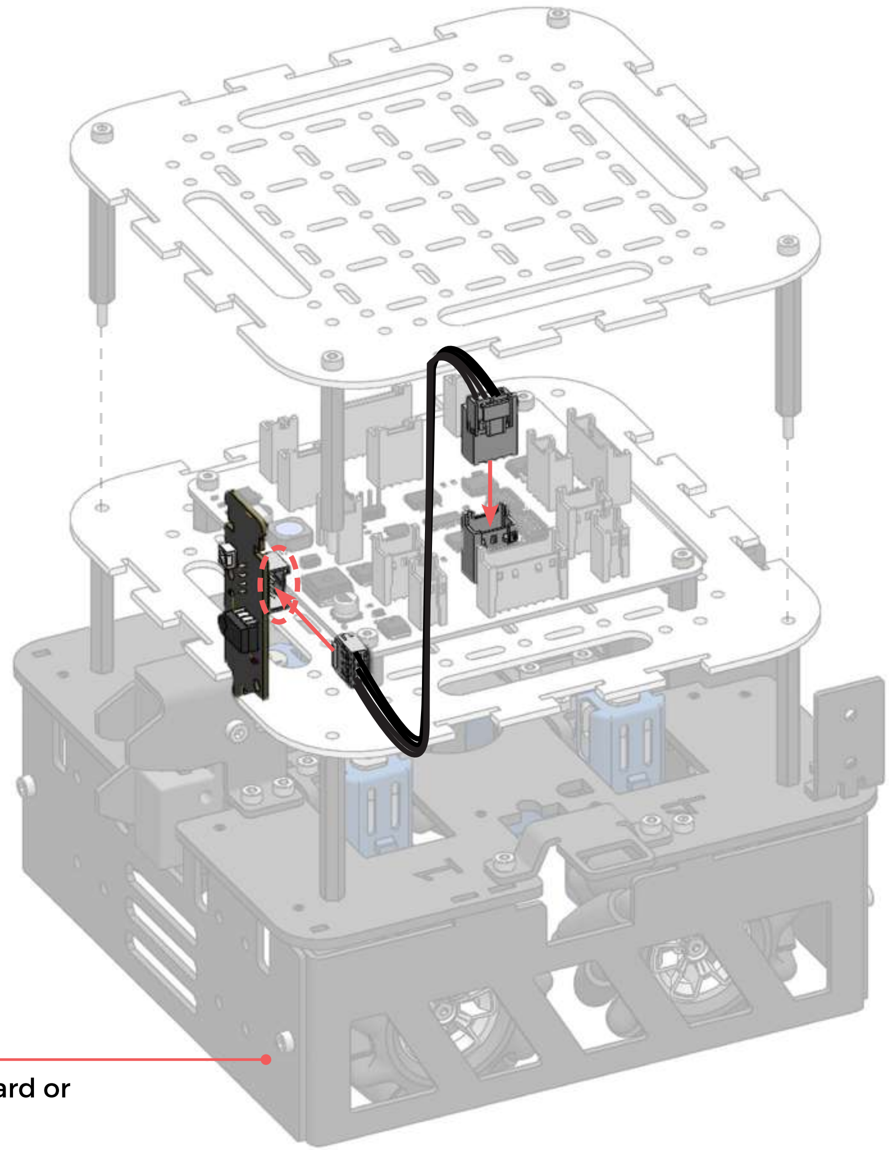
For sensor related activities and implementation, please refer to the online class exercises by following the link below:
https://github.com/WefaaRobotics/Smorphi/wiki/Robot-Exercises

* Tutorial on how to operate the potentiometer can be found by following the link below:
https://github.com/WefaaRobotics/Smorphi/wiki/Exercise-6

sound/IR sensor wiring





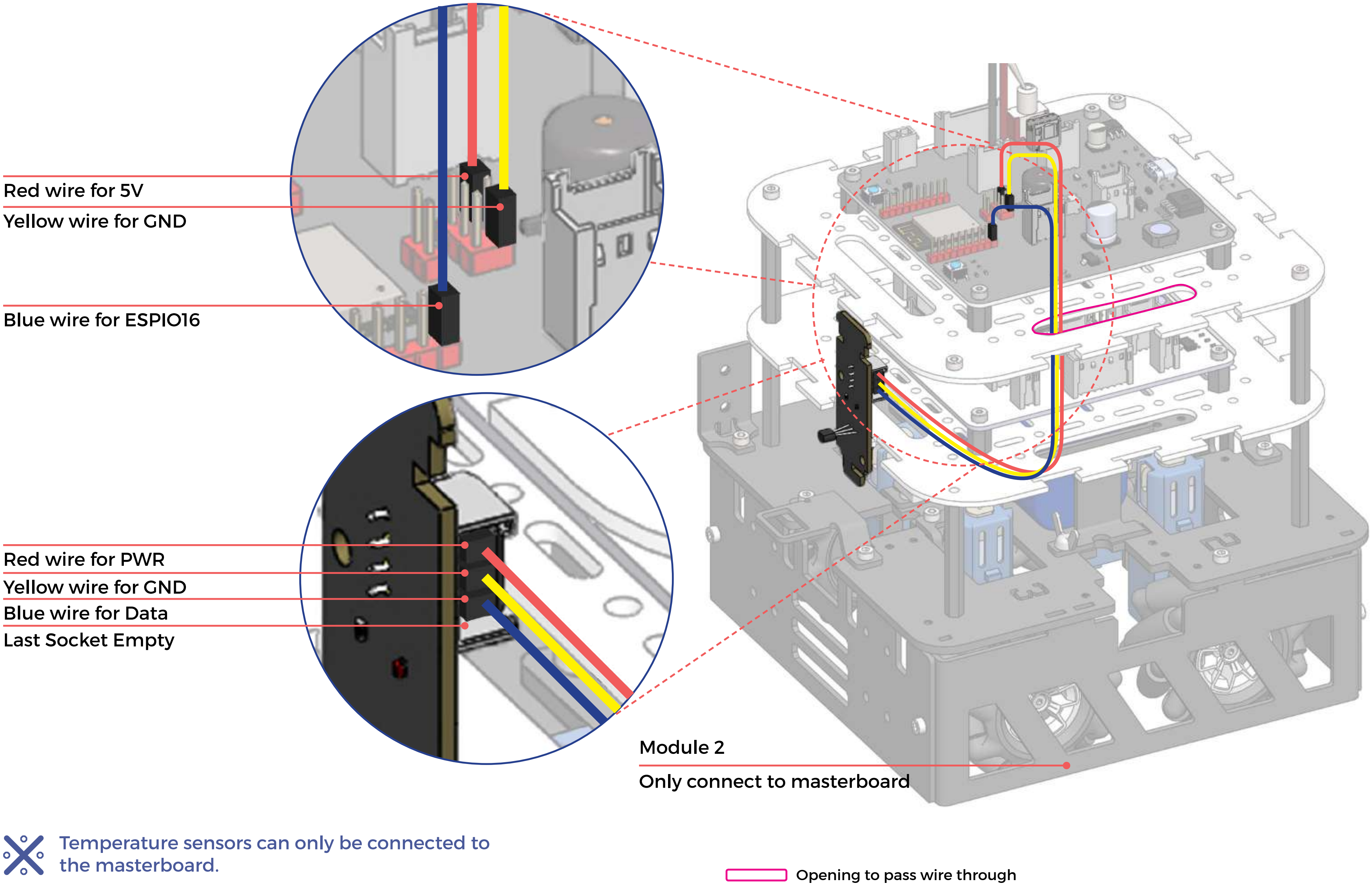


Module 1/2/3/4

Connect to masterboard or slaveboard

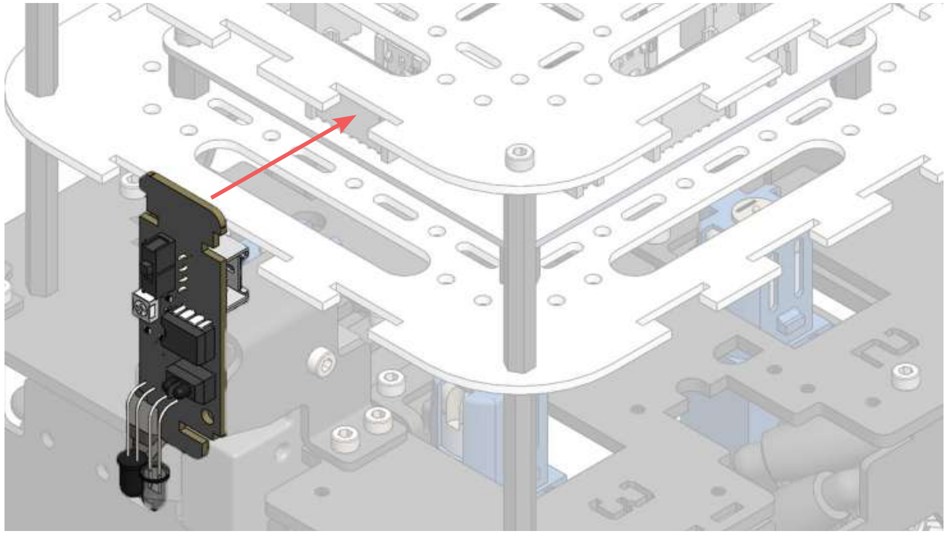
Sound/IR sensor can be connected to any of the 6 sensor ports on the slaveboard (top) or any of the 4 sensor ports on the masterboard (bottom).

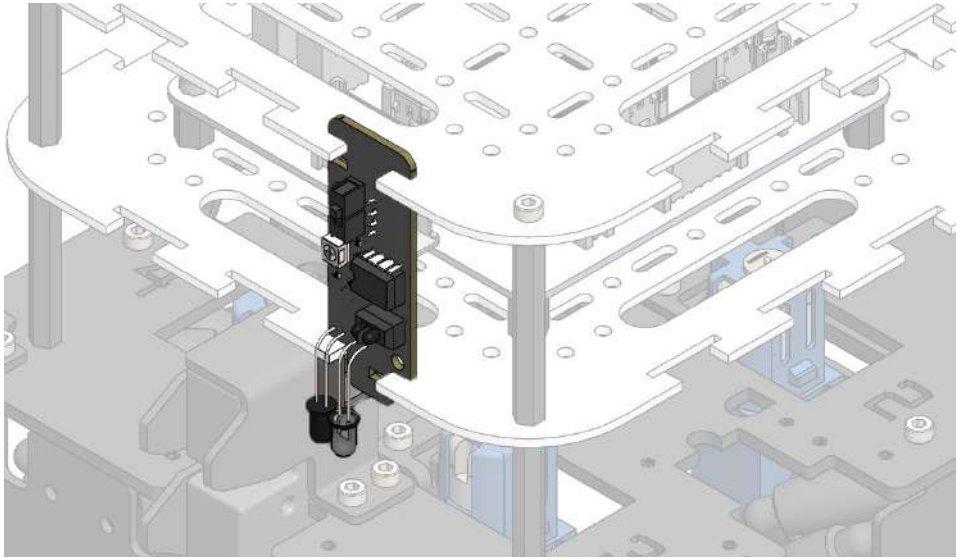
(temperature sensor wiring)

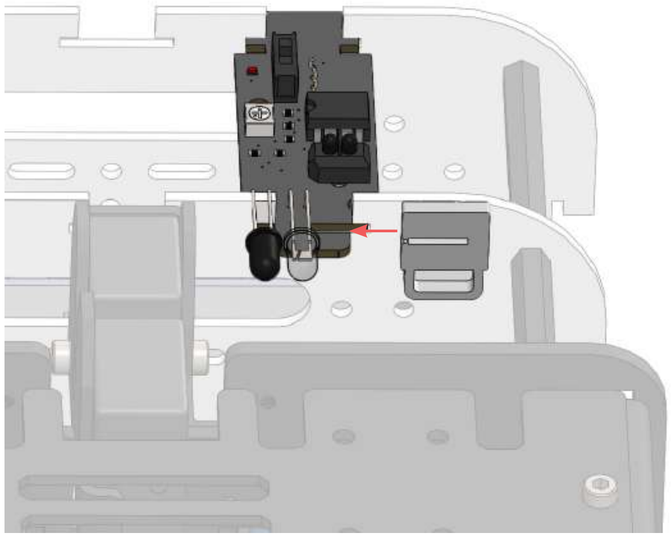


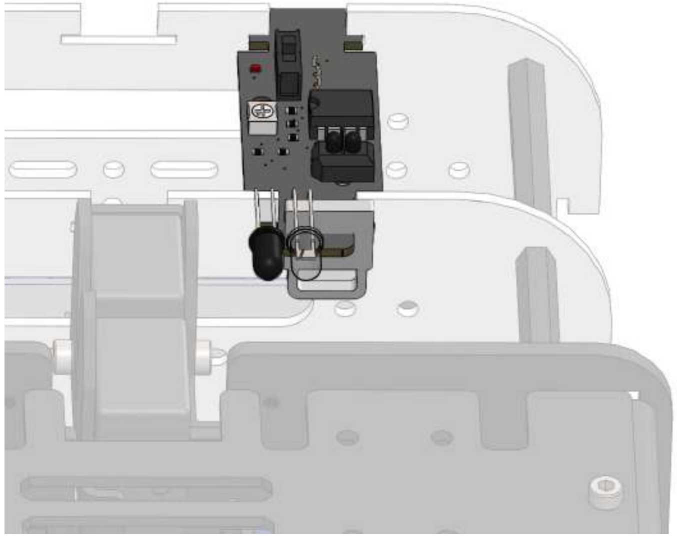
Temperature sensors can only be connected to the masterboard.

(sensor position 1)



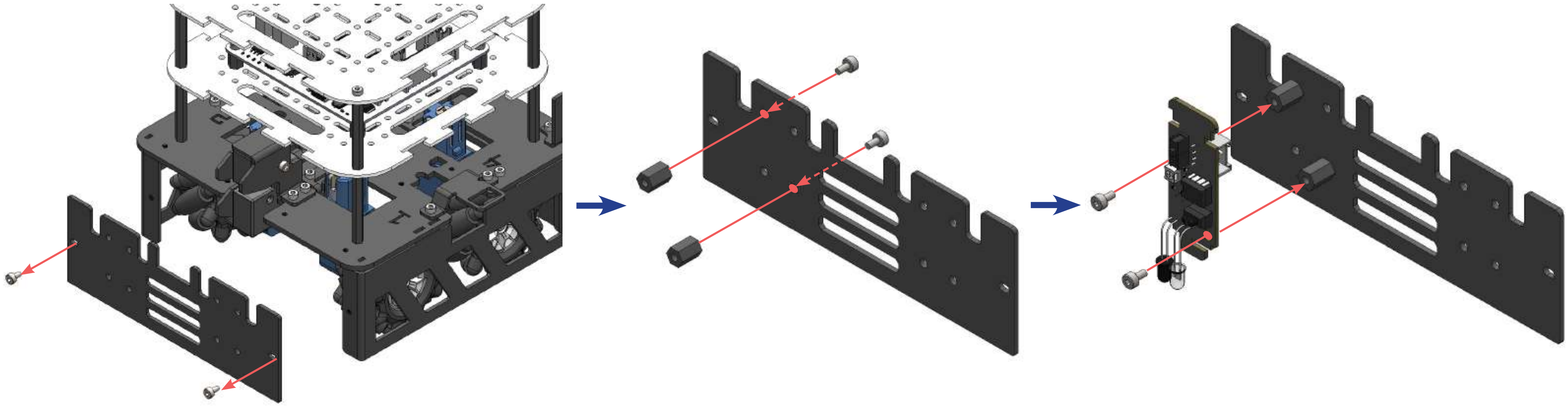


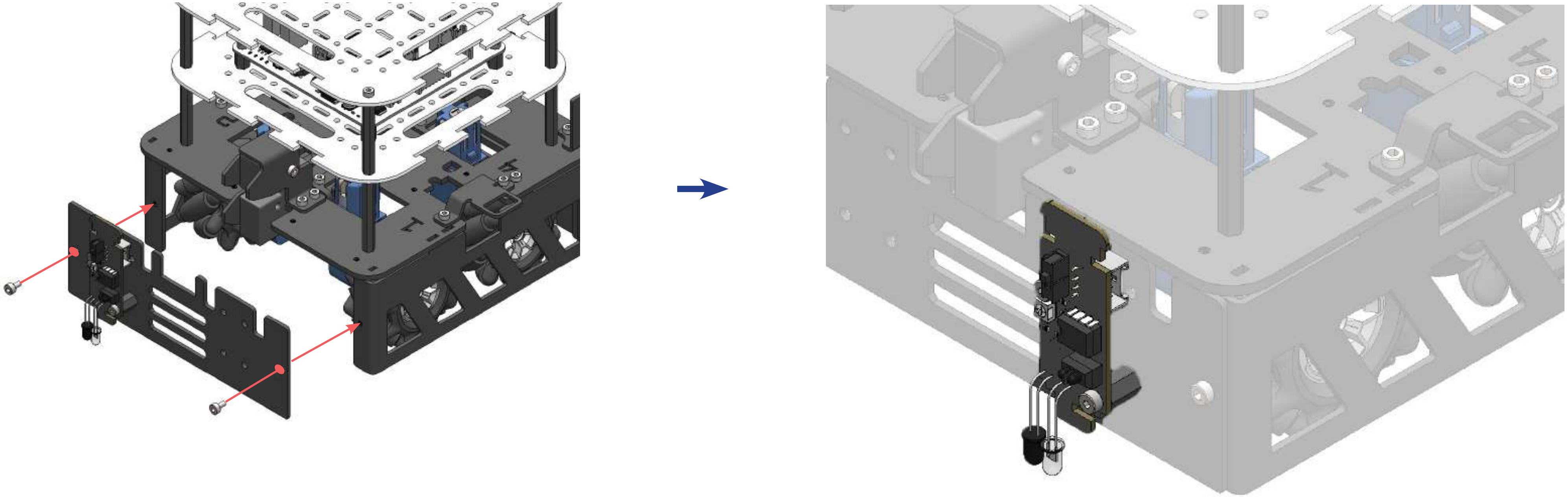




Sensors can be secured to the robot using the sensor lock provided.

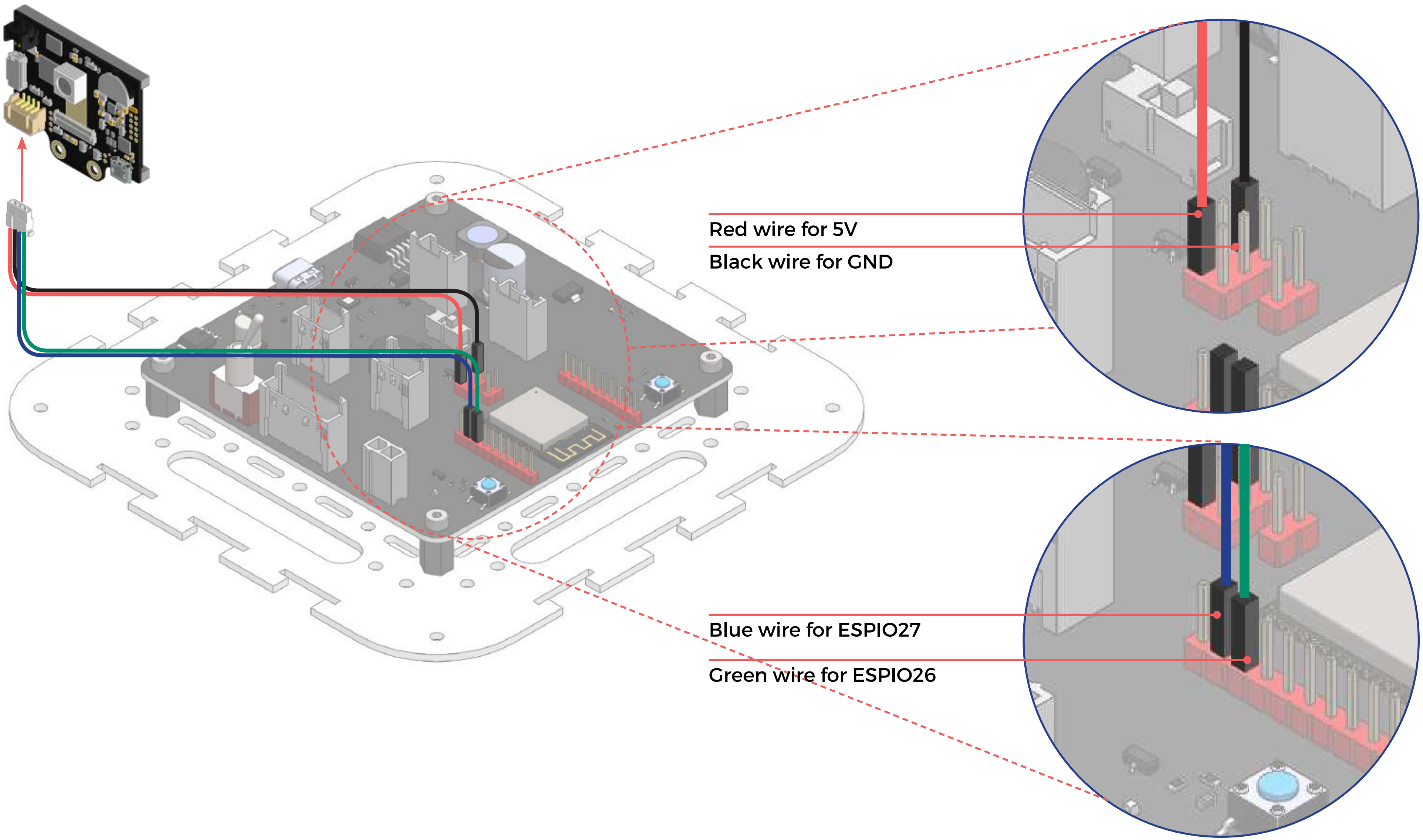
sensor position 2



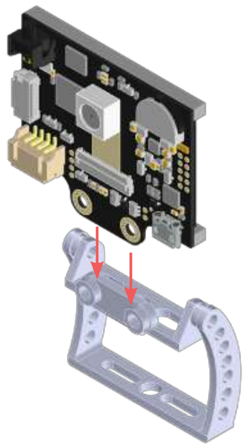


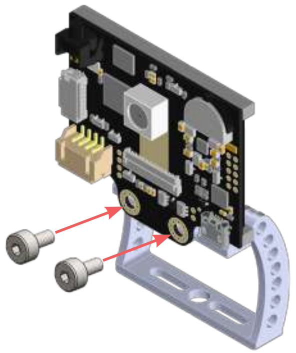
Sensors can also be attached at the skirt panel of the robot.

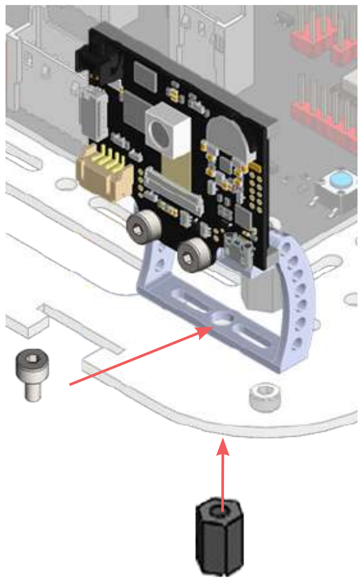
huskycam wiring

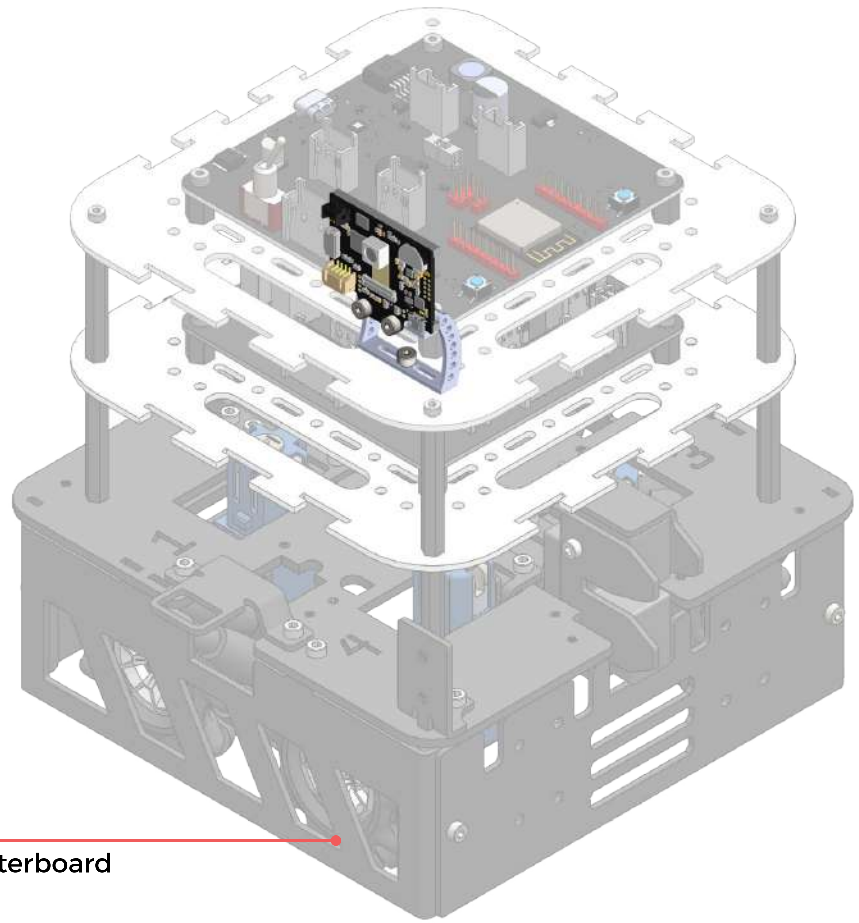


(mounting the huskycam)







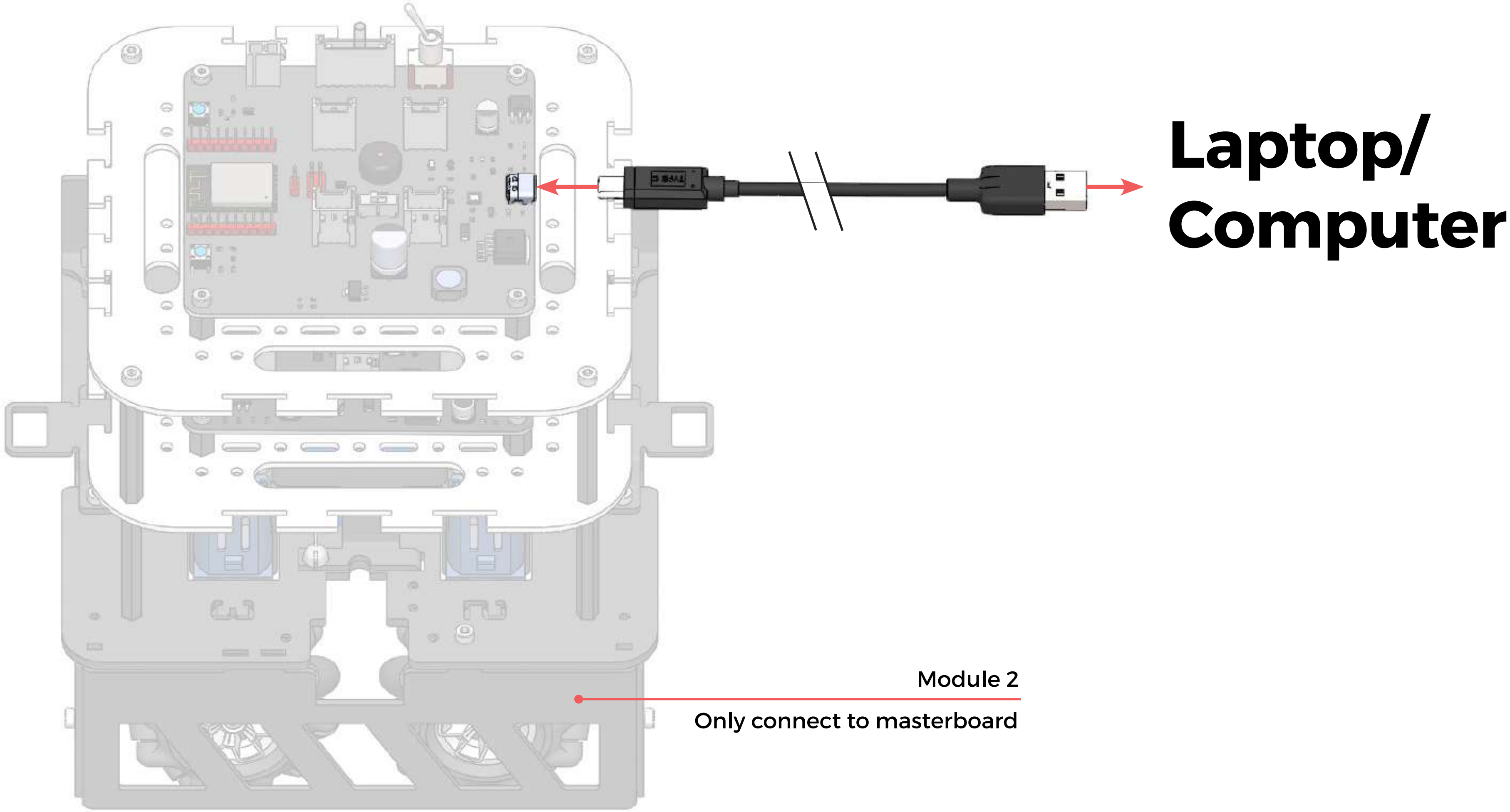


Module 2

Only connect to masterboard

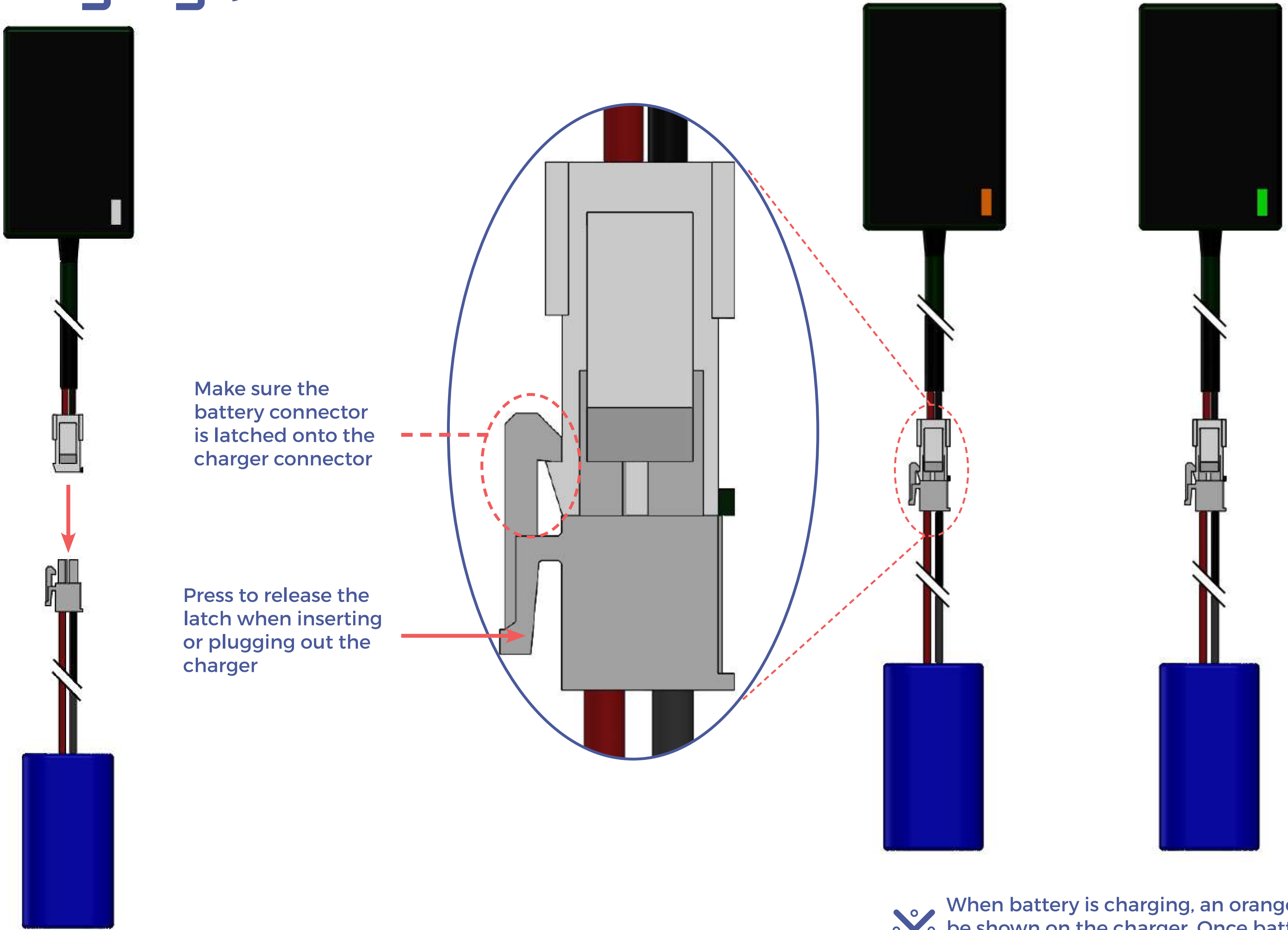
Huskycam can be mounted on anywhere on the acrylic board.

(connect to laptop)



Plug in the USB-C cable as shown above to connect the masterboard to the computer.
It allows us to upload our code from our computer onto the masterboard.

(charging)



When battery is charging, an orange light will be shown on the charger. Once battery is fully charged, a green light will be shown

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